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List of authors

Francesco Bailo is a senior lecturer at The University of Sydney, where he is also deputy director of the Centre for AI Trust and Governance. He is a social scientist interested in deliberation and opinion diffusion on social media. He is an expert in quantitative research methods applied to digital and spatio-temporal data. Over the last ten years, he has acquired substantial experience working in interdisciplinary and international research groups involving qualitative and quantitative researchers.

Emily Booth is a research fellow whose research spans young adults' reading practices, the relationship between their leisure reading and encounters with online misinformation, Australian high school curriculum design, and diversity in the Australian book publishing context. Her recent projects explore toxic masculinity and the manosphere in the Australian context, and she was a 2024 and 2025 finalist for the Australian Defence and National Security Workforce Awards for Best Female Technical Specialist Under 35.

Jennifer Williams is a lecturer in cybersecurity analysis, specialising in behavioural cybersecurity, at Macquarie University in the School of International Studies. She has both technical and nontechnical cybersecurity skills and a background in quantitative experimental cognitive science with a focus on information processing and decision-making. She has been involved in several multidisciplinary Defence-funded projects ranging from identifying and tracking foreign influence operations to developing a spatio-temporal crime prediction algorithm for the New South Wales Police Force to improve police resourcing.

Marian-Andrei Rizoïu is an associate professor leading the Behavioural Data Science Lab at the University of Technology Sydney. His interdisciplinary research crosses computer and social sciences, blending psycholinguistics, digital communication, and stochastic modelling to understand human attention dynamics in the online environment, the emergence of influence, and opinion polarization. He is the winner of the Excellence Award and Academic of the Year at the 2023 Australian Defence Industry Awards, and a two-time

finalist (2024, 2025) for the Australian Museum Eureka Prizes for Outstanding Science in Safeguarding Australia.

Lanqin Yuan is a PhD candidate in the Behavioral Data Science Lab at the University of Technology Sydney. He is a computer scientist with broad research interests in machine learning, data mining, natural language processing, and computational social science. His research examines the use of computational models and machine learning in addressing the spread of problematic speech online.

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Glossary

Term	Definition
Application programming interface (API)	A set of protocols and tools that allows different software applications to communicate with each other. Wikibase provides APIs for programmatic access to data.
Artifact	In research documentation, self-contained pieces of content that can be referenced and updated throughout a project.
Class	In ontology design, a category or type that defines groups of similar items. Classes help organise items hierarchically (e.g., ‘MP’ as a subclass of ‘Person’).
Command-line interface (CLI)	A text-based interface for interacting with computer programs, such as Terminal on a Mac or PowerShell on a Windows computer, used for installing and managing Wikibase on the remote machine.
Codebook	A document that defines variables, their meanings, and coding schemes used in research. In Wikibase, this becomes part of the project ontology.
Comma-separated values (CSV)	A simple file format that stores tabular data in plain text, with each row representing a record and columns separated by commas, commonly used for data exchange between different software applications.
Containerisation	A method of software deployment using Docker containers, recommended for Wikibase installation.
Data type	The kind of value a property can have in Wikibase (e.g., Item, String, URL, Time, Quantity).
Description	In Wikibase, a short explanatory text that clarifies what an item represents, helping to disambiguate items with similar labels.
Docker	A platform for developing, shipping, and running applications in containers, used in Wikibase deployment.
Docker compose	A tool for defining and running multicontainer Docker applications, used to orchestrate Wikibase components.

Term	Definition
Domain name system (DNS) record	A database entry that maps domain names (like ‘myproject.example.com’) to IP addresses, enabling users to access websites using memorable names rather than numerical addresses.
Entity	A general term for items, properties, or other objects stored in Wikibase that represent distinct concepts, ideas or things.
Extension	Additional software modules that add functionality to MediaWiki, such as Wikibase itself.
Federated properties	A feature allowing a Wikibase instance to use properties defined in another Wikibase (typically Wikidata).
GitHub	A web-based platform for version control and collaboration, where Wikibase source code and documentation are hosted.
Instance of	A fundamental property that links an item to its class, defining what type of thing it is.
Interdisciplinary research	Research that integrates views, methods, concepts, and theories from multiple disciplines to address complex problems.
Item	The basic unit of data in Wikibase, representing a distinct concept, object, or entity. Items are identified by Q numbers (e.g., Q1, Q2).
Iterative process	A cyclical approach to research where findings from one phase inform and refine subsequent phases.
Knowledge base	A structured repository of information that can be queried and analysed. Wikibase creates knowledge bases from linked data.
Knowledge graph	A network of interconnected entities and their relationships, forming a web of structured knowledge.
Label	The primary name or title given to an item or property in Wikibase, which can be translated into multiple languages.
Linked data	A method of publishing structured data where items are interlinked, allowing data to be connected and read across different sources.
Linux	An open-source operating system recommended for hosting Wikibase servers, with Ubuntu being a popular distribution.
MariaDB	An open-source SQL database system (and an alternative to MySQL) that can be used by MediaWiki and Wikibase to store data.

Term	Definition
MediaWiki	The free, open-source wiki software that powers Wikipedia and serves as the host software for the Wikibase extension.
Mixed-methods research	Research that combines qualitative and quantitative approaches to provide comprehensive understanding of research questions.
MySQL	See MariaDB.
Ontology	A formal representation of knowledge that defines entities, their properties, and relationships within a domain. The conceptual framework for organising data.
Persistent identifier	A permanent, unchanging reference to an item (Q number) or property (P number) that remains constant, even if labels change.
Point of interface	The stage in mixed-methods research where qualitative and quantitative data strands are meaningfully integrated, which may occur during data collection, analysis, or interpretation phases.
Prefix	In SPARQL queries, abbreviations (like ‘wd:’ or ‘wdt:’) that represent longer URL strings for more concise query writing.
Property	Attributes or relationships that describe items in Wikibase. Properties are identified by P numbers (e.g., P1, P2).
Q number	The unique identifier for items in Wikibase, consisting of the letter Q followed by a number (e.g., Q42).
Qualifier	Additional information that modifies or provides context for a statement, such as dates or sources.
Query service	See WDQS.
Resource description framework (RDF)	A standard model for data interchange on the web, used for representing linked data.
Schema	The technical structure defining how data is organised in a database, including entities and their relationships.
Semantic Web	An extension of the World Wide Web that enables machines to understand and interpret data meaningfully.
SPARQL	A query language for retrieving and manipulating data stored in RDF format, used to query Wikibase data.
Special page	Pages in MediaWiki that provide specific functionality, such as item pages or property pages, distinct from regular wiki pages.
Structured query language (SQL)	A programming language for managing and querying data stored in relational databases such as MySQL or MariaDB.

Term	Definition
Secure shell protocol (SSH)	A cryptographic network protocol for secure remote login and command execution on servers.
Statement	In Wikibase, a piece of information about an item, consisting of a property-value pair, potentially with qualifiers and references.
Structured data	Information organised in a predefined format that can be easily searched, analysed, and processed by machines.
Subclass of	A property indicating hierarchical relationships between classes (e.g., ‘MP’ is a subclass of ‘Person’).
Template	In MediaWiki, reusable pieces of wiki markup that can be inserted into multiple pages for consistent formatting or functionality.
Term	A collective name for labels, descriptions, and aliases associated with an item or property.
Thematic coding	A qualitative research method for identifying patterns or themes within data, which can be structured in Wikibase.
Triple	The basic unit of RDF data, consisting of subject-predicate-object (e.g., ‘MP-has-speech’).
Unstructured data	Information without a predefined data model, such as free text, images, or videos.
Uniform resource locator (URL)	A web address used to identify resources, including MediaWiki pages and external content.
Virtual machine	A software emulation of a computer system, often used to host Wikibase installations in cloud environments.
Visualisation	Graphical representation of data, which can be generated from SPARQL queries (e.g., timelines, charts, maps).
Watch list	A MediaWiki feature allowing users to monitor changes to specific pages or items of interest.
Web 3.0	The third generation of the World Wide Web, characterised by decentralised technologies, semantic data processing, and intelligent machine interpretation of content. See also Semantic Web.
Wikidata Query Service (WDQS)	The query engine that enables searching Wikibase data using SPARQL, allowing complex data retrieval and analysis.
Wikibase	Open-source software for creating and maintaining structured data repositories, originally developed for Wikidata.
Wikidata	The largest Wikibase instance, serving as a free, collaborative knowledge base that anyone can edit.

Term	Definition
WikiMedia Foundation	The non-profit organisation that operates Wikipedia and develops MediaWiki and Wikibase software.
Wikipedia	The free, multilingual online encyclopaedia that inspired the development of MediaWiki and Wikibase.
Wikitext	The markup language used in MediaWiki for formatting text and creating page elements like links and tables.

1. Introduction: the case for interdisciplinary mixed-methods research

Francesco Bailo, Emily Booth, and Jennifer Williams

It's day one of your new project, and everything is a fresh, blank slate to work on. Amidst the excitement, there are a few twinges of concern: since you and your team designed the project, one of your colleagues moved to a different university; you've discovered another colleague isn't familiar with your discipline's favourite everyday software; and most challenging of all, you have a brand new research assistant who needs to be instructed in both the project as a whole and a range of technical skills they've yet to learn. Caught up in the chaos, you focus on filling the gaps wherever you see them. Keen to make up for time lost due to delays, you power on ahead through the project and start collecting your data. Later—be it months or even years later—it comes time to write up what you've found. It's then that you discover your colleagues can't all access or use the system your data is stored in, and worse, your research assistant wasn't exactly clear on how to organise it, so they just—didn't. The thrills of research and working with a team can quickly be dulled by these all-too-common pain points of location, discipline-specific technical skills, and data management. Surely there's some kind of solution that can address these issues. Right?

For us, the answer to that question has been Wikibase. While we stop short of suggesting it is a universal cure-all to the challenges of project management and interdisciplinary research in teams, it has proven itself to us across a series of projects conducted since 2020. Each of these projects has involved an interdisciplinary team with members working from multiple locations, and with different skill sets befitting their range of scholarly backgrounds and research methods. Furthermore, each project has required multiple different types of data to enable both qualitative and quantitative analysis at the hands of humans and computers alike. In each scenario—including one project that directly overlapped with the 2020 COVID-19 pandemic—Wikibase has had a

vital role as a system for data management and analysis that is accessible to all members of the team. We would like to share how we made this approach work for us, in the hopes it will work for you, too.

INTERDISCIPLINARY RESEARCH

Interdisciplinary research is, at its simplest, any research conducted by scholars who identify with different disciplines. The research that emerges from an interdisciplinary project integrates different perspectives and values as well as cultural, theoretical, and methodological frameworks. At the end, the frameworks developed through the project will not squarely fit within the boundaries of any one discipline but instead will occupy a region intersecting these boundaries. As the project moves outside and criss-crosses disciplinary boundaries developed through decades of research practice and research communication (think about journals, conferences, job descriptions, etc.), so do the researchers. This means that interdisciplinary research is uncomfortable because it requires academics to step beyond the comfortable boundaries of their disciplinary traditions and collaborate in an intellectual space that belongs to no single field. And yet interdisciplinary research is necessary exactly because of the reasons that make it uncomfortable: some research problems are just too multifaceted to be comprehensively addressed using the toolbox from a single discipline.

If you are part of an interdisciplinary team, you probably have come to the conclusion that although your discipline could cut through some of the aspects of the research answer, it could not, by itself, provide the expansive answer required for connecting all the dots and providing a coherent interpretation and explanation of what is going on with the issue at hand. But when do you know that what you are doing is in fact interdisciplinary research?

The key concept of interdisciplinary research is *integration*. An interdisciplinary team must work from the very first day in the same room and around the same table, integrating assumptions, approaches, methodologies, design, and methods. And, yes, as you might have guessed, Wikibase can provide this required interdisciplinary table—a shared digital workspace where all team members collaborate simultaneously. Alternatively, in a *multidisciplinary* team, researchers can work in different rooms, each within their disciplinary boundaries, while contributing to the project only when everybody's work is done.

For example, and to simplify, climatologists, economists, and political scientists can collaborate on a project on the impacts of climate change where each discipline contributes its distinct expertise without necessarily integrating its theoretical frameworks or methodologies. The climatologists would provide findings on temperature increases, precipitation patterns, and extreme

weather events using established climate modelling techniques. Economists would separately calculate the financial costs of climate impacts—such as agricultural losses, infrastructure damage, and adaptation expenses—using their own economic forecasting models and cost-benefit analysis methods. Political scientists would independently examine policy responses, governance challenges, and international and subnational conflicts through the lenses of policy process, political, and international relations theories. In this multidisciplinary approach, the final research output would combine these separate disciplinary contributions into a comprehensive report, with each section reflecting the distinct methodological approaches and theoretical foundations of each field. The climatologists wouldn't modify their models based on political considerations, nor would the economists integrate climate tipping points into their financial projections in ways that fundamentally alter their analytical frameworks. Instead, each discipline maintains its autonomy while independently contributing valuable but separate pieces to understanding the broader climate challenge.

Comparatively, in an *interdisciplinary* team, contributions are *interdependent*. Climatologists might modify their models to incorporate political stability indicators identified by political scientists as factors affecting climate vulnerability, recognising that political instability can amplify climate risks. Economists would integrate these politically informed climate projections into their financial models, developing new economic frameworks that account for both physical climate risks and governance failures. Political scientists, in turn, would incorporate climate tipping points and economic disruption thresholds into their analysis of policy feasibility and conflict potential. This creates a shared analytical framework where climate science, economic impacts, and political responses are understood as interconnected systems, requiring new methodological approaches that transcend traditional disciplinary boundaries.

Another example is the analysis of the impact of communication technologies such as social media on social and political dimensions. An interdisciplinary team will develop integrated approaches, as demonstrated in our research on online problematic speech and Internet-mediated influence operations. In these projects, social scientists specialising in qualitative analysis didn't work in isolation from computer scientists developing computational methods. Instead, the team created shared frameworks that could accommodate both nuanced qualitative observations about hate speech and misinformation patterns and the structured data requirements needed for algorithmic analysis. The collaboration operated through iterative cycles where qualitative researchers' manual coding became training data for machine learning models, while algorithmic results were fed back to qualitative analysts for validation and refinement. Importantly, the entire interdisciplinary team collaborated through Wikibase as a shared workspace, where social scientists could enter and code

observations using familiar web-based interfaces, while computer scientists could access the same data programmatically for algorithmic processing, ensuring that insights from both qualitative interpretation and computational analysis remained connected and accessible to all team members throughout the research process.

A successful interdisciplinary project requires creating from day one an *interdisciplinary working space* that is accessible to every member of the team independently from their background and technical expertise. Information buried in technical jargon or complicated data packages only accessible through programming means or complicated user interfaces will not do. There are many examples of data sources that are critical for interdisciplinary research that are mostly not accessible outside of the discipline that created them. A good example of this is the climate data created by public agencies. Although this data is nominally downloadable by everyone with an Internet connection, in practice, only those with sufficient skills about the specific data format (e.g., GRIB, netCDF, HDF) and with the hardware and software tools required to read it can access it.

In addition to creating a shared working space, interdisciplinarity is also about developing a *common working language* to give each researcher the freedom to move independently within the interdisciplinary working space without requiring cross-disciplinary interpreters. Wikibase can provide this common, shared working space and facilitate the development of a shared working language that enables genuine interdisciplinary collaboration while maintaining the rigour and insights that each discipline brings to complex research problems. If this shared working space is the website where Wikibase and the knowledge graph can be created and navigated, the common language emerges from the capacity to layer structured and unstructured information in consistent ways, without sacrificing the inner complexity of the data. This is achieved as qualitative information becomes systematically structured through well-defined and meaningful properties and links, creating a unified framework (which we'll discuss in Chapter 6 as the project's 'ontology') that allows researchers from different disciplines to contribute, access, and analyse data independently from the disciplinary tradition of the research creating the data.

MIXED-METHODS RESEARCH

If interdisciplinarity defines the intellectual territory where your research team will develop its approach and establish a common language for collaboration, mixed-methods research defines your research *design*, research *methods*, and, critically, the data integration strategy—how different types of evidence will be collected, structured, and analysed together to address research questions that require both qualitative depth and quantitative breadth.

While interdisciplinarity addresses the challenge of bringing together different disciplinary perspectives, mixed-methods research provides the methodological framework for systematically combining and analysing diverse data types within that collaborative space. Not every mixed-methods research is going to be interdisciplinary, but, arguably, most interdisciplinary research is going to use mixed methods.

Mixed-methods research involves using both qualitative and quantitative data and methods within a single research project. But as interdisciplinary research is not just about piling up research findings from different disciplines but coherently integrating different perspectives to tackle complex research challenges, mixed methods is about coherently integrating the analysis of both types of data, qualitative and quantitative, into the same project to address a single problem. In other words, although the terminology can be confusing here, mixed-methods research is not merely about combining different *methods*—understood as the practical procedures for data collection, analysis, and interpretation. Mixed-methods research is fundamentally about adopting a comprehensive research *design* strategy that systematically integrates these methods throughout the entire research process, from initial conceptualisation through data collection, analysis, and interpretation, ensuring that qualitative and quantitative components work together synergistically rather than simply coexisting within the same project (see Creswell & Creswell, 2022).

Critically, whether your research project requires a mixed-methods approach or rather a quantitative or qualitative approach is one of the foundational questions of your project. How do we answer this question? We suggest breaking down this question into three preliminary questions.

First, can your research problem be convincingly addressed using knowledge generated from the analysis of only one type of data? If the answer to this question is positive, then you don't need to consider mixed methods. For example, if your research problem is about estimating the vaccination uptake patterns by age, income, and geographic location, you can effectively address it only by collecting and statistically analysing quantitative data. But if you want to understand not only what kind of people do not vaccinate their kids but what motivates them, then you could consider, for example, also collecting interview data and analysing it through thematic analysis.

Second, can you practically and ethically collect only one type of data? Again, if the answer is positive, you can't use a mixed-methods approach.¹ For example, studying sensitive topics like domestic violence or illegal drug use

¹ It should be noted that research teams cannot independently determine ethical appropriateness when human participants are involved. Any research involving human participants must undergo review by an independent ethics committee or institutional review board.

may make it practically impossible to conduct face-to-face interviews or surveys, as potential participants may be unwilling to discuss these experiences directly due to safety concerns, legal implications, or social stigma. Researchers might only be able to collect quantitative data from hospital records, police reports, or anonymous online surveys. Conversely, studying certain vulnerable populations—such as children in crisis situations or individuals with severe mental health conditions—may make it ethically inappropriate to burden them with lengthy quantitative instruments, leaving researchers to rely primarily on qualitative observational data or brief, carefully designed interviews. In historical research, practical constraints often dictate available data types: studying eighteenth-century social movements might only allow access to qualitative archival documents, letters, and newspaper accounts, as quantitative data collection methods simply didn't exist, or records have been lost over time.

Third, do you and your team have the analytical capabilities to work effectively with only one type of data? Mixed-methods research requires that your team possesses or can acquire the capabilities to collect and analyse multiple data types. These capabilities encompass both human expertise—such as whether team members can conduct qualitative interviews, perform statistical analysis, or interpret computational results—and technical infrastructure, including access to sufficiently powerful computers for quantitative analysis, appropriate software for qualitative coding, and technical support staff to maintain these systems. If your team lacks these diverse capabilities, attempting a mixed methods approach may compromise the quality of either the qualitative or quantitative components, ultimately weakening rather than strengthening your research.

If your answers indicate that a mixed-methods approach is possible for your project, it is now time to address possibly the most critical questions when designing mixed-methods research: where are the points of interface between qualitative and quantitative analysis, and how can they be integrated? We will discuss how Wikibase will help you and your team operationalise, in practice, the integration between different types of data in Chapter 2. But before your team starts collecting different types of data, it is critical to already have a good idea of at which point (or points) of the research process your mixed-methods data will be integrated and what the integration is going to look like. Failing to integrate data that you have already collected is not only a waste of time and resources; it can also have serious ethical implications and compromise the interdisciplinarity of your project. Ethical assessment of research projects depends not only on the risks posed to participants and researchers, but also involves a risk-benefit analysis where higher foreseeable benefits from research outcomes (such as reducing the likelihood of civil conflict or improving public health interventions) can justify higher levels of risk to participants. However, if researchers fail to properly use the data they collect, they create

only risks without generating any corresponding benefits. This violates the fundamental ethical principle that research should maximise benefit while minimising harm. In this sense, researchers must avoid replicating what digital platforms routinely do: collecting vast amounts of user data simply because they can, without clear purpose or justification.

In an age in which unprecedented volumes of personal data are readily available, it might be tempting to just collect it before designing an integration strategy. This approach is what Fuller (2023) refers to as ‘data surfacing’—also exemplified by data analytics company Palantir’s strategic premise that ‘all available data is potentially valuable’—which prioritises data acquisition over purposeful research design. But although this might be a sound business strategy, we strongly believe it is not a good *research* strategy. Mixed-methods research clearly opens opportunities to work with more diverse data types—in fact, potentially any data about people ever produced, regardless of format or content. However, because of the vast combinatorial possibilities that these opportunities offer, especially when data can be transformed into more variables than any person could meaningfully interpret, researchers could practically find support for any conclusion they wish to draw (we discuss how to use Wikibase to align a project to open science principles in Chapter 6). It is therefore critical that research design establishes clear boundaries around what data will be collected and how different data types will be meaningfully integrated to address specific research questions, rather than simply collecting everything available and hoping coherent insights will emerge.

OVERVIEW OF THE BOOK

Here we outline the structure of this book, so you can best determine which chapters are most relevant for your needs. In this introduction, we have established our experience with interdisciplinary mixed-methods research and highlighted why robust data management tools are essential for such collaborative endeavours. From here on out, the chapters become progressively more specific and practical.

Chapter 2, ‘What Is Wikibase and Who Needs It,’ introduces Wikibase in detail, covering its development history within the Wikimedia ecosystem, its key features for research applications, and the types of projects and teams best suited to benefit from its capabilities. We conclude this chapter by comparing Wikibase to alternative software solutions you might be familiar with—such as NVivo for qualitative analysis or Neo4j for graph databases—highlighting the unique affordances that make Wikibase particularly effective for interdisciplinary mixed-methods research.

Chapter 3, ‘How to Install and Maintain Wikibase,’ covers the technical implementation of Wikibase, providing a comprehensive guide to installation,

configuration, and ongoing maintenance. Although this chapter is primarily aimed at team members with advanced computer skills, we have made every effort to make it accessible to researchers with varying levels of technical expertise. The chapter includes system requirements, step-by-step installation procedures, backup strategies, and troubleshooting guidance, along with references to more detailed technical documentation available online.

Chapter 4, ‘Structuring and Linking Research Data,’ explores the theoretical foundations and practical applications of linked data within the Wikibase environment. This chapter introduces the principles of the Semantic Web and explains how research data can be structured to support the translation and transformation across the qualitative-quantitative divide that is central to mixed-methods research. We provide practical examples of how to organise data effectively to facilitate comprehensive analysis and demonstrate how structured linkages can create opportunities for both human interpretation and computational processing. The chapter also addresses publishing research data for reusability and scientific integrity.

Chapter 5, ‘Designing and Maintaining Project Ontologies with Wikibase,’ takes a step back from the technical implementation to focus on the conceptual framework that will guide your project. We discuss the creation of project ontologies—the conceptual models that define what entities exist in your research domain and how they relate to each other. While ontology development can be approached using traditional analogue methods and shares considerable overlap with familiar tools like qualitative codebooks, we provide specific guidance on how to implement and maintain ontologies within Wikibase’s infrastructure. This chapter emphasises the iterative nature of ontology development in interdisciplinary research, where conceptual frameworks must evolve as teams’ understanding of their data and research questions develops.

Chapter 6, ‘Designing and Running Collaborative Research Projects with Wikibase’ returns to practical implementation, focusing on the day-to-day workflows that make collaborative research possible. This chapter covers data entry procedures for different research designs—convergent, explanatory, and exploratory—and demonstrates how teams can navigate and query their data using both the web interface and SPARQL query language. We explore computational access to data through application programming interfaces (APIs), enabling automated data processing and large-scale analysis. The chapter includes detailed examples of human-in-the-loop machine-learning approaches that scale manual coding processes whilst maintaining research rigour. We discuss how to use Wikibase to align projects with open science principles, promoting transparency and reproducibility in collaborative workflows.

Chapter 7, ‘Collecting, Mapping, and Modelling Social Media Data with Wikibase’ presents detailed case studies from our own interdisciplinary

research projects investigating online problematic speech and Internet-mediated influence operations. These real-world examples demonstrate how teams comprising social scientists, computer scientists, terrorism studies experts, and digital media scholars successfully used Wikibase to manage complex, multimodal data involving textual content, multimedia files, network relationships, and temporal dynamics. The chapter illustrates how theoretical concepts from earlier chapters translate into practical research workflows, showing how ontologies evolve iteratively, how qualitative coding integrates with computational analysis, and how collaborative workflows enable genuine interdisciplinary integration.

The ‘Glossary’ section of the book provides a comprehensive reference for key terms and concepts used throughout the book. As this book is designed to be a resource for teams comprised of members from diverse disciplinary backgrounds, this glossary serves not only as a definitional reference but also as a bridge for understanding terminologies and concepts used by colleagues from different fields, supporting long-term collaborative success.

HOW TO USE THIS BOOK

This book is designed with flexibility in mind, recognising that different readers will have different needs and entry points:

- For research team leaders and project managers, Chapters 1, 2, 5, and 7 provide the conceptual framework and real-world examples needed to understand whether and how Wikibase might benefit your projects.
- For technical team members responsible for implementation, Chapters 3, 4, and 6 offer detailed practical guidance on installation, data structuring, and workflow development.
- For researchers new to interdisciplinary or mixed-method approaches, the entire book provides a comprehensive introduction, but Chapters 1, 2, and 7 will be particularly valuable for understanding the theoretical foundations and seeing practical applications.
- For experienced researchers exploring new collaborative tools, Chapters 2, 4, 5, and 6 focus on Wikibase’s specific capabilities and how they can enhance existing research practices.

While the chapters build logically upon each other, each is designed to be comprehensible on its own, allowing readers to focus on the sections most relevant to their immediate needs whilst providing clear references to related concepts discussed elsewhere in the book.

2. What is Wikibase, and who needs it?

Francesco Bailo, Emily Booth, and Jennifer Williams

WIKIBASE: DEVELOPMENT AND USE

BOX 2.1 KEY SOFTWARE COMPONENTS AND TERMINOLOGY

MediaWiki

- **Website:** <https://www.mediawiki.org>
- **What it is:** Free, open-source wiki software platform that powers websites like Wikipedia.
- **Primary function:** Enables collaborative creation and editing of web pages using wikitext markup language.
- **For researchers:** Provides the foundational platform for collaborative websites and basic documentation.

Wikibase

- **Websites:** <https://www.mediawiki.org/wiki/Wikibase>, <https://wikiba.se>
- **What it is:** MediaWiki extension that transforms a standard wiki into a structured data repository.
- **Primary function:** Enables creation, management, and querying of structured, linked data within a MediaWiki environment.
- **For researchers:** Your primary tool for building structured research databases, managing complex relationships between data points, and supporting mixed-methods research approaches.

Wikidata

- **Website:** <https://www.wikidata.org>
- **What it is:** The world's largest collaborative structured knowledge base, powered by Wikibase software.
- **Primary function:** Centralises and structures factual information from Wikipedia and other Wikimedia projects in a machine-readable format.

- **For researchers:** Serves as both an inspiration and a resource—you can reference its data structures, import relevant entities, and learn collaborative editing practices for your own projects.

Wikimedia Foundation

- **Website:** <https://www.wikimedia.org>
- **What it is:** Non-profit organisation responsible for running MediaWiki, Wikibase (through Wikimedia Deutschland, <https://www.wikimedia.de>), and Wikidata projects, along with the Wikipedia project.
- **Primary function:** Develops and maintains the open-source software ecosystem that powers collaborative knowledge creation.
- **For researchers:** Provides the organisational backing, infrastructure, and community governance models that make large-scale collaborative projects possible.

Key Historical Developments

The development of Wikibase is part of the history of Wikipedia, the online encyclopaedia. With millions of pages written in dozens of languages, Wikipedia is one of the largest websites on the Internet. Launched in 2001, Wikipedia is one of (if not the) most successful crowdsourcing projects, with hundreds of thousands of editors, actively editing and improving its content, and more than 100 million registered users.

Wikipedia is made possible by a sophisticated piece of free open-source wiki software called MediaWiki, originally created in 2002 for Wikipedia. If a wiki is a web document edited collaboratively, wiki software is the architecture that makes serving the page on the Internet and collaborating on and maintaining its content possible. Wikibase, on the other hand, is mostly about helping Wikipedia to order the content created by its editors.

The importance of Wikipedia (and of all the other projects run by the not-for-profit Wikimedia Foundation) is derived not only from its size, but also from the critical central role it plays across the Internet more broadly (Ford & Iliadis, 2023). When we search for information using a search engine or a chatbot (let's say, 'what is the capital of France'), it is very likely that the information that we will get was typed in by a Wikipedian, one of the millions of volunteer editors maintaining Wikipedia. But how was this information encoded to be consumed by the Internet platforms that we are interrogating every day? Between 2001 and 2012, the fact that Paris is the capital of France was encoded and presented in dozens of different Wikipedia pages across multiple languages. This, of course, created several limitations. First, facts presented on one page were not necessarily consistent with facts presented on

a different page (the population of France could be 68,373,433 on one page and 67,749,632 on a different page). Second, facts were not easily readable by machines, and therefore, they couldn't be easily integrated into other services. To address these limitations, in 2012 the Wikimedia Foundation launched a project called *Wikidata*. The objective of Wikidata was to centralise all the facts presented across all the versions of Wikipedia into a single knowledge graph. The Wikidata project allowed users to first create data objects that could be embedded across Wikipedia, and second, to *describe* all these data objects through meaningful relationships. This was done with the aim of facilitating the *discovery* (by people and machines alike) of new knowledge from the information contained in Wikipedia, hopping from one data object to the next—so from the COVID-19 pandemic to the COVID-19 misinformation to conspiracy theory to anxiety. The knowledge graph software developed by the Wikidata project to achieve these goals was named *Wikibase* and integrated into the MediaWiki software.

What Is Wikibase?

Wikibase is not a standalone software but an extension of the free and open-source collaborative MediaWiki software. In other words, to take advantage of the features of Wikibase, a full installation of MediaWiki is required (details on how to install MediaWiki and Wikibase are provided in Chapter 3).

Although MediaWiki and Wikibase are closely integrated, the two provide different functionalities. MediaWiki allows you and your team to run a full-fledged website, to be accessed privately or by the entire Internet, and to create and edit wiki pages through an intuitive user interface. These pages can contain everything you find in a regular web page: text, tables, images, videos, and links to other pages, and are written using *wikitext*, a popular markup language used to style text and specify page elements such as hyperlinks. The Wikibase extension instead allows users to create and edit 'Special Pages' representing a well-defined item, such as a document or a person, while specifying information about the item and its relationship with other items in the form of precise statements as *structured data*. The distinction between regular wiki pages created with MediaWiki and item pages and their properties created with the Wikibase extension is important, and we will come back to this later in the book.

MAIN FEATURES OF WIKIBASE FOR RESEARCH

Wikibase offers several useful features to facilitate collaborative research projects that are especially useful for interdisciplinary research. Wikibase enables the creation and management of a research data repository, transforming it

into a knowledge base that can be easily accessed by multiple users through a single website—hosted and served locally or globally via the MediaWiki software. But Wikibase also helps researchers work with both quantitative and qualitative data, regardless of their specific research methodology or disciplinary approach.

Types of Data

One of the most attractive features of Wikibase is that it allows users to store (and share with people and machines alike) any type of digital data and file produced through a research project. Your Wikibase will be, first, a website using wikitext, which allows your team to create and edit web pages using a regular web browser. These web pages can, of course, contain anything from text, images, and links to more specialised files, which are then accessible to everyone surfing the site from their own web browser. Second, your Wikibase will allow you to create special data items of your choice that can be highly structured through internal links pointing to properties and then to other items. As you store your research data in your Wikibase website, it is therefore useful to think about it not only as qualitative or quantitative information but as research data layers, which become progressively more structured as they progress from top to bottom.

Traditionally, research data is categorised as either *quantitative* or *qualitative*. Simply put, quantitative data is information that can be unambiguously expressed through numbers and that can therefore be analysed through statistical methods. For example, data collected with a multiple-choice survey question about voting in an election can be unambiguously counted and summarised, with the total number of respondents who have voted for each one of the parties available among the choices presented in the questionnaire. If quantitative data is expressed through numerical values, qualitative data is any type of data that contains information that cannot be unambiguously expressed with a number. Probably the most common type of qualitative research data is textual documents (e.g., news articles, policy documents, interview transcripts, social media posts, research notes). But qualitative data can also be collected as images, videos, audio, or any type of artifact.

By definition, mixed-methods research will collect and use both quantitative and qualitative data. Wikibase can easily and effectively store and serve through a web interface research data of any type, as long as it is available in a digital format. But Wikibase not only allows your team to store and retrieve data. In fact, Wikibase provides powerful features to flexibly add structure to your data, without compromising or transforming the original qualitative or quantitative characteristics of your data. Refer to Chapter 4 for more practical guidance on integrating qualitative and quantitative data into your Wikibase.

Layering Research Data, from Unstructured to Structured

If noncomputational researchers like to talk about qualitative and quantitative data, researchers such as computer scientists, on the other hand, like to refer to structured and unstructured data. In simple terms, *structured data* refers to data that has been ordered in some manner, whereas *unstructured data* has not been ordered. Therefore, it is imprecise to equate qualitative data with unstructured data and quantitative data with structured data. Just as data can be translated from qualitative to quantitative or vice versa, depending on different analytical methods, data more broadly can span the spectrum from structured to unstructured, to the semistructured space between.

What is the difference between structured and unstructured data? If you ask a computer scientist, they will probably explain that structured data is data that you can fit into some sort of table. In structured data, the elements contained in your data are explicitly related to each other and to properties that make sense in the real world. But then what are the elements of your data? The elements of your data are whatever is described by your data. In this book, using the terminology of Wikibase, we refer to these elements as *items*. So, for example, an element (i.e., an item) can be a person you have surveyed using a questionnaire. The structure of your data is given by the relationships that link that person, an item in your data, to other information also contained in your data, such as this person's age, whether this person voted (or not) for party X, or maybe the textual transcription of their response to an open-ended question (e.g., 'what do you think is democracy?'). Importantly, while machines process these relationships as simple links connecting a uniquely identifiable item (your respondent) to values, for human researchers, they carry precise real-world meanings: this person is 64 years old and voted for party X. We will come back to this in Chapter 5 when we discuss the difference between a schema and an ontology.

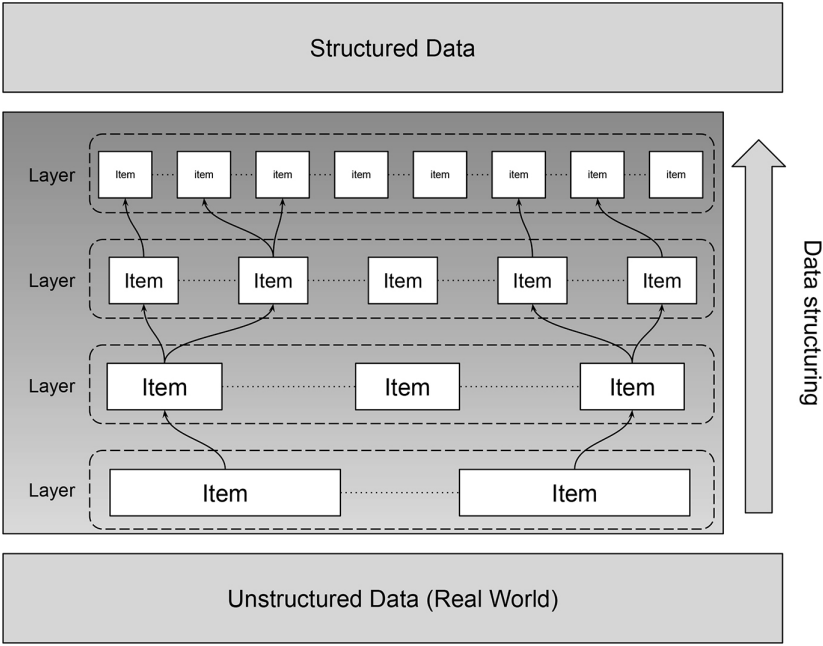
In comparison, unstructured data is data without such explicitly defined relationships. A classic example of unstructured data is a text document without any annotations. Another common example of unstructured data can be an image without any metadata. Semistructured data sits somewhere in the middle between structured data, where all the information is presented according to some pre- and well-defined format or model, and unstructured data, where information is presented without a format or model. An example of semistructured data can be a list of interview transcripts with the header of the documents always presented (in a standardised way), information such as the date of the interview, the names of the interviewee and interviewer, and the place of the interview. In a continuum from unstructured to structured data, these interview transcripts will probably sit closer to an unstructured data type. Yet, this can change during the life cycle of a research project. For example, researchers

can start collaboratively annotating these transcripts by linking parts in the text to themes. These annotations and relationships (between the document and some themes) should be intended as layering some additional structure to our research data. This layer of structure can, of course, be captured by the Wikibase features to make our data searchable and discoverable based on the relationships and structures we decide to overlay onto it.

The layering metaphor is useful to understand where and how our data should be stored in a Wikibase site. Wikibase offers you and your team sophisticated functionalities for your research data but does not dictate how you should use it. This flexibility gives you a significant margin for defining and designing how to best adapt it to your research needs. Let's again imagine that your project involves the collection of a series of interview transcripts. This data is clearly qualitative and, at least initially, mostly unstructured. Researchers would probably want to store each transcript as an individual wiki page within the Wikibase site. Again, this is a mostly textual document rendered as a traditional web page, with the only twist being that you can use wikitext to style your text and embed images or hyperlinks. This first external layer of data is therefore stored as unstructured data—or mostly unstructured, because by storing each transcript as an individual web page on our site, we are, of course, giving our data some order and structure already. In some sense, we are already defining what some important items in our research project are: our interviews.

As visualised in Figure 2.1, we can then quickly layer additional structure to our data by creating for each interview a new *item* and linking it through specific properties to other items and values (which in Wikibase are called qualifiers). As we will discuss in more detail in Chapter 4, in Wikibase an item is created as a special page—to differentiate it from regular wiki pages—with another special page being a new property page. By defining each interview as an item within our Wikibase site, we can start adding structure to our data. For example, our interview item special page can contain a link (also intended as a property and relationship) to the wiki page that contains the full transcript of the interview. Although item pages can store text, it is good practice to store long texts somewhere accessible as unstructured data (structured data, while machine-readable, isn't the most user-friendly format for reading lengthy texts like full interview transcripts). Therefore, your interview item page does not contain the actual interview transcript itself, but instead links to it using a relationship—what Wikibase calls a 'property'—that we might label 'interview transcript'.

We can also define additional properties and qualifiers for our interview. For example, if the same interviewer conducted more than one interview, and if we think it is important to be able to relate each interview to its interviewer, we can define our interviewer as a new item in the Wikibase site and then link



Note: Research data progressively gains structure through collaborative annotation and linking. Each layer preserves the original data while adding semantic relationships. Vertical links connect items to their properties, while horizontal relationships enable comparative analysis across items at the same structural level, supporting both qualitative and quantitative research approaches.

Source: Authors.

Figure 2.1 Data layering structure in Wikibase

the interview item to it with the property ‘interviewer’. Moreover, if we want to specify the date and place of the interview, we can add them with properties we can call ‘date’ and ‘place’, which will then point to a date and text qualifier on the interview item page. But these structure layers are likely only the very preliminary steps of a research project. What do we do when we start analysing the interviews? We can just keep adding structure or ‘order’ to our data.

At this stage, we have a simple wiki page containing the full transcripts of the interview (maybe also enriched with the video recording of the interview), which everybody in our team can access concurrently as a regular web page from their browser, but also a special page, giving some basic and preliminary structure to our interview item, such as the interviewer, the date, and the

place. Where can our team start annotating the interview transcript? Again, Wikibase is not prescriptive in terms of how you should use it, and the answer probably depends on where you want to be on the continuum between fully unstructured data and fully structured data. Your team has now two options. First, one or more members of your team could start adding their annotations, comments, or notes directly in the wiki page containing the transcript. As Wikibase is part of MediaWiki, the software that powers Wikipedia, it is perfectly able to deal with concurrent users and versioning of text documents—in addition to complex styling of text. By adding annotations directly to the text of the interview, maybe leveraging the markup features offered by wikitext, we add an additional thin layer of (semi)structure to our data. We refer here to a semistructure because in adding these annotations our research team is not necessarily following a predefined standard (something we will later refer to as ontology). As a consequence, our annotation can be effective as a collaborative preliminary step but won't allow our team (or a machine) to easily compare and analyse *across* interviews.

A solution to this would be, as a second option, to add a thicker layer of structure to our data, fully leveraging the functionalities offered by Wikibase. Again, there are probably as many ways to do this as there are researchers, but the simplest possible approach would be to add new relations between the item interview and items representing, for example, themes emerging in the interviews (and relevant to our research problem) or, as other examples, places and people mentioned by the interviewees. How many layers to add or how much to structure our data depends on our research design and, more broadly, on the type of questions we want to ask our data and the methods we want to apply to answer them.

Querying the Knowledge Graph

As we keep adding layers of structure to our data, we create a *graph*. A graph is here intended as a mathematical network-like structure linking items (also known as vertices or nodes) through relations (also known as edges). A graph can be visualised, providing an insightful overview of how our interview items are linked to common items such as themes. Because we attach a meaning to the relations of our graph (after all, we have created and labelled them!) we can also refer to it as a knowledge graph. The knowledge graph is an effective research tool for mixed-methods research because if a machine only needs to see the graph structure to traverse it while querying our data, humans can also see and learn from the *semantic* patterns emerging from the same graph. Because of this, the idea of knowledge graphs is strongly related to what we refer to as the Semantic Web. The Semantic Web is about developing a set of standards to make the information served across the Internet 'understandable'

by machines, as humans explicitly link this information through relations that are meaningful to people.

Because these standards are also implemented by Wikibase, we can use a semantic query language called SPARQL to construct complex queries about our data. For example, we could construct a query to return all the interviews conducted within a time frame and in which a specific theme appears. As we will see, the Wikibase query engine can then return the results from our queries as a table but also a graph. The Wikibase query engine (which we refer to as the Wikidata Query Service [WDQS]) is a powerful feature to navigate and monitor our data in real time as we keep adding complexity to its structure.

In conclusion, Wikibase, as part of the MediaWiki collaborative software, provides research teams with a powerful tool not only to efficiently and effectively store their digital data but also to work on it by flexibly and collaboratively adding order to the data in the form of successive layers of structure that can be navigated to seek insight and ‘extract knowledge’ for the entire database. Let’s now briefly discuss how Wikibase could be used under different research goals and designs.

WIKIBASE FOR MIXED-METHODS RESEARCH

Different Research Goals

Wikibase is an extremely flexible tool to accommodate a wide range of data research needs and goals. Indeed, researchers might have different goals for their projects when using Wikibase. Researchers might have a research question to answer. In this case, Wikibase will help collect, organise, and analyse their data while shaping the structure of the data, whether qualitative or quantitative, in support of their specific approach. But researchers might also use Wikibase if their project is only about creating and sharing research data—that is, without addressing any project-specific research question but instead with the aim of creating a knowledge base to be used by other researchers or to be easily accessed by a nonacademic audience.

Indeed, it is quite common nowadays for the primary (or sole) output of a research project to be a database usable by the general public, policymakers, or other researchers. In other words, a research project can be about the systematic process of collecting, organising, cleaning, and annotating data, with the primary aim of creating a high-quality, well-documented, and reusable database. In this situation, the focus will be on ensuring data integrity, accuracy, and relevance to facilitate further research, replication, and knowledge dissemination. Wikibase is an ideal software companion, as it is able not only to collect and structure qualitative and quantitative data but also to easily and

effectively serve the resulting knowledge base through an intuitive website to a generalist public without further work or additional software components.

Rhizome is a non-profit affiliated with New York's New Museum of Contemporary Art.¹ Its ArtBase project (https://artbase.rhizome.org/wiki/Main_Page) represents an exemplary case of Wikibase managing inherently multimodal research. The project's archive contains 'born-digital' artworks spanning net art, software, code, websites, moving images, games, and browsers—data types that resist traditional categorisation and require flexible structural relationships. Rhizome adopted linked data standards and Wikibase for this project specifically because digital culture 'involves quickly changing materials and terminology, driven by the pace of software development and the rapid rise and fall of cultural practices', requiring a knowledge base that doesn't rely on fixed metadata standards. This implementation demonstrates precisely why Wikibase excels for mixed-methods research: it enables collaborative knowledge building across diverse data types whilst preserving complex contextual relationships. The platform allows 'input from various user communities and the broader digital preservation field' through iterative development, mirroring how interdisciplinary research teams need systems where qualitative researchers, quantitative analysts, and domain experts can contribute without being constrained by rigid schemas. Most critically, Wikibase captures the 'interrelationships among artifacts and their contexts'—exactly what mixed-methods researchers require when linking quantitative data points with rich qualitative context, ensuring that different types of evidence remain connected rather than siloed in separate systems.

Different Research Designs

Wikibase can support different types of research design choices. Wikibase offers a very valid tool to projects that only deal with either qualitative or quantitative data. But it also offers powerful features when it comes to *integrating* qualitative and quantitative data in the context of so-called mixed-methods research.

According to Creswell and Clark (2018), in mixed-methods research, there are three 'core' types of design—which of course can then be declined in many different specific types of design. Which core types a mixed-methods research team is using depends on the order in which data collection and analysis of qualitative and quantitative data are conducted. If qualitative and quantitative data are collected and analysed at the same time, either by analysing the two types of data independently and then comparing the results, or by transforming

¹ <https://artbase.rhizome.org/wiki/About>.

one type into the other so that qualitative and quantitative data can (at least partially) be analysed together, we say we have a *convergent research design*. If quantitative data is collected and analysed first, then the results and qualitative data are used to further explore in more detail some of the dimensions of the patterns highlighted in the quantitative analysis. We refer to this as an *explanatory research design*. Finally, if we collect and analyse qualitative data first and then transform some of the characteristics of the qualitative data into quantitative variables to be applied in the analysis of additional quantitative data, we refer to this as *exploratory research design*.

We refer to Creswell and Clark's (2018) excellent textbook and also to Chapter 6 for more details on designing your mixed-methods research. In general, designing mixed-methods research, therefore using both qualitative and quantitative data and methods, requires a precise and methodologically sound definition of the '*point of interface*, or the point in the research process in which the two components [that is, the qualitative and the quantitative] meet' (Morse & Niehaus, 2009, 55). Critically, the identification of data as either qualitative or quantitative is dictated not only by practical considerations (e.g., I can't analyse these documents qualitatively because there are too many) but by an ontologically and epistemologically well-defined research idea about what the data represents and how researchers should access this representation. This is particularly relevant as new, powerful generative AI tools are becoming available, giving researchers the possibility to access *qualitative* data that, without sampling, would have been in the past only analysable *quantitatively* (i.e., large collections of texts and images).

The point (or points) of interface can occur during the data analysis, as the qualitative and quantitative data are at least partially integrated, or after the data analysis, as the *results* (and not the data) from the two research components are integrated and used in the explanation or description of the subject of the study. An example of qualitative-to-quantitative integration is the identification of *qualitative* themes that can then be identified and counted across quantitative data. An example of quantitative-to-qualitative integration is instead the *quantitative* identification of patterns that are then more deeply explained through the qualitative analysis of the same patterns—either through sampling additional qualitative data or a qualitative investigation of the original dataset.

Wikibase helps explicitly define and communicate the points of interface based on your research design. These can be expressed through the definition of structured references, as one data item links to another through some semantic link. Yet, Wikibase will not prescribe or necessarily inform how to define the process and the sequencing of your research. Indeed, because of its characteristics and features, Wikibase should be intended as fundamentally a data-centred research tool to be used during the data phase (collection, analysis, and

reporting) of a research project and not to shape or define its research design and processes (which should always precede the data collection!). This is why we find it useful to term the relationships that Wikibase allows you to define as the *layers* of a structure that you progressively lay over your data, turning it from unstructured to progressively more structured. Although this structure should, of course, be compatible with the worldviews behind the design of your research project, we think that it is also useful to allow for the definition and emergence of this structure during the research project independently from the sequencing and overall design of your research.

ALTERNATIVES TO WIKIBASE FOR RESEARCH

In this section, we list and briefly discuss alternative software options that utilise similar features to those of Wikibase and Mediawiki.

Graph Databases

Neo4j (<https://neo4j.com/>) is a powerful native graph database that stores data as nodes and relationships rather than tables. It excels at managing complex, interconnected datasets and has been used in high-profile investigations such as the Panama Papers analysis, which involved the systematic investigation of 11 million documents (Srivastava & Singh, 2018). However, Neo4j requires significant technical expertise and lacks the collaborative wiki-style interface that makes Wikibase accessible to nontechnical team members.

Semantic Wiki

Semantic MediaWiki (<https://www.semantic-mediawiki.org/>) extends standard MediaWiki functionality with semantic capabilities, allowing users to add structured data annotations to wiki pages. Whilst it shares MediaWiki's collaborative foundation with Wikibase, it focuses more on enhancing traditional wiki pages rather than creating dedicated knowledge graphs.

Qualitative Data Analysis Software (QDAS)

NVivo (<https://lumivero.com/products/nvivo/>) remains the research industry standard for qualitative data analysis, offering sophisticated coding capabilities and support for multiple data types, including text, audio, and video, and for collaborative workflows. However, NVivo is proprietary software that requires, in most scenarios, a licence and paying subscription fees.

Among other paid QDAS are Atlas.ti (<https://atlasti.com/>), which provides similar qualitative analysis capabilities to NVivo, with strong coding and

memo features, and MAXQDA (<https://www.maxqda.com/>), which offers mixed-methods analysis tools that integrate quantitative and qualitative approaches within a single platform. Taguette (<https://www.taguette.org/>), on the other hand, represents an open-source alternative that supports collaborative tagging and coding of text documents through a web interface.

3. How to install and maintain Wikibase

**Lanqin Yuan, Emily Booth, Jennifer Williams,
and Francesco Bailo**

Wikibase is a website, and therefore it requires installation on a computer (or better, a server, since you want it to always be switched on) that is connected to the Internet. This chapter provides a practical guide to installing and maintaining Wikibase using the ‘containerised’ approach recommended at the time of writing by Wikimedia Deutschland, which has been leading the development of Wikibase since the project’s inception. We begin by outlining the technical competencies and system requirements needed for a successful installation. The chapter then walks through the complete installation process, from setting up the prerequisites to configuring your Wikibase instance for research use.

For many researchers, particularly those working within university environments, you may need to share this chapter with your institution’s IT staff to collaborate on setting up a working instance of MediaWiki and Wikibase, inclusive of the query engine that enables you to search your data using SPARQL. University IT departments often have specific policies regarding server deployment, security requirements, and resource allocation that will need to be considered during the installation process.

The Wikibase ecosystem evolves rapidly, with regular updates to both the software and deployment methods. Whilst this chapter provides current best practices as of 2025, we strongly recommend always consulting the most up-to-date official documentation (<https://github.com/wmde/wikibase-release-pipeline/tree/main/deploy>) before beginning any installation. The official documentation will contain the latest installation instructions, system requirements, and any important changes that may have occurred since this book’s publication.

Before installing Wikibase on your machine, consider whether the service offered by Wikibase Cloud (<https://www.wikibase.cloud/>) meets your project requirements, particularly regarding public access and data compatibility. Wikibase Cloud provides a hosted solution that allows you to create your own Wikibase instance without the technical overhead of local installation and maintenance. This cloud-based option can be particularly valuable for research

teams who lack technical expertise or institutional IT support, as it eliminates the need to manage servers, security updates, and system administration tasks. Check with your university IT system and ethics department to ensure this option aligns with your institution's data security and ethics policies.

BASIC TECHNICAL COMPETENCIES

We begin with an overview of the technical competencies recommended to follow the installation guide.

To proceed, the user must have full network, administrative, and access privileges on the computer. Specifically, they should be able to configure network settings access all system files and install new software on the machine. These privileges are typically available when working on a fresh machine. However, if the machine is managed by an external entity (e.g., a university IT department), the user should consult the administrator to ensure they have the necessary permissions.

Basic competency with a command-line interface (CLI), such as Terminal or PowerShell, is recommended. This familiarity is important since much of the installation process takes place in a CLI.

Some knowledge of basic networking concepts and terminology is also beneficial. Key terms include internet protocol (IP) address, ports, the distinction between local and external IP addresses, proxies, and domain name system (DNS). These concepts appear frequently in this chapter, and a general understanding of them will aid in following the guide.

Finally, discussion regarding the installation of Wikibase will focus on the installation process on a computer running the Linux operating system. Linux is a free and open-source operating system available in many different flavours, with the Ubuntu distribution being most popular as of 2024. We chose to focus on Linux, as it is the most common operating system used by server hosting services. Wikipedia itself is run on a Linux server machine. It is also significantly easier to access documentation and troubleshooting support for Wikibase on Linux than on other operating systems.

SYSTEM REQUIREMENTS FOR WIKIBASE

Hardware Requirement

The exact system specifications required to run Wikibase changes subject to updates to the Wikibase software. Yet, as of 2025, what is required in terms of hardware is the following.

1. A computer connected to the Internet through a public IP address. This simply means that your computer (i.e., server) can be reached by any other computer also connected to the Internet using an address such as 192.0.2.1 using a browser or a terminal.
2. The central processing unit (CPU) of your computer should be x86_64 (AMD64). This is the most prevalent architecture across both desktop computers and servers.
3. Finally, your computer should have at least 8 GB RAM and 4 GB free disk space. As a general rule of thumb, bigger projects will require more system resources, specifically more disk space.

Wikibase will need to be run on a computer in order to work and deploy your website throughout the Internet and to your team members. However, maintaining a physical machine 24/7 comes with several challenges. The machine must meet the aforementioned system requirements to run Wikibase, which can be costly if you do not have a spare suitable computer lying around. Furthermore, the machine will need to be exposed to the Internet at large, which brings along many security-related challenges that must be addressed, as the machine can be attacked by malicious actors from across the globe.

Due to these factors, we recommend running Wikibase on a virtual machine (VM) hosted by a cloud service provider like Amazon AWS, Google Cloud, or Microsoft Azure, rather than maintaining a physical machine locally, such as a laptop or a desktop computer (especially if this machine is not offered and maintained by your IT department). A virtual machine is a software-based emulation of a physical computer. Multiple virtual machines may be run on a single physical machine. In essence, virtual machines can be understood as having a computer within a computer.

Shifting the responsibilities of maintaining a physical computer to a third-party host has several benefits. First, cloud providers typically offer flexible, pay-as-you-go pricing models. This allows you to avoid the high upfront cost of buying hardware, which can be particularly useful if your Wikibase deployment is only needed for a specific project or time period. Furthermore, cloud providers typically offer a variety of security features such as firewalls, distributed denial-of-service (DDoS) attack protection, and automated machine backups. These features would have to be manually implemented on a local physical machine. Finally, the cloud-hosted VMs are typically flexible in terms of their computational resources, meaning you can easily adjust resources based on your project needs, should they change. For example, if your Wikibase project grows over time and requires more storage or computing power, you can add these resources on demand without the need to physically upgrade hardware.

Domain Name and DNS Records

Although it is not strictly required, it is also recommended that you purchase an Internet domain (e.g., ‘example.com’) that allows you to define two domain name system (DNS) records: one for the website of MediaWiki and Wikibase (e.g., ‘myproject.example’) and one for the website serving the user interface to the query engine (e.g., ‘query.myproject.example’). There are many registrars that allow you to purchase and manage your domains and DNS records for (depending on the name and the suffix) less than 5 USD per year. It is important to have a stable domain name because you can hardwire it into every uniform resource locator (URL) internal to your project. The URL that contains your domain name will not need to be rewritten if you happen to change the IP (that is, the public number) of your server, for example, because you have moved your site across machines.

Software Requirement

Once you have secured a computer and a domain name for your project, you are now in the position to install and run Wikibase, which, as free and open-source software, requires no additional software costs.

Wikibase can be installed in two ways: directly on the machine’s file system or using containerisation software such as Docker, which we recommend. Docker is a program that enables containerisation. One can understand containerisation, and by extension Docker, as isolating different pieces of software that are installed on a single machine. An analogy would be to think of software as fruits and vegetables in the fridge, where the fridge represents the computer on which the software is installed. Using containerisation would be to store each piece of produce in a separate plastic food container in the fridge, whereas not using containerisation would be to store all produce in the fridge directly. This has several benefits, with the most obvious being that one bad apple does not spoil the whole fridge. This extends to computer security, meaning that should the Wikibase be compromised or crash, the containerised program would not affect other processes on the machine, hence making the machine more secure from errors and attacks. Another benefit of Docker is that the installation process is significantly more straightforward than manual installation. Going back to the food analogy, one can think of Wikibase as a meal. Manual installation would be to buy the ingredients bought from the supermarket, cutting and preparing the vegetables yourself, then finally cooking the meal before storing it in the fridge. Installing through Docker is more similar to getting a prepackaged meal box, where all the ingredients are already sliced and in the right portion for the meal.

For the purposes of this guide, we will focus on the installation using Docker. Therefore, you will need the following:

1. Docker 22.0 (or greater) and Docker Compose 2.10 (or greater). Docker is a containerisation software that allows you to package applications and their dependencies into lightweight, portable containers. Docker Compose is a tool for defining and running multicontainer Docker applications, which is essential for Wikibase, as it requires multiple interconnected services (database, web server, query service, etc.) to work together.
2. Git. Git is a version control system that we use to download the Wikibase Suite Deploy files from the official repository. It allows you to easily obtain the latest deployment configuration and update it when new versions are released.
3. MediaWiki, which of course will contain Wikibase. MediaWiki is the wiki software that powers Wikipedia and serves as the foundation for Wikibase. When using WBS Deploy, MediaWiki comes prepackaged with Wikibase and other necessary extensions, so you don't need to install or configure these components separately.

HOW TO INSTALL WIKIBASE FOR INTERNET ACCESS

We now give a step-by-step guide on how to install Wikibase on a Linux machine using Docker. We are here referring to and explaining the list of instructions presented at <https://github.com/wmde/wikibase-release-pipeline/tree/main/deploy> by the Wikimedia Deutschland team.

The Wikibase is a convenient way to install in one go all the software components and services you will need to run Wikibase. Among these, the critical components are

1. MediaWiki, including the Wikibase extension.
2. MariaDB, the structured query language (SQL) database service where MediaWiki can store all the data it needs to operate and where the data you are going to enter into the site are also going to be stored.
3. The backend, frontend, and updater for the Wikidata Query Service (WDQS), which is the engine that will allow you to query your structured data using SPARQL (more on this in Chapter 4).

If you have secured access to your Linux machine, you can now log into that machine using a CLI, which will likely be either Terminal if you are accessing your Linux machine from a Mac computer or PowerShell if you are Windows computer. In this process, your Mac or Windows computer—the one you're

physically typing on—serves as your *local* machine. The Linux machine where we will install Wikibase becomes the *remote* machine. Using Terminal or PowerShell on your local, you can use secure shell protocol (SSH) to log into the remote machine. The command to enter in your CLI will look like this:

```
ssh myusername@192.168.1.1
```

In this, ‘myusername’ should actually be replaced by your username on the remote machine and ‘192.168.1.1’ with the IP of the remote machine. After entering your password, you will now be able to execute commands on the remote machines.

To begin, ensure Docker and Docker Compose are installed on your remote machine (from this point forward, all commands will refer to the remote machine unless otherwise specified). Verify that Docker, Docker Compose, and Git are installed by running the following commands in a CLI:

```
docker --version
docker compose version
git --version
```

If you get an error because some of these are not installed, use your package manager to install them. On Ubuntu and Debian, this can be done using the following commands (although the exact command to label the software might vary):

```
apt install docker.io docker-compose-v2 git
```

Next, obtain the official Wikibase file for installation. This can be downloaded on your remote machine using Git from the official GitHub repository of Wikimedia Deutschland (this will also ensure that you will always download the most recent version of the software).

```
git clone https://github.com/wmde/wikibase-release-pipeline
```

The Git command ‘clone’ will simply make a copy of the software package on your server. Now we can move into the package we just downloaded using their folder structure with the command ‘cd’ (for ‘change directory’).

```
cd wikibase-release-pipeline/deploy
```

The Wikibase Suite Deploy installation is controlled mainly by two files: ‘docker-compose.yml’, which specifies parameters of the Wikibase installation, and ‘template.env’, a template environment file, which Docker will read to set the Wikibase environment parameters. These are plaintext files in this directory that we must modify using a command-line interface text editor like GNU nano (or an alternative, Vim) before installing Wikibase.

First, let’s make a copy with the command cp of the configuration template file ‘template.env’ to a new file, ‘.env’.

```
cp template.env .env
```

We will now edit the newly created file with a text editor (in this case, GNU nano)

```
nano .env
```

We will begin by defining the admin account on your Wikibase. This account will be the administrator account and have access to all of the Wikibase. Set the admin email, username, and password (longer than ten characters) by editing the following lines to the desired values after the equal sign:

```
MW_ADMIN_PASS=change-this-password  
MW_ADMIN_NAME=admin  
MW_ADMIN_EMAIL=admin@example.com
```

Now set the database name, admin account username, and password by editing the following lines to the desired values after the equal sign:

```
DB_NAME=my_wiki
DB_USER=sqluser
DB_PASS=change-this-sqlpassword
```

Finally, set the URL for where you will be hosting Wikibase by editing the following line:

```
WIKIBASE_PUBLIC_HOST=wikibase.example
WDQS_PUBLIC_HOST=query.wikibase.example
```

Here is where you will specify the two DNS records from the domain you have purchased, as explained above.

You have now successfully configured your environment file for Wikibase! You can now exit your text editor (in GNU nano you can do so by pressing Ctrl and X on your keyboard) after confirming you want to save it with Y.

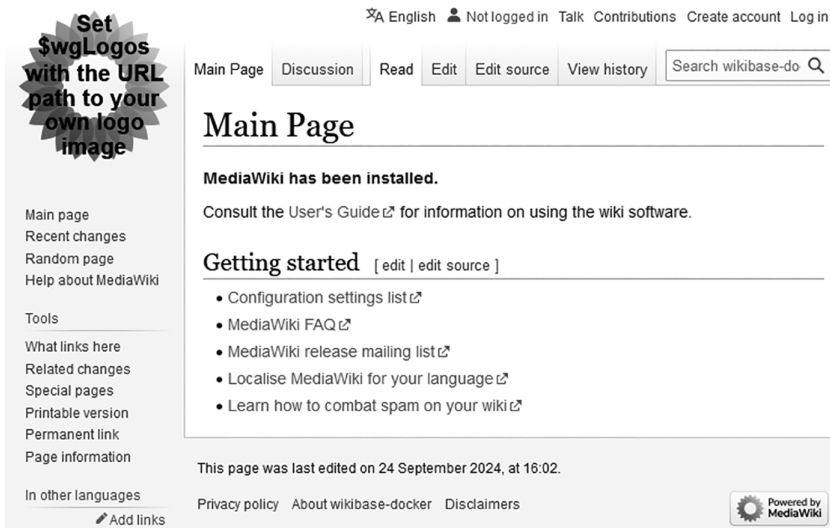
If additional Wikibase services and networking configuration are required (e.g., changing port numbers or reverse proxy settings), custom settings should be specified in the Docker configuration file ‘docker-compose.yml’. Assuming no additional configuration is required, we are now ready to get our Wikibase running and online.

To start Wikibase, run the following command from the ‘wikibase-release-pipeline/deploy’ (where you should still be if you haven’t changed folders):

```
docker compose up -d
```

Note that if you get a ‘permission denied’ error, you must use the ‘sudo’ command before ‘docker compose’, as (depending on your system) this operation might require superuser privileges. If you don’t have these privileges, you must reach out to whoever helped you set up your server (i.e., your remote machine).

The command you just entered tells Docker (a tool of the Docker software) to follow the instructions contained in the ‘docker-compose.yml’ in the current directory and set up (i.e., compose) all the services specified there and required by Wikibase (Wikibase, database, query engine, etc.). The option -d simply tells Docker to return to the prompt to run other commands after the containers have started.



Source: Authors.

Figure 3.1 Wikibase home page

This operation can take from a few seconds to a few minutes. If this was done successfully, you should see no error messages and a list of all the containers that have successfully ‘started’ or are ‘healthy’.

In case you want to shut down Wikibase, you can run the following command from the ‘deploy’ directory:

```
docker compose down
```

If the installation was successful, Wikibase is now ready to be used from your web browser.

By entering the domain you have purchased for Wikibase (and the one you have specified in your ‘.env’ file), you should see a web page similar to that shown in Figure 3.1.

Log in as the admin account using the credentials you defined in your ‘.env’ file to ensure the configuration was successful. Congratulations, you have just successfully set up Wikibase!

CONFIGURING WIKIBASE AFTER THE FIRST LAUNCH

At this stage Docker has set up a working version of Wikibase on your remote machine. The first time you execute Docker Compose, the installer script will create a folder called ‘config’ in the ‘deploy’ folder. Within this folder there is an important MediaWiki configuration file you might want to edit to fine-tune how it works. ‘LocalSettings.php’ is the MediaWiki (and Wikibase) primary configuration file. This file contains all site-specific settings for the Wikibase instance, allowing you to customise various aspects of your installation, such as including language settings, access permissions, and enabling extensions.¹

Before editing this file, it is always advisable to make a copy of the original configuration as a reference and fallback option in case we accidentally break something. From the ‘deploy’ folder, we can move to its ‘config’ subfolder and make a copy of the original configuration file with

```
cd config
cp LocalSettings.php LocalSettings.php.backup
```

We are now ready to edit the configuration file with

```
nano LocalSettings.php
```

As before, we might require adding the command ‘sudo’ before ‘nano’ to be able to save the ‘LocalSettings.php’ file after editing it. Also, the ‘LocalSettings.php’ created by the installer during the first launch of Wikibase will only contain a selection of variables to configure (in PHP variables are defined using the dollar sign prefix ['\$'], followed by the variable name of the assignment operator [=], followed by the new value you want to store, which is quoted in case of textual values and followed by a semicolon [;] to terminate the statement). When a variable to redefine is not already included in ‘LocalSettings.php’, the corresponding statement can be added at the bottom of the file.

Change the Name and Name Space of Your Wikibase Site

The variables we want to change to personalise our site are ‘\$wgSitename’ and ‘\$wgMetaNamespace’ by indicating the project name there.

¹ We refer to the official documentation at <https://www.mediawiki.org/wiki/Manual:LocalSettings.php> for details on this configuration file.

```
$wgSitename = 'help_outlinemy-project-name';  
$wgMetaNamespace = 'my-project-name';
```

The ‘\$wgSitename’ variable sets the name of your wiki that appears throughout the interface—in browser tabs, page titles, the top-left corner of pages, and in email notifications—essentially branding your Wikibase with your project’s identity. The ‘\$wgMetaNamespace’ variable creates a special namespace for project-related administrative pages (like guidelines, help documentation, and project information) that are separate from your main research content, allowing you to create pages such as ‘My-Project-Name:About’ or ‘My-Project-Name:Guidelines’ that help organise project documentation.

It should be noted that any changes to the configuration file are not immediate and require a restart of the Wikibase instance, which can be done using the Docker commands described in the previous section (with ‘docker compose down’ followed by ‘docker compose up -d’).

Limit Access to Registered Users

We will now discuss how to make the Wiki private, as in many use cases, it may not be desirable for the Wikibase instance to be accessible and editable by everyone. As previously noted, Wikibase is hosted as a website potentially accessible to any user with an Internet connection.

To restrict access to the wiki to only logged-in users,

```
$wgGroupPermissions['*']['read'] = false;  
$wgGroupPermissions['user']['read'] = true;
```

The first line removes read permissions for everyone, whereas the second line reinstates read permissions for registered users. If these configurations are set, a user landing on the Wikibase homepage will be required to log in, not to be able to navigate the knowledge base and its content. Importantly, this configuration only restricts access through MediaWiki, but it won’t restrict access to the query service interface (WDQS) (which is an independent service with a different domain name), and through the query interface, any user will have unrestricted access to the entire knowledge base without logging in! We will see below how to restrict access also to the WDQS.

If you wish to allow the wiki to be read by anyone but only want edits to be enabled for registered users, you can set the permissions in a similar way, shown below:

```
$wgGroupPermissions['*']['edit'] = false;
$wgGroupPermissions['user']['edit'] = true;
```

Allowing for Images to Be Uploaded and Stored

By default, the site doesn't allow for images or other files to be uploaded to the site. Of course, depending on the project requirements, it might be important to allow for this. To enable uploads we must set the '\$wgEnableUploads' variable to 'true'.

```
$wgEnableUploads = true;
```

Also, by default, only a limited number of file types will be allowed to be uploaded to the site. These are probably enough for most situations, but again, further configurations might be required for specific configurations, which are not covered here.

Changing the Character Limits for String Values

Another useful variable to redefine is the one controlling the length of strings in Wikibase. As we will see in the remainder of the book, Wikibase allows us to structure our data through relationships where items are either linked to other items through a property, or they are linked to some values (e.g., with the item 'Napoleon' linked through the property 'died on' to the value '5 May 1821' of date type).

By adding the following variable definition to the bottom of 'LocalSettings.php', we increase the string limit of the value type ('VT') 'string' to 1,500 characters.

```
$wgWBRepoSettings['string-limits'] = [
    'VT:string' => ['length' => 1500]
];
```

Importantly, the length of accepted string values should be increased carefully because if too large, they might compromise the underlying database where the data is physically stored.

Formatting URL Properties

A final configuration for ‘LocalSettings.php’ that we propose is the possibility of adding a URL formatter for specific property values. Imagine you are building a research database about politicians, and you want to connect each politician in your database to their profile on external websites like Twitter, LinkedIn, or parliamentary websites. Without URL formatters, you’d have to manually type out the complete web address for each politician’s profile—a time-consuming and error-prone process. URL formatters solve this problem by creating a template or pattern that automatically generates the correct web link when you only need to provide the external identifier.

Let’s say you’re studying members of parliament and want to link each MP to their official parliamentary profile. Instead of storing the full URL like: https://www.aph.gov.au/Senators_and_Members/Parliamentarian?MPID=R36

You can set up a URL formatter with the pattern: ‘[https://www.aph.gov.au/Senators_and_Members/Parliamentarian?MPID=\\$1](https://www.aph.gov.au/Senators_and_Members/Parliamentarian?MPID=$1)’.

Then, for each MP in your database, you only need to store their external identifier (like ‘R36’), and Wikibase automatically creates the clickable link by substituting that number where the \$1 appears.

In order for this to work on our site, we must stop the site through these steps.

First, in our Wikibase we must create a property (as we explain in Chapter 4) of type ‘string’ that we can call ‘formatter URL’. Once this property has been created, it is important to remember its ID (let’s assume it is ‘P1’).

Second, we must add the following variable definition in our ‘LocalSettings.php’ (and, as always, stop and restart the Wikibase containers):

```
$wgWBRepoSettings['formatterUrlProperty'] = 'P1';
```

Third, we must create a property of type ‘external identifier’ to deal with this specific URL. In this example, the property can be, for example, labelled ‘Australian Member of Parliament ID’.

Fourth, on the property page ‘Australian Member of Parliament ID’ we must add a statement indicating as property ‘formatter URL’ and as value ‘[https://www.aph.gov.au/Senators_and_Members/Parliamentarian?MPID=\\$1](https://www.aph.gov.au/Senators_and_Members/Parliamentarian?MPID=$1)’.

It might be necessary at this stage to stop and restart Wikibase again. Once Wikibase is back online, we are then able to add an ‘Australian Member of Parliament ID’ property to any item, and the value will be clickable to open the corresponding page on <https://www.aph.gov.au>.

Setting Up a Reverse Proxy to Restrict Access to WDQS

The WDQS is the query service associated with Wikibase. It allows users to query and retrieve data from the Wikibase database. This enables a variety of tasks, such as performing complex queries and visualizing data.

The WDQS operates independently from the main Wikibase web page service. As such, a separate port must be exposed to allow access to the WDQS interface associated with a different DNS record. This also means that Wikibase set to private does not make the WDQS interface private. By default, the WDQS web interface has no security features associated with it. It is not recommended to expose the web interface without adding additional security, as anyone with access to the web page will have the ability to access your data without authentication and authorisation.

There are different ways to restrict access to your WDQS interface, including adding a virtual private network (VPN) layer between your remote machine and the open Internet, so that you can only connect to it using a VPN client (and a username and a password), by adding a reverse proxy, or by a combination of both for extra security. In this section we explain how to quickly set up a reverse proxy using a web server to add basic security authentication to the WDQS web interface. Yet, it is important to note that if your data is highly confidential or sensitive, or if you have reason to believe that your project knowledge base might be targeted to expose your data, you must seek professional advice to protect your data. In layman's terms, we set up a web page to act as a middleman that checks the credentials of the user trying to access the web interface. We will now provide a brief guide on how to achieve this.

To begin, install the *nginx* and *apache2-utils* packages with the following commands on your remote machine:

```
sudo apt install nginx
sudo apt install apache2-utils
```

Start *nginx* with the following commands:

```
sudo systemctl start nginx
sudo systemctl enable nginx
```

Next, we need to set a username and a password for a secure account. This account is independent of the Wikibase and will only be used for accessing WDQS through the *nginx* reverse proxy. Run the following command,

replacing 'yourusername' with your desired username and setting the password when prompted:

```
sudo htpasswd -c /etc/nginx/.htpasswd yourusername
```

We now need to set up the configuration file for nginx to enable the reverse proxy. Open the nginx configuration file using GNU nano with the following command:

```
sudo nano /etc/nginx/sites-available/wdqs
```

Add the following configuration to set up the reverse proxy, replacing 'query.myproject.example' with the DNS record you purchased and assigned to the WDQS interface (not the main Wikibase site) and 'PORT' with the allocated port for WDQS:

```
server {
    listen 80; server_name query.myproject.example;
    location / {
        proxy_pass http://localhost:PORT; # Replace PORT
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # Enable basic authentication
        auth_basic "Restricted Access";
        auth_basic_user_file /etc/nginx/.htpasswd;
    }
}
```

Having set up the configuration file, we now need to apply the configuration. First, link the configuration file to nginx using the following command:

```
sudo ln -s /etc/nginx/sites-available/wdqs /etc/nginx/sites-enabled/
```

Next, test the configuration file for any syntax errors using the following command:

```
sudo nginx -t
```

If there are no errors, apply the configuration and reload nginx to apply the changes:

```
sudo systemctl reload nginx
```

Finally, visit your WDQS domain using a web browser. If successfully configured, you should be prompted for a login. Enter the username and password you set. If successful, you should see the WDQS interface.

HOW TO MAINTAIN WIKIBASE

Connecting to the Mail Server

Wikibase requires a mail server to send emails. While Wikibase can be run without a mail server, several functionalities will not work without an operational mail server, such as password recovery for users. Although a password can be reset by directly accessing the server, email functionalities can be useful for large research teams. Setting up a mail server is beyond the scope of this chapter; however, we will now focus on how to connect to an existing mail server.

Email functionalities can be enabled by setting these variables in the 'LocalSettings.php' and adding the following lines, replacing them with your desired email addresses:

```
$wgEnableEmail = true;  
$wgEmergencyContact = 'admin@example.com';  
$wgPasswordSender = 'no-reply@example.com';
```

The first line enables email functionalities for Wikibase, whereas the second line specifies the admin contact email. The third line specifies the sender address for emails sent for password resets.

Next, configure the SMTP mail server by adding the following in 'LocalSettings.php', filling in the fields with your web server address, domain, port number, username, and password:

```
$wgSMTP = [  
'host' => 'ssl://smtp.example.com', // SMTP server address 'IDHost' =>  
'example.com', // Your domain  
'port' => ###, // Port number  
'auth' => true, // SMTP authentication  
'username' => 'your-email@example.com', // Your email address  
'password' => 'your-email-password', // Your email password  
];
```

Creating Users and Maintaining Their Credentials

The easiest way to create a new user and define their privileges is through the user interface of your Wikibase site. After logging in as an administrator, navigate to the 'Create account' page, which can be found among the site's 'Special pages', accessible from the Main Page. Alternatively, you can access this page directly at myproject.example/wiki/Special:CreateAccount.

When creating the account, you will need to enter the username for the new user. You then have two options for setting up their login credentials:

1. If you have configured a functioning email server: Enter the user's email address. The system will automatically send them a temporary password that they can use to log in and then change to their preferred password.
2. If you do not have an email server configured: You will need to manually set a password for the user. You can then share these login credentials with them through a secure communication method.

After creating the user account, you may need to adjust their user permissions, depending on their role in your project. This can be done through the 'User rights' page (Special: UserRights), where you can assign the appropriate user groups and permissions based on what level of access they require within your Wikibase instance. Importantly, by default, a user should not have the right to delete a page or an item.

If you have set up an email server, a user can request at any moment to reset their password in case they forget it. But for security reasons, a password can't be reset from the user interface. If you haven't set up an email server, the only way to reset the password is by accessing the server through the command line.

In order to do it, you will need to execute a script inside the Docker container of your Wikibase. The full command should look like this:

```
docker exec CONTAINER_NAME php /var/www/html/maintenance/
run.php changePassword --user='username' --password='newpassword'
```

In this command you will need to replace ‘CONTAINER_NAME’ with the actual name of your container running Wikibase. If you don’t know this name, you can run

```
docker ps
```

Backing Up and Exporting Your Data

When it comes to exporting your data produced by your Wikibase instance, there are different options, depending on whether you want to back up your data to allow you to restore your full project on a different machine or after a data loss, or if you want to share it in a specific format.

Backing up all containers

Backing up your data is critical for maintaining your Wikibase installation. Since Docker Compose sets up multiple containers—each running different services required by your Wikibase—it’s important to understand that each container generates specific types of data that must be backed up to fully restore your instance. As explained in the documentation of Wikibase Suite Deploy, to back up all the data from your containers, you can run

```
docker compose down
for v in \
  wbs-deploy_wikibase-image-data \
  wbs-deploy_mysql-data \
  wbs-deploy_wdqs-data \
  wbs-deploy_elasticsearch-data \
  wbs-deploy_quickstatements-data \
  wbs-deploy_traefik-letsencrypt-data \
; do
```

```
docker run --rm --volume $v:/backup debian:12-slim tar cz backup > $v
.tar.gz
done
```

This will produce for each container a tar.gz file that can be restored into a new installation with the command

```
docker compose down
for v in \
  wbs-deploy_wikibase-image-data \
  wbs-deploy_mysql-data \
  wbs-deploy_wdqs-data \
  wbs-deploy_elasticsearch-data \
  wbs-deploy_quickstatements-data \
  wbs-deploy_traefik-letsencrypt-data \
; do
  docker volume rm $v 2> /dev/null
  docker volume create $v
  docker run -i --rm --volume $v:/backup debian:12-slim tar xz < $v.tar.gz
done
```

Backing up the MediaWiki database

To instead only generate an SQL dump of the MediaWiki database, this command can be used:

```
docker exec CONTAINER_NAME mysqldump -u DB_USER -DB_
PASS DB_NAME > sql_backup.sql
```

In this case the CONTAINER_NAME will be the name of the container running the database (likely indicating ‘mysql’ in the name), with the database name (‘DB_NAME’), username (‘DB_USER’), and password (‘DB_PASS’) specified in the .env file, as explained above.

Exporting your data as RDF

A final option to export your data is through the resource description framework (RDF) format. RDF is a standard model for data interchange, which can be understood as a standard that defines how data should be structured.

Dumping the RDF is often useful for a variety of use cases, such as visualisation of the Wikibase structured data. The command to export all your Wikibase statements as an RDF dump is the following:

```
docker exec CONTAINER_NAME php /var/www/html/extensions/  
Wikibase/repo/maintenance/dumpRdf.php > rdf_backup.rdf
```

with `CONTAINER_NAME` being the name of the Wikibase container.

Alternatively, it is also possible to export the knowledge graph in JSON format, replacing `dumpRdf.php` with `dumpJson.php` in the command above.

TROUBLESHOOTING AND WHERE TO GET HELP

Common Debugging Scenarios

Many different types of errors and crashes can occur when using Wikibase and Docker. While exhaustively covering all possible errors is impossible, we will provide some common checks to identify the cause of common issues.

A common problem is a Docker container crashing, causing a service to be unavailable. A useful command to check the state of the Docker containers running the Wikibase services is

```
docker ps
```

This command lists all the containers currently running in Docker. A Docker container not being able to start can be caused by a variety of errors but is often related to invalid configurations or incompatibilities.

An important debugging step is to examine the error logs of the Docker containers. This should be the first place to check for the cause of any errors. This can be done using the command (replacing `CONTAINER_NAME` with the name of the interested container)

```
docker logs CONTAINER_NAME
```

The error messages will contain information regarding any potential crashes or errors encountered by the container, which can then be used for further debugging.

Where to Get Help

When seeking assistance with Wikibase-related problems, the best starting point is the Wikibase wiki (<https://www.mediawiki.org/wiki/Wikibase>). This provides official documentation maintained by developers and community members. However, as with many open-source projects, documentation and examples can often lag behind the latest changes and may not reflect recent updates to the software. In such cases, reaching out to the broader Wikibase community may be more helpful. The Wikibase GitHub repository (<https://github.com/wmde/wikibase-release-pipeline/>) maintains an issue tracker for reporting bugs or reviewing known problems. Chatbots based on large language models, such as Anthropic's Claude, OpenAI's ChatGPT and Google Gemini, can also offer assistance when debugging. Whilst these tools can be valuable, any advice they offer should always be verified against official documentation. Finally, the broader programming and technology communities on platforms like Stack Overflow (<https://stackoverflow.com/>) are excellent resources for finding solutions and advice.

4. Structuring and linking research data

**Jennifer Williams, Emily Booth, and
Francesco Bailo**

LINKED DATA AND THE SEMANTIC WEB

What Is Linked Data and the Semantic Web?

Wikibase is built upon the foundational principles of the Semantic Web, a vision for transforming how information is structured and shared across the Internet (Allemang, 2011) first proposed by Tim Berners-Lee, the inventor of the World Wide Web. To understand how Wikibase facilitates research, it is essential to grasp these underlying concepts and their practical implications for data management.

The Semantic Web represents an ambitious extension of the traditional World Wide Web, designed to create a more intelligent, machine-readable digital environment. Rather than simply displaying information for human readers, the Semantic Web aims to structure data in ways that computers can understand, process, and reason about automatically (Berners-Lee & Lassila, 2001). This transformation relies on shared understandings—formal definitions called ontologies—and standardised database formats, such as resource description framework (RDF), that enable machines to interpret the meaning and relationships within datasets.

Whilst the broader vision of Web 3.0 continues to evolve and develop (Nabben, 2023), the Semantic Web approach forms the technological foundation of Wikibase. This creates a versatile data environment where researchers can construct knowledge graphs—sophisticated databases that organise information as networks of interconnected entities. These knowledge graphs excel at representing complex relationships by creating linkages between different types of data, whether fully structured (like numerical measurements) or semistructured (such as categorised interview themes) (Hogan et al., 2022).

At its core, Wikibase implements the principles of linked data, a methodology for connecting information across the web through standardised protocols and unique identifiers (Färber et al., 2017; Possemato & Guerrini, 2013). Think

of linked data as creating a vast web of knowledge where each piece of information has a unique address (like a postal code) and explicit connections to related information. This allows both humans and machines to navigate seamlessly between related concepts, discovering connections that might not be immediately obvious when data is stored in isolated databases or documents.

For research teams, this interconnected approach offers significant advantages. When a researcher creates an item representing an interview participant, that item can be linked not only to the interview transcript but also to themes discussed, geographical locations mentioned, policy documents referenced, and connections to other participants who discussed similar topics. These relationships become part of the knowledge graph structure, enabling sophisticated queries and analysis that would be difficult or impossible with traditional database approaches.

It is crucial to understand that a knowledge graph refers to the underlying network structure of interconnected data entities, rather than a literal visual diagram. Whilst knowledge graphs can certainly be visualised as network diagrams (as discussed later in this chapter), the visualisation is merely one way of representing the data relationships—it is not the knowledge graph itself. The actual knowledge graph exists as a structured database where entities, properties, and relationships are formally defined and machine-readable, regardless of whether anyone creates a visual representation of these connections.

What Are Structured, Unstructured, and Semistructured Data?

Understanding the distinctions between structured, unstructured, and semi-structured data is fundamental to appreciating how Wikibase organises and manages research information. As established in Chapter 2, these categories represent different approaches to data organisation that complement the traditional qualitative-quantitative divide, offering researchers greater flexibility in how they conceptualise and work with their materials.

Structured data refers to information that has been organised according to a specific, clearly defined format, with explicit relationships between data elements (Lemus-Rojas & Odell, 2018). In Chapter 2, we described how structured data can be understood as information that ‘fits into some sort of table’, where elements are explicitly related to each other and to properties that carry real-world meaning. The most familiar examples include Excel spreadsheets (.xls) or comma-separated values files (.csv), where columns represent specific attributes (such as participant age, survey responses, or geographic location), and rows contain individual records. In a research context, structured data might include demographic information from survey participants, voting tallies, or standardised interview metadata where each interview record contains the same set of fields: date, interviewer name, duration, and location.

Unstructured data, as introduced in Chapter 2, lacks such explicitly defined relationships and formal organisational structure. This type of data conveys information primarily through natural language in formats that are immediately human-readable but traditionally challenging for machines to process automatically (Salman et al., 2024). Classic examples include news articles, interview transcripts, written essays, field notes, or open-ended survey responses. We illustrated this with the example of a text document without annotations or an image without metadata—information that humans can easily interpret but that lacks the formal structure necessary for systematic computational analysis.

Semistructured data occupies the middle ground between these two extremes, combining organisational elements with the flexibility of unstructured formats. Semistructured data might include interview transcripts where document headers consistently present standardised information (date, participants, location), whilst the main content remains in natural language form. Another example would be open-ended survey responses where each participant receives identical questions but provides natural-language answers that cannot be easily categorised. This approach allows researchers to maintain some organisational consistency whilst preserving the richness and complexity of qualitative responses. Common semistructured data formats include JavaScript Object Notation (JSON), YAML, and XML. These formats enable complex relationships to be represented in ways that are both human- and machine-readable, providing significant advantages for data entry, processing, and analysis. As seen in Chapter 3, Wikibase allows us to export the content of a knowledge graph in the JSON format or in the RDF format, which is also a semistructured format.

The power of semistructured data becomes particularly evident in its analytical potential. Machine learning algorithms can be implemented to expand the knowledge base or detect errors in its construction (Cimiano & Paulheim, 2017), as discussed further in the section on automation later in this chapter. Additionally, semistructured data can be visualised and explored through graphical knowledge maps, although computational power may become a limiting factor when working with very large databases (see the section on data visualisation below).

As established in the discussion of data layering in Chapter 2, Wikibase functions primarily by creating connections between structured and unstructured data elements through what we term ‘progressive layering of structure’. Rather than forcing researchers to choose between data types, Wikibase enables teams to work across the entire spectrum—storing unstructured content (such as full interview transcripts) as regular wiki pages whilst simultaneously building structured relationships through the creation of items, properties, and semantic links.

This approach aligns with the insight that the same research material can span multiple positions on the structured-unstructured continuum, depending on how it is processed and analysed. An interview transcript might begin as purely unstructured text but gain semistructured elements when researchers add standardised metadata, and develop additional structured dimensions when themes are systematically coded and linked to other interviews discussing similar topics. Wikibase accommodates this evolution, allowing research teams to add layers of structure organically as their analysis progresses whilst preserving the original unstructured content for continued qualitative investigation.

ENTERING AND LINKING DATA FOR RESEARCH

Entities are connected via data linkage to form a complex knowledge graph. Data linkage is the fundamental concept that underpins Wikidata as a knowledge graph database. Fundamentally, all data linkage is characterised through a basic structure known as a triple statement (Haller et al., 2022), as so:

Item -> property -> object

Here, the property is a characteristic of the item, and the object is the value of that property. The value of an object can be qualitative (e.g., text) or quantitative data (e.g., datetimes, social media metrics, etc.), or another item within the database (e.g., a person, event, or location). The flexibility of this basic statement structure is what allows the representation of complex datasets based on relationships between entities.

Wikibase, by default, contains many useful special pages that serve specific database management functions, including the creation of new items and properties. As such, these pages are the backbone of Wikibase. Here, we will explore the most useful special pages to assist in data entry and management. A list of all special pages can be found in the `yourdomain.example/wiki/Special:SpecialPages` URL path of your site. It might be a good idea to add the most common among these pages (e.g., ‘create new item’ and ‘create new property’) in the MediaWiki sidebar for ease of access and efficient data entry (you can edit the sidebar by visiting the special page `wiki/MediaWiki:Sidebar`).

Items

Items are used to represent entities in a structured data format. Items can be created to represent social media groups, pages, people, or user accounts as

Special page

Create a new Item

Make sure to check if the Item already exists!
You should create a label and a description for all new items.
By clicking "Create", you agree to the terms of use, and you irrevocably agree to release your contribution under the [].

Create a new Item

Language:
en

Label:
enter a label in English

Description:
enter a description in English

Aliases, pipe-separated:
enter some aliases in English

Create

Source: Authors.

Figure 4.1 Create a new Item page

well as specific events or political groups. Each item is assigned a unique persistent identifier beginning with ‘Q’ (e.g., Q42 may represent the item labelled ‘Douglas Adams’). When creating an item, there is an option to set several terms, including the ‘label’ of the item and its ‘description’. Linkages are done through the unique identifier for each item. Therefore, linkages will be maintained even if you change the label. Additionally, it is optional to input information about aliases, key statements from the item page, and links to external sites and other databases.

To create a new item, navigate to the ‘Create a new Item’ special page (Figure 4.1). Ensure that you input a short but somewhat descriptive label and description. These elements are editable after the item has been created.

If a duplicate item is created, the two items can easily be merged with the ‘Merge two Items’ special page (Figure 4.2). All you need to do is enter each item’s persistent identifier (ID) and click ‘Merge Items’!

Special page

Merge two Items

Merge two Items

This is a special page, and it cannot be edited

If you merge two Items, all labels, descriptions, aliases, sitelinks and statements will be moved from one Item to the other.

The ID to merge from

The ID to merge to

Merge Items

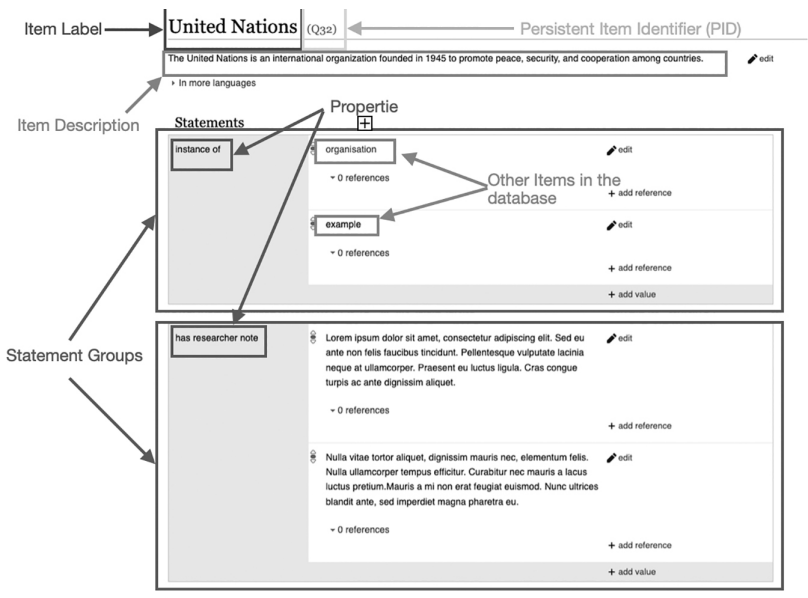
Source: Authors.

Figure 4.2 Merge two Items page

Properties

Properties are the predicators of the triple statement (Haller et al., 2022), providing a descriptive link between items and other items or data, such as a text string, date, or file. Properties are assigned a specific datatype that determines what values they can represent. For example, the property ‘Number of Likes’ would be limited to representing a number. Like items, they are also assigned unique persistent identifiers beginning with ‘P’ (e.g., P4 may represent the property ‘Published On’, which is used to specify which page or group a post is published on). Of note, a property can be used multiple times on an item to reflect multiple linkages, which allows for a more realistic reflection of real-world dynamics in which an entity may have a specific relationship with several other entities. For example, social media posts tend to have multiple comments, companies develop several different products, and a person often has more than one friend, may own multiple cars, or have lived at several addresses. The block in which a property is used to define several linkages is referred to as a ‘statement group’.

Below (and in late chapters), we explore a few useful properties for social media research. However, which properties you choose to include will be dependent on the scope and goals of your project. They are extremely flexible and can be created to represent just about any relationship between two pieces of data—consider what metrics and relationships are valuable to your project and get creative!



Source: Authors.

Figure 4.3 Wikibase Item page

Once you have decided what properties to include, you can create them through the ‘Create a new Property’ special page (Figure 4.4). The property label should be succinct yet clear, with the description detailing the property usage more thoroughly to ensure the property is easily understood by members of the research team to ensure a shared understanding of how to use this new property.

Now that you know how items and properties are created and entered on the front end of Wikibase, here is an example of how this data would be reflected in a JSON file connecting the structured and unstructured data to form semi-structured data:

Special page

Create a new Property

Make sure to check if the Property already exists!
You should create a label and a description for all new Properties, and in addition a valid Property type.
By clicking "Create", you agree to the terms of use, and you irrevocably agree to release your contribution under the [].

Create a new Property

Language:

en

Label:

enter a label in English

Description:

enter a description in English

Aliases, pipe-separated:

enter some aliases in English

Data type:

(pick a data type)

Create

Source: Authors.

Figure 4.4 Create new Property page

```
{
  "name": "Q42"
  "nameLabel": "Douglas Adams",
  "birthDate": "1952-03-11",
  "works": ["Q43", "Q44"],
  "worksLabel": ["The Hitchhiker's Guide to the Galaxy", "Dirk Gently's
Holistic Detective Agency"],
  "notes": "Douglas Adams once signed my T-shirt!"
}
```

Here, ‘notes’ demonstrates unstructured data, whereas ‘name’ and ‘birthdate’ are structured. Altogether, this file creates a Wikibase entity (explored in detail in the next section). Here, the property ‘works’ connects the entity Douglas

Adams to the entities associated with ‘The Hitchhiker’s Guide to the Galaxy’ and ‘Dirk Gently’s Holistic Detective Agency’.

Useful Properties

Datetime

Recording the date and time (‘datetime’ in programming terms) that the content was created may be beneficial for research purposes. For example, examining fluctuations of content over time or comparing changes in content to co-occurring events. There are two relevant existing methods of recording time: ‘point in time’ and ‘Extended Date/Time Format’ (EDTF).

A point-in-time property is used to reflect when some event took place (e.g., when something was recorded, created, or destroyed).¹ Notably, point in time does not allow the entry of time stamps, nor does it have the capacity to reflect time zones—both of which may be important pieces of information in your project. In cases where time stamps and time zones are valuable data (e.g., social media data), using EDTF is more appropriate. EDTF allows the inclusion of the date, time stamp, and coordinated universal time (UTC) time zone. Time zone may be an important metric, as datetimes on many social media sites will be relative to the user viewing and collecting the data, rather than the original creator. Once the data has been recorded in the appropriate UTC time zone, it can easily be converted to different time zones to examine patterns of behaviour in the locally relevant time. EDTF is a specific data type and, as such, needs to be specified when creating a property to handle it.

Point in time and EDTF are useful for recording details about a single fleeting moment in time. However, some entries may require a range of datetimes rather than a single point in time. For example, significant political and social events can span several hours, days, or months (e.g., protests, elections, festivals, military actions, etc.). Using two separate properties may be appropriate in this case: ‘start date’ and ‘end date’. These properties can be coded as either point in time or EDTF data types, depending on the precision required.

Metrics

Recording engagement metrics for social media posts, such as likes, shares, and comments, may benefit projects where the interaction rate with content is important. These can easily be encoded as custom properties to supply as much or as little detail as you wish. For example, you may want to record all reactions on a post with the numerical ‘has number of reactions’ property. Alternatively, you may want to record each type of reaction independently

¹ <https://www.wikidata.org/wiki/Property:P585>.

(e.g., ‘has number of love reactions’). The same can be done with comments, shares, and views. If you are interested in how content is disseminated, you could encode properties such as ‘shared to’ to record what other pages and groups the content is disseminated to.

Strings

Of course, properties can also be used to store written content, such as the text of a social media post or the transcript of a video. In technical fields, when discussing how data is coded and stored, this qualitative written data is referred to as a ‘string’. The relationship between an item and its corresponding written content (i.e., a ‘string’) can be represented as so:

This is the post label -> Has Text -> 'This is the content of a social media post'

Additionally, string properties can be used to hold researcher notes, allowing notes to be kept in an intuitive and easily accessible location relative to the data.

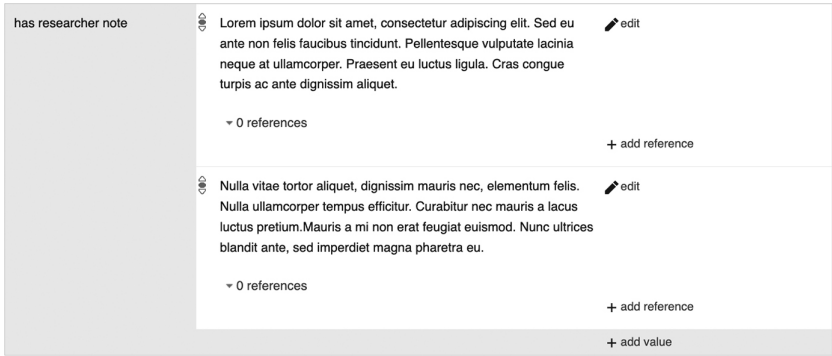
A Post -> Has Note -> 'This is a great example!'

There is one important limitation to storing textual data on an item: Wikibase has a default character limit of 400 characters at the time of writing for each textual property entry. This setting can be changed in the Wikibase settings as explained in Chapter 3. However, this is an advanced setting, and modifications are discouraged as they may cause issues in the Wikibase.² Conveniently, this default setting can also be circumvented without modifying the Wikibase settings by including multiple property entries to capture the entire textual entry (Figure 4.5).

Files

The data of interest may not always be purely text-based. MediaWiki comes with the ability to upload files as items and link them to items of interest (Figure 4.6). For example, a social media post may contain a meme in either

² [https://www.mediawiki.org/wiki/Wikibase/Wikibase.cloud/Wikibase_settings#:~:text=Changing%20them%20may%20cause%20problems,400%20\(default\)%20to%202500](https://www.mediawiki.org/wiki/Wikibase/Wikibase.cloud/Wikibase_settings#:~:text=Changing%20them%20may%20cause%20problems,400%20(default)%20to%202500).



Source: Authors.

Figure 4.5 Multiple character fields

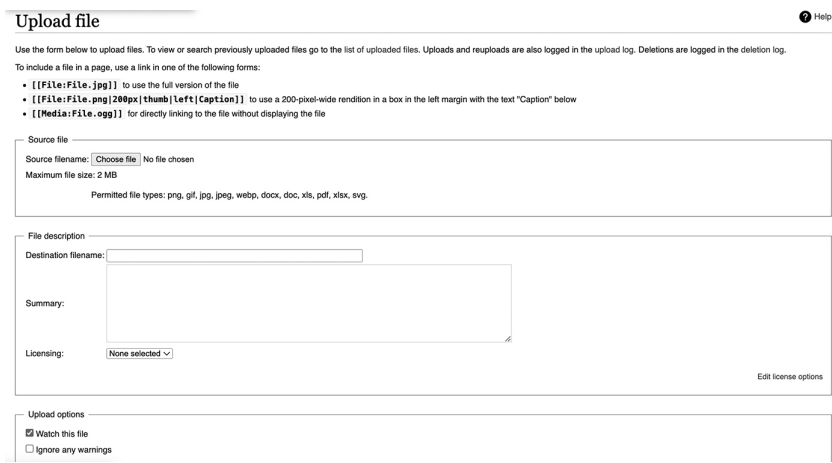


Figure 4.6 Upload file page

image or video format. This file can be downloaded from the social media webpage, uploaded into the MediaWiki, and linked to the relevant Wikibase entity. This may be particularly valuable where the content of a post is important, and you want to preserve it indefinitely, as valuable research data may be lost if the posts are taken down.

Media files are entered as their own entity and can be linked to multiple items like any other entity. For example, you may see the same image in several different posts, yet you only need to upload a single file and link that file to each separate post. By default, MediaWiki will not allow you to upload files, but as discussed in Chapter 3, we know that this default behaviour can be changed by editing the `LocalSettings.php` file. Notably, the file itself is not stored in Wikibase's knowledge graph but rather as a MediaWiki page, with only the relationship to that page captured in the knowledge base.

TRANSLATING, TRANSFORMING, AND VISUALISING RESEARCH DATA FOR MIXED-METHODS RESEARCH

Wikibase serves as an optimal platform for mixed-methods research due to its dual nature as both human- and machine-readable infrastructure. Unlike traditional research environments that require separate systems for different data types, Wikibase inherently facilitates the integration of quantitative and qualitative data whilst preserving the detailed characteristics of each. Most significantly, whilst conventional mixed methods approaches typically employ sequential methodologies (Morse & Niehaus, 2009), Wikibase's capacity to store quantitative and qualitative data simultaneously eliminates the need for artificial differentiation between data types. This unified approach enables researchers to conduct inherently linked analyses in parallel, creating what might be termed 'convergent research design' where both analytical streams inform each other continuously (for a comprehensive exploration of mixed-methods research frameworks, see Chapter 2). The following sections demonstrate how Wikibase can be employed to collate, structure, and analyse both quantitative and qualitative data for integrated research outcomes and dynamic visualisation.

Quantitative Analysis and Data Visualisation

Wikibase data can easily be visualised using SPARQL, the query language for the Semantic Web. Quantitative analysis in Wikibase can begin simply with straightforward aggregation of your structured data. Consider two common scenarios: first, when you've linked textual documents to thematic categories, you might count how frequently each theme appears across different groups or time periods. Second, if you've connected your items to geographical locations such as countries, you can easily filter and analyse data by specific regions. Wikibase enables these analyses directly through SPARQL queries (explored comprehensively in Chapter 6), eliminating the need for external statistical software. For instance, a simple query can count items associated with each country in your dataset, whilst more sophisticated queries can generate

dynamic visualisations—from bar charts and timelines to interactive maps—all within the Wikibase interface.

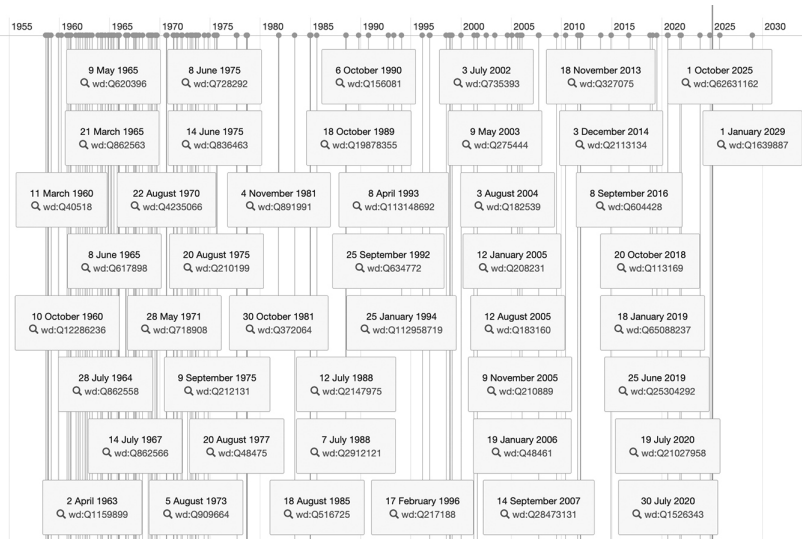
For example, the following query counts the number of items associated with each country:

```
SELECT (COUNT (*) AS ?count) ?countryLabel
WHERE {
  ?item wdt:P31 ?country.
  ?country rdfs:label ?countryLabel
}
GROUP BY ?countryLabel
```

This query, which assumes that the property linking items to ‘country’ has the internal database identifier ‘P31’, can be adjusted to group by other labels with relative ease by replacing the property and country labels with your chosen data (e.g., engagement metrics).

You may want to plot a visualised time sequence of your data. While the default search returns a simple table of results, queries can be modified to return visualisations such as bar charts, bubble charts, image grids, OpenStreetMap markers, timelines, tree maps, and many more. Thus, data can be explored with relative ease within the user interface without having to export and manipulate the information in another external program or coding language. Once you have crafted a query to examine your chosen data (e.g., aggregated data, time series, etc.), you can simply select the appropriate graph from the drop-down menu. Alternatively, adding a default view setting to your query will ensure it is displayed in that format:

```
#defaultView:Timeline
SELECT ?item ?itemLabel ?launchdate
WHERE
{
  ?item wdt:P31 wd:Q26529 .
  ?item wdt:P619 ?launchdate .
}
GROUP BY ?item ?itemLabel ?launchdate
```



Source: Authors.

Figure 4.7 Timeline plot

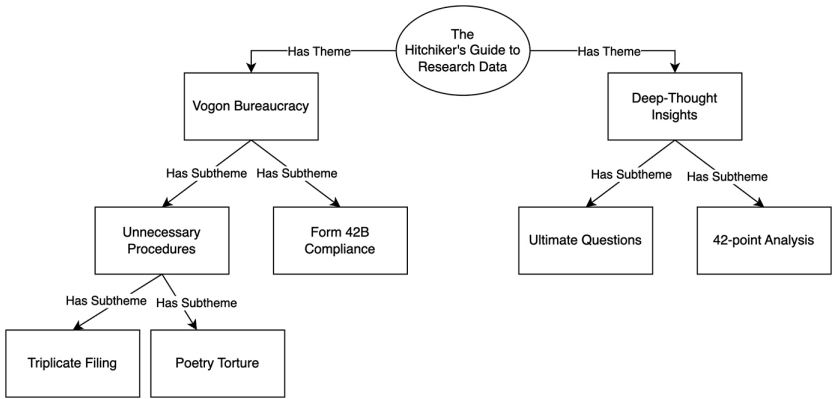
This query examines all instances of space probes in the database and plots them along a timeline by their launch date, as shown in Figure 4.7.³

Thematic Coding and Analysis

The ability to encode relational linkages between data is particularly advantageous for thematic coding. A complex codebook (or ontology, as we will discuss in Chapter 5) of themes, subthemes, and other thematic characteristics can be created and stored organically within the database and applied to the data as needed (see Figure 4.8 for an example codebook structure). There is no limit to the number of themes, subthemes, or even codebooks that can be created and applied to the data.

Themes can be input into the knowledge base as entities and linked to items using a custom property, as so:

³ This example was drawn from the Wikibase exemplar page. See this page for more examples: https://www.wikidata.org/wiki/Wikidata:SPARQL_query_service/Wikidata_Query_Help/Result_Views.



Source: Authors.

Figure 4.8 Thematic coding example

Post #1 -> Has Theme -> Space Travel

Using this method, data can be coded natively within the Wikibase at any stage. For example, data can be coded during the entry process using a pre-defined codebook. Alternatively, if using a grounded theory approach, data can be coded all at once after the collection and entry process, and emerging themes can be created in the Wikibase as needed. Regardless of whether a particular theme is input into the system as a theme, subtheme, or high-level characteristic, all themes should be linked to the relevant entities using the same property (e.g., ‘Has Theme’) to simplify data entry and analysis while still preserving the complexity of the codebook.

After coding, data can either be analysed within Wikibase using the Wikibase query language, SPARQL, or it can be extracted and analysed externally. This utility is particularly advantageous for improving efficiency and data management when coding and analysing very large qualitative datasets.

Comparative and Between-groups Analysis

Data can also easily be grouped to facilitate comparative analyses (in qualitative methods) or between-groups statistical analyses (for quantitative methods). For example, for qualitative comparative analyses, you may want to examine how the presence and potency of themes differ between locations

(either physical or digital), over time, or in different phases of data collection. Alternatively, thematic patterns may differ between posts and comments. In between-groups analyses, you may want to examine how post or comment engagement differs between locations, groups, or pages.

Analysis can be made more complex by applying statistical methods to compare instances between groups (e.g., independent-sample t-tests, analysis of variance) or over time (dependent sample t-test or repeated analysis of variance). For example, you may want to use a quasi-experimental design to assess whether there is greater engagement on posts in one Facebook group compared to another. In this case, you could create a query for all posts from each group across a set period and conduct a t-test on each metric (likes, comments, and shares). You could even input the number of group members as a confounding variable to examine whether there is a true change in engagement between these groups.

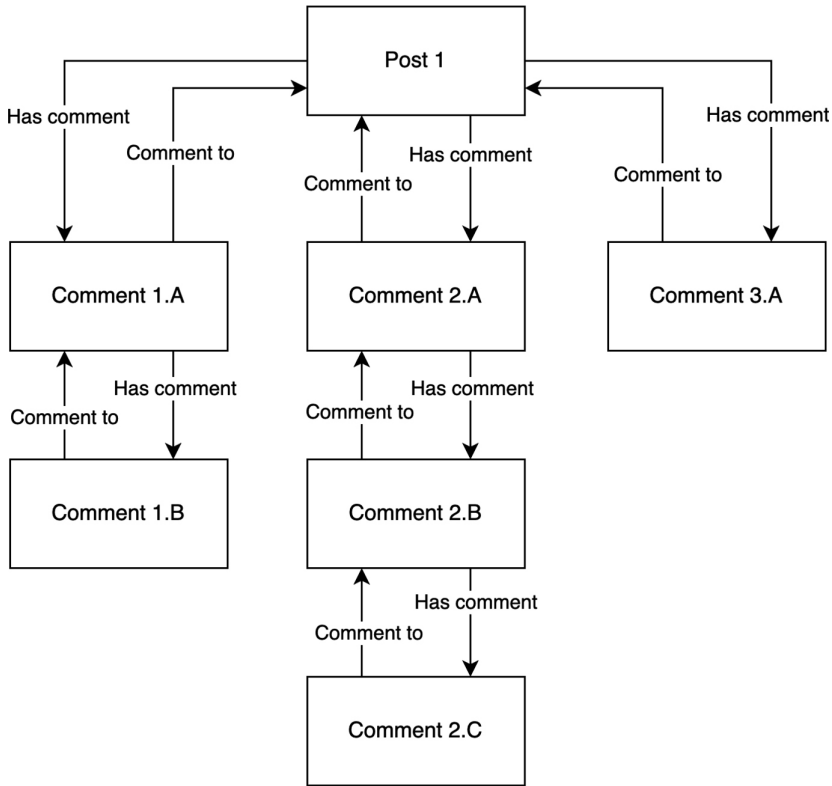
Alternatively, you might be interested in some other type of categorical variable rather than simply which groups the posts were made in. In this case, entities can easily be grouped for these comparative analyses with customisable properties (e.g., the platform, location, or phase of the study in which the data was collected), demonstrating how Wikibase offers extreme flexibility and customisability to facilitate these kinds of group-based analyses.

Practical Considerations

In this section, we discuss several useful practical considerations you can employ in your research project. These considerations are drawn from our personal experiences in managing a social media project within a large research team consisting of nine main team members in addition to several research assistants.

Backward and forward linking

There are some instances in which you want to be able to do ‘click through’ exploration in Wikibase—that is, exploring the linkages between items manually. Unfortunately, when linking an item to another through a property, the link is only visually apparent on the page in which you specified the link; that is, it does not ‘back link’ to the other page automatically. Therefore, manually including a ‘back link’ may be useful in some instances to facilitate manual exploration. For example, if you are examining social media exchanges in comment sections on a post, you may have a chain of comments stemming from a single post. Including the ‘Comment to’ property allows you to demonstrate how each comment is linked to a previous comment or post (see Figure 4.9), and no matter which item you start with, you can explore its relationships with other posts and comments within the graphical user interface.



Source: Authors.

Figure 4.9 Back-link diagram

Alternatively, Wikibase also comes with a special page (i.e., ‘What Links Here’) to show you a list of items linked to the current item or page. Note that this option will provide you with all items linked to the current item.

Orphaned data and missing properties

Manual data entry can be tedious and subject to human errors. One such error is orphaned data. The Wikipedia glossary defines orphaned data in Wikibase as ‘an article with no links from other pages in the main article namespace’ (Wikipedia, 2024). When data is orphaned, it may be left out of vital analysis or deleted if using automated database cleaning tools, ultimately undoing your hard work! It can also be difficult to locate through manual searches, as it may

not be linked to any other data intuitively. Thankfully, the Wikibase query language SPARQL offers functionality (with the set ‘MINUS’) to identify orphans easily and efficiently using the following query:

```
SELECT ?item ?itemLabel
WHERE {
  ?item rdfs:label ?itemLabel.
  MINUS { ?item wdt:P31 ?instanceOf }
  FILTER(CONTAINS(STR(?item), '/entity/Q'))
}
```

This query returns a table of items that do not contain the ‘instance of’ property (denoted here by wdt:P31) and the labels associated with each item while filtering out non-items (i.e., properties). Thus, orphaned data can be identified and manually edited. This query can be easily adapted to search for any instances of missing properties by simply modifying which property the query is searching for. For example, you may want to identify which items have not yet been thematically coded by querying items without the ‘Has Theme’ property:

```
SELECT ?item ?itemLabel
WHERE {
  ?item rdfs:label ?itemLabel.
  MINUS { ?item wdt:P27 ?HasTheme }
  FILTER(CONTAINS(STR(?item), '/entity/Q'))
}
```

Missing-property detection can also be automated, which is discussed in Chapter 6.

DATA MANAGEMENT AND SCIENTIFIC INTEGRITY

Good data management has become a foundational requirement for good research (Wilkinson et al., 2016). Depending on the scale of the project, manual data entry in Wikibase can be a colossal undertaking, and most projects will likely have multiple team members entering data into the knowledge base. On larger long-term projects, you may want to bring in research assistants or new team members. Therefore, to avoid error and miscommunication, it is important to develop a shared understanding of how data should be entered into the system to make onboarding smoother, avoid inconsistencies and

subsequent complications in analysis, and ensure that data is of high quality and accessible for reusability and replicability. To do this, firstly, we advise developing a detailed and shared understanding of the ontology of the project (the theoretical background of ontologies is explored in detail in Chapter 5). Second, we advise keeping detailed notes on methodological decision-making (e.g., in the form of field notes, meeting minutes, visualisations, and summaries) to keep track of how the data collection and entry protocol develops over time. These records are particularly useful for projects following an exploratory iterative approach in which methods may change several times as new information comes to light.

Finally, we also advise the inclusion of this information on your Wikibase site so that (1) team members can access it easily during data entry, and (2) there is transparency when the data is made publicly accessible (discussed below). This information can be reflected on the main landing page of the Wikibase or a custom methodology page indexed in the sidebar for easy access. In paying careful consideration to these practical elements, it will improve both research workflow and scientific integrity. As a bonus, the information is already collated for writing up a detailed method section in subsequent publications!

Following the replication crisis in the psychological and social sciences, many journals, grant funders, and human ethics committees have mandated open-access data to ensure proper storage, reuse, verification, and preservation of scientific materials (Korbmacher et al., 2023; Wilkinson et al., 2016). Additionally, these entities often require robust data version control to maintain data integrity over time. Thus, depending on the needs of the project, it may be important to have the ability to version control, share, and store your data in a public manner. Here, we explore Wikibase's version control features, alongside two strategies for publicising your wiki data: via a public-facing web page and through established public data repositories.

Version Control

Version control systems—widely used in the computer science community to keep track of code modifications and enable smooth collaboration between many groups and individuals—refer to any kind of ‘system that manages the development of an evolving object’ (Zolkifli et al., 2018). Wikibase incorporates a robust version control system that meticulously tracks data modifications—by whom, when, and what changes, including automated updates (Nalepa, 2010; Rossenova et al., 2020). Should issues arise, it is straightforward to revert the page to a previous state (Rossenova et al., 2020). Moreover, the use of version control in Wikibase enhances accountability, transparency, and data integrity, bolstering the scientific rigor of research projects (Leible et al., 2019; Steinwandter & Herwig, 2019).



Source: Authors.

Figure 4.10 History page

For qualitative work, version control is particularly valuable for codebook generation and thematic analysis. This utility provides a native and automated record of how the codebook has been developed and refined through an iterative process, making it easy to share with other members of the team or with other researchers and external entities (See Chapter 5 for more information on codebook and ontology development for thematic coding). Version history can be viewed easily through the graphic user interface (see Figure 4.10).

Web Page Functionality

Wikibase can easily be converted into a public-facing web page by changing the access control settings. This can be done during the set-up process, as outlined in Chapter 3, to ensure that the database is publicly accessible for the duration of the project. Alternatively, the settings can be changed at the completion of the project so only the finalised dataset is made available. Deciding when to make this data available will be subject to the methodological and ethical requirements of the project. It is important to note that the web page will require constant maintenance to ensure that it stays publicly available. For examples of public-facing web pages that use Wikibase, check out FactGrid⁴ or the ArtBase project mentioned in Chapter 2.

⁴ <https://blog.factgrid.de/>.

Data Exportation for Public Data Repositories

Alternatively, public data repositories that are hosted by a third party may be a more suitable option. Using public data repositories negates the need to host and maintain a server, unlike for a web page, and many repositories offer storage for free to incentivise scientific integrity and data reuse. Wikibase data is stored as a relational database (MySQL or MariaDB) that can be exported and stored in an online data repository (see Chapter 3 on how to export RDF data). Alternatively, for less tech-savvy users, select items can be exported using the ‘Export’ special page, where you can input the label for each page you want to export into a list, which will produce an XML file of said pages.⁵ However, this can be cumbersome for larger amounts of data. Here is another area where the SPARQL query system can be an asset. After crafting an effective SPARQL query (see Chapter 6), the queried data can be exported in a variety of different file formats, such as a JSON, TVS, CSV, or HTML file, through the ‘Download’ menu on the right-hand side of the query engine. This method is much less tedious than the ‘Export’ special page for larger selections of data. How you choose to export and store your data will depend on your specific needs and use cases.

Once you have exported your data in an appropriate format, you will need to store it somewhere. There may be limitations in external repository storage, depending on the size of your wiki database. For example, at the time of writing, the Open Science Framework (OSF) limits private projects to 5GB per component and public projects to 50GB per component.⁶ While 5GB should be ample for small-to-moderate datasets of simple text, including bulkier files like images and videos, may balloon the required storage space. If required, these limitations can be circumvented by connecting the OSF repository to another storage service, such as Dropbox, OneDrive, or GitHub.⁷ However, these options may conflict with ethics and data management requirements that mandate data to be stored in a specific manner or location. Overall, if your project is large, these logistical factors may be limiting or require additional consideration.

⁵ Consider exploring the Wikibase Export Extension or a more user-friendly approach to exporting specific pages or filtering pages by set values: https://export.wikibase.wiki/wiki/Main_Page.

⁶ <https://help.osf.io/article/137-osf-storage-caps>.

⁷ <https://help.osf.io/article/395-storage-add-ons>.

5. Designing and maintaining project ontologies with Wikibase

**Emily Booth, Jennifer Williams, and
Francesco Bailo**

CODEBOOKS, SCHEMAS, OR ONTOLOGIES?

What Is a Codebook?

Codebooks are widely used among qualitative researchers as a way of guiding and structuring the data analysis process. The term ‘codebook’ refers to an organised collection of codes and the supplementary materials created to interpret them. A standard codebook typically has six key components: ‘the code, a brief definition, a full definition, guidelines for when to use the code, guidelines for when not to use the code, and examples (of data to which the code could be applied)’ (MacQueen et al., 1998, 2). However, a ‘code’ is only an initial building block in the analysis process and not a theme itself. Codes become themes once they are meaningfully grouped so as to contribute a response or engagement with a project’s research question(s) (Braun & Clarke, 2006, 82). Codes can be added, merged, split, and collated as needed during the research process; due to the potentially large volume of codes that one could end up with, a codebook is a useful strategy for maintaining clarity and structure.

A codebook not only allows for the organisation of one’s initial phases of data analysis; it enables comfortable collaboration in the analytical stage by ensuring there is a shared understanding of how the data is being interpreted. Due to the inherent subjectivity involved in interpreting data, research teams where members have diverse opinions and disciplinary backgrounds can face challenges at this stage; a codebook resolves this by guiding interpretation and thereby creating greater consistency in the interpretation of the data (Oliveira, 2023, 301). This allows multiple researchers to be involved in the analysis process, without the concern that data will be coded inconsistently. While codebooks have historically been organised on paper—a method that may

still work for some researchers and datasets today—computer programs commonly used by qualitative researchers include NVivo or, for smaller datasets, spreadsheets such as those by Microsoft Excel.

Building a codebook

To build a codebook, one must first develop a series of codes. Coding can be defined as ‘the attribution of meaningful labels to bits of data (e.g., interview transcripts) that will support its subsequent analysis’ (Oliveira, 2023, 300). There are a number of coding strategies that exist, but two options that can be used to approach coding data are *deductive coding* and *inductive coding*. In deductive coding, a researcher takes a ‘top-down’ approach to the analysis that is structured by their theoretical concerns and/or research questions, and looks for data that specifically connects to these prompts (Oliveira, 2023, 301). In contrast, inductive coding involves ‘coding the data without trying to fit it into a pre-existing coding frame, or the researcher’s analytic preconceptions’ (Braun & Clarke, 2006, 83). In other words, a researcher might code whatever stands out as potentially interesting or relevant, regardless of whether it fits into the existing project concerns, and the codes will be labelled with terms that are found within the data itself, such as a certain phrase a participant uttered. Researchers often use a combination of both approaches, and the complexity of the process depends on both a researcher’s level of experience and the complexity of the data itself.

The nuances to building a codebook will vary based on every individual involved, whether that be research team members or a student’s thesis advisor. The anticipated method of analysis to be undertaken once the codes are developed will also shape the approach, whether that be thematic analysis (see Braun & Clarke, 2006), grounded theory (Urquhart, 2013), or any other option. However, generally, the process of building a codebook involves the following:

1. The researchers who will be conducting the coding immerse themselves in the data through multiple rereads of the dataset.
2. The researchers undertake an initial round of coding of a representative sample of the dataset, including developing the supplementary materials for each code.
3. The researchers involved in the analysis collectively review the codebook and discuss any disagreements until agreement is reached.
4. The team members involved in the analysis independently code the same representative sample of the data, using the codebook as it exists and adding new codes as needed.
5. The team members regroup to compare analyses and refine the codebook further.

6. The team members completing the coding return to the data to complete coding it.

Stages 3–6 of the above may be iteratively repeated any number of times, depending on the size and complexity of the dataset. Once all data is coded, team members regroup to bring the codes together to form the themes that will inform the analysis of the data.

Codebook strengths and weaknesses

As outlined above, the codebook has several strengths for qualitative researchers and is frequently a basic component of any qualitative research project. The codebook organises and records the interpretation of the data, ensuring clarity and consistency in how it is interpreted. This is beneficial for solitary researchers, including students, but it is especially seen as beneficial for teams, where disagreement could otherwise bring the analysis process grinding to a halt. It also records the vital intermediary stage between the initial interpretation of the data and the development of sound themes that respond to a project's research questions. It can be revisited and modified at any time, allowing researchers the flexibility and fluidity that is an inherent benefit to qualitative research.

However, there have also been criticisms about the use of codebooks. There is a risk of favouring prominent codes—that is, codes that appear in a greater quantity—in the analysis, which can lead to interesting findings being overlooked if they do not appear frequently in the dataset. Rather than prominence, codes should only be discarded if they are eventually found to bear no value to the research (Ballesteros & Mata-Benito, 2018). There is also a need for multiple revisions of a codebook as a project develops, which can be time-consuming. There is disagreement about the extent to which a revision should entail the removal of old codes and notes: some hold that a 'codebook should be a distillation, not an historical document' (MacQueen et al., 1998, 6), and so irrelevant material should be discarded, while more recently, there have been calls for codebooks that preserve older codes and notes for the sake of research transparency (Reyes et al., 2021, 90). Lastly, and perhaps most significantly, is the concern that the need for agreement about codes and codebooks can inadvertently constrain research by suggesting that it is 'possible—and desirable—for qualitative coding to be "accurate" and "objective"', when the very notion of a single fixed truth is antithetical to the fluidity that gives qualitative research its strength (Gareth et al., 2017, 6). Despite these criticisms, the widespread use of codebooks is a testament to their value for qualitative research.

Data Schemas

What is a data schema?

A data schema can be understood as that which ‘defines the structure of a database in a formal language’ (Uschold, 2015, 243), with ‘formal’ here referring to the specific technical programming language used in that database. Data schemas are predominantly used by scholars in ‘Knowledge Engineering, Artificial Intelligence and Computer Science’, as well as for functional purposes like e-commerce (Suárez-Figueroa et al., 2011, 10). Schemas and ontologies share a fundamental similarity in that they both express and organise concepts, as a result of many researchers working on the development of both tools at the same time in the late twentieth century (Uschold, 2015, 244). However, whereas an ontology aims to articulate meaning that can be shared with others, a database schema has a narrower remit, aiming to simply ‘describe data’ and aid with its efficient storage and retrieval (Uschold, 2015, 245–6). Schemas, therefore, are less semantically rich than ontologies, as they do not require the same richness of meaning as an ontology seeks to convey (Uschold, 2015, 247).

While schemas lack complexity and qualitative depth, making their development simpler, this compromise means that, unlike an ontology, a schema never becomes data itself (Noy & Klein, 2004, 430). This lack of richness caused by schemas having ‘very little explicit semantics’ for their data can cause more challenges when a schema needs to be updated (Noy and Klein, 2004, 431).

Ontologies

What is an ontology?

The definition of *ontology* varies depending on the field. In philosophy, *Ontology* (with a capital O) refers to the study of things that exist, and how we know they exist. However, the term *an ontology* (small o) is used by computer scientists, borrowing its name from philosophy, to refer to a kind of fixed and interconnected vocabulary that describes a particular field. The classic definition, proposed by Tom Gruber, is, ‘An *ontology* is an explicit specification of a conceptualization’ (1993, 199). This ‘conceptualization is an abstract, simplified view of the world’, which can be illustrated through the use of the ontology (Gruber, 1993). A more updated definition of an ontology is that it is ‘a formal, explicit specification of a shared conceptualisation’ (Studer et al., 1998, 184), recognising that the understanding is both formalised and shared—and sharable—between people.

To put it into plain terms, an ontology is a set of vocabulary with which to talk about a particular domain or field. Like a glossary, all items in the

vocabulary are accompanied by a description; however, unlike a glossary, this description comes in the form of a set of fixed traits that are also connected to other items in the ontology through a ‘common structure’ of relations (Sarraipa et al., 2010, 5). These relations are often expressed in simple phrases like ‘instance of’, which could be attached to the entry for a book to note that it is an *instance of* a larger category called *paper-based object*. What is distinct about an ontology as used by computer scientists, compared to a codebook (encoding meaning) or a data schema (encoding relations), is that an ontology encodes both of these aspects. Furthermore, it does so in a way that allows the contents of an ontology to be shared between humans and computers (Guarino et al., 2009, 14). This human–machine communication is essential for projects that will involve computational processes, ranging from collecting and labelling data to modelling future predictions.

Building an ontology

Since their emergence in computer science in the late twentieth century, a key consideration for the development of ontologies was ensuring their shareability across domains or fields. This is to say that, due to the amount of time and money an individual ontology can cost to build, a goal among researchers was to build ontologies that could be adapted across different contexts (Fensel, 2001, 2; Neches et al., 1991, 37). However, this emphasis on reuse comes with its own problems: the more an ontology is designed to be reusable, the less tailored it is to its original context, thereby making it less valuable to create in the first place (Swartout et al., 1997, 138). Whether one uses an established ontology or creates their own as part of the research process depends on the overall research design and team members’ capabilities. This chapter will assume that readers want to build their own from scratch, but even if you adopt an existing ontology, it is possible that this will still need to be adapted to specific research contexts and your needs.

Although an ontology should be routinely revised over the course of a project, a basic ontology should be established at the outset. This is possible because, unlike a codebook or a schema, an ontology does not require data to exist (Noy and Klein, 2004, 431). This is due to ontologies being primarily about meaning, in terms of identifying what a concept *is*, and what that concept’s qualities *are* in concrete terms (Uschold, 2015, p. 246). As Tom Gruber has noted, “‘non-technical people’ must also be involved in an ontology design process” (Gruber, 2004, 3) if the ontology is to be used by people beyond these fields, and therefore, this is essential for interdisciplinary teams. Wikibase assists with this by providing an intuitive web interface that enables collaborative editing without requiring programming expertise, allowing team members from diverse disciplinary backgrounds to contribute directly to ontology development through familiar collaborative editing functions. This

means that qualitative researchers, domain experts, and other nontechnical team members can participate meaningfully in defining concepts, relationships, and project-specific terminology through the same user-friendly interface used by Wikipedia editors worldwide, whilst the underlying structured data architecture ensures these contributions are systematically organised and machine-readable for computational analysis.

A basic starter ontology will need to contain both the discrete concepts, including all their traits, and their relations to each other. To this end, the initial ontology can be developed by inviting team members to contribute a list of concepts they believe will be essential to the project. This can include possible data types (e.g., varieties of social media posts) and key themes that relate to the research question. Other means of categorising the concepts will also need to be included: for example, the date an interview was conducted; a URL to a specific webpage; any associated images or metrics, and other details. Once these concepts have been consolidated, which could be done by a single person, such as the lead chief investigator, or by the entire team, the relations will also need to be determined either by using specialised software like Protégé (Musen & Protégé Team, 2015) and Wikibase, or by more generic software, including diagramming tools and simple text editors.

A diagram of some kind will need to be assembled to enable the navigation and review of the ontology throughout the project. The format this takes will depend on the nature of the research team, as well as the technical capabilities of the team members. One simple method for outlining an ontology, particularly if it is small in its early stages, is an indented list, which can be created in any word processing software and is thus not technically demanding. The hierarchy is indicated by the position of each item to its parent and child entries; however, a limitation to this format is that only these vertical relations can be depicted easily (Dudáš et al., 2018, 6). Alternatively, an ontology can be visualised through the highly common ‘node-link’ method, in which labelled nodes representing the discrete concepts or individuals in the ontology are connected by ‘links [that] represent [the] relations between them’ (Dudáš et al., 2018, 6). This format, not unlike a 2D mind map of the ontology, allows for multiple relations beyond the vertical to be depicted, thus enabling a team to gain a birds-eye view of their ontology. While this can assist with the development and review process, it is important that the appropriate software is used to ensure access to functions like zooming in, should the ontology grow quite large.

Ontology strengths and weaknesses

Ontologies, just like codebooks and schemas, have their own strengths and weaknesses. The greatest strength of an ontology is, of course, its linked nature, which allows researchers to capture deep knowledge about a concept,

including its relation to other concepts. This rich mapping of a domain is beneficial in its own right, but it can also be deployed to computational ends, such as for data scraping or modelling. Depending on how context-specific an ontology is, it may be sharable, or at least adaptable, across different domains, meaning it could save time if someone is working on multiple projects in related areas. If used in a program such as Wikibase, it will also provide a networked way of managing data that is accessible to researchers in different locations, making it possible for different people to work across different tasks simultaneously.

On the flip side, an ontology also brings with it its own challenges. The entry point for building and using an ontology is higher due to the greater technical complexity, particularly if the end goal is to share the ontology with a computer system. Ontologies can also easily become unwieldy if they are not properly and regularly refined to ensure they contain only what is needed, and enough of it to make it meaningful. An incomplete ontology, where important concepts are not represented, could lead to an incomplete understanding of a particular domain. Lastly, an ontology developed in a team requires consistent agreement on how to define each concept, and this may bring challenges if team members have divergent views about a particular entry.

DESIGNING A PROJECT ONTOLOGY FOR INTERDISCIPLINARY RESEARCH

Interdisciplinary Research and Ontologies

Ontologies, as we have outlined them here, are perfect for mixed-methods interdisciplinary research as they are usable by both qualitative and quantitative researchers: by their very nature, they are designed to bridge gaps in communication between the means of interpretation in each field (human and machine, respectively). In this section, we provide some practical advice about how to design a project ontology that is specifically tailored for interdisciplinary research.

Interdisciplinary research is distinguished from standard multidisciplinary research, in which researchers' contributions to a project remain separable by their field (Pennington, 2016, 301). In interdisciplinary research, the team members coming from multiple backgrounds build knowledge connections between disciplines to produce their novel insights (Pennington, 2016, 301), developing projects and knowledge that 'cannot be realized within existing disciplines' (Kajikawa et al., 2014, 432).

Interdisciplinary projects are also particularly valuable to the investigation of issues where there are high levels of uncertainty and complexity, for which a singular disciplinary approach might lack the perspectives and approaches

needed (Barker et al., 2023, 11). Interdisciplinary research, due to its embracing of different methods and perspectives, also has more potential for engaging practitioners and industry stakeholders who may wish to contribute their own insights to the solution they are seeking (Boulton et al., 2005, p. 2021; Katoh et al., 2021, 14). Interdisciplinary research, therefore, offers distinct potential benefits for research teams and those invested in the outcomes of the research.

Ontologies are well-suited to interdisciplinary research. The process of building them addresses some of the challenges that can arise in interdisciplinary teams, namely, the issue of explaining one's own knowledge background to peers, and the risk of misconceptions if it is not sufficiently explained (Pennington, 2010, 5). To resolve this, interdisciplinary teams must develop shared understandings of relevant concepts and issues (Wenger, 2000, 234), and this is precisely the purpose of constructing an ontology. Although creating the shared conceptualisations of a concept for an ontology is perhaps more detailed, its taking place at the outset of a project can potentially resolve tensions that could arise later in an interdisciplinary team. Thus, an ontology is useful for such an interdisciplinary project, even in a data-less state. These benefits are extendable based on the disciplinary backgrounds within the team, and whether or not the ontology will be required for computational tasks. In this way, an ontology can be more useful than either a codebook or schema for such interdisciplinary teams.

Establishing Team Requirements and Data Needs

The practical construction of an ontology in an interdisciplinary team, as noted above, is a complementary process to the negotiation of meaning that is required at the outset of any interdisciplinary project. Teams, at the start of the project, must establish the shared understandings that will allow them to communicate throughout its duration. When designing an ontology, members should submit a list of the concepts and data types they expect will be relevant to the project, as well as potential relations between the items. This process will allow an interdisciplinary team to simultaneously work towards two goals of their own shared understanding and the construction of the ontology itself.

Part of this process should also include a discussion of each team member's data needs. This will inform what is included in the ontology as well as give an idea of how much data may need to be collected for the ontology to become useful. These needs will differ depending on individual research backgrounds and the goals of the project. At the start of a project, all team members should outline the amount of data they expect to need and the type of data, as well as any other requirements. This could be done by completing a simple form set up by the data manager. Included in this could be the project timeline, with individuals indicating the deadline by which they will need certain data in

order to progress, as well as the amount of time they will require at each stage to work with the data.

For example, qualitative researchers may only need a small (but deep) sample size, meaning a project might have only 30 pieces of primary data in the form of interviews. This could be supplemented as needed by data from other sources, such as entries for items mentioned by interviewees that are within the scope of the project, for example, television shows. Therefore, the ontology would be built to contain these categories—with room to expand later—but likely remain small and easily manageable. On the contrary, if quantitative methods are to be used, the volume and type of data required should be established early on, as large amounts are required for some methods of analysis. Likewise, for some quantitative methods like modelling, there may be a need for negative examples of data (e.g. a news article that is *not* fake news), which is not normally collected by qualitative researchers. Computationally collected data may need manual review and labelling, which can be a lengthy and time-consuming task, so this could also be outlined.

CREATING AND MAINTAINING PROJECT ONTOLOGIES WITH WIKIBASE

Establishing a Basic Framework

A basic ontology requires the following entries: a means of labelling a concept type, a means of labelling the attributes of data types (as they relate to the project, and in more general terms), and a means of indicating the relations between items. For example, you could use newspaper articles as data, which would use the relation ‘instance of’ applied to the data type of ‘newspaper article’ to define it. Certain attributes could include the topic it was about and the date it was published. Other key entries could be a label for specific individuals or entities that relate to the data type, such as the ‘media company’ that owns the ‘newspaper company’, which employs the ‘journalist’ who wrote the ‘newspaper article’. As an ontology is ideally a complete picture of a domain, these contextual entries are essential to give full meaning to the entries for the data.

Additionally, if your project will involve automated data processing or computational analysis beyond manual data entry, you will need to consider what types of concept entries (the ontology individuals represented as Wikibase *items*) and relationships (defined as Wikibase *properties*) will be necessary for algorithmic processing. For example, if newspaper articles are published online, the entry for each article item should include its URL as a property, along with quantitative metrics such as the number of comments, to enable automated data collection and analysis workflows. This would require

an additional entry defining what a ‘URL’ is and how URLs are used in the Wikibase. It would also require an additional property (such as ‘has URL’) to add URLs to the larger entries for each item. All ontologies should grow as a project progresses, and so existing entries and relations may change, and new ones may be added over the course of the research. However, preparing well in advance by understanding the needs of all researchers will reduce unnecessary stress.

Recording Data Collection to Inform an Ontology

As a project progresses, the ontology will very likely grow and change with it, especially if the research involves qualitative components. To facilitate this growth, records will need to be kept of the data collection process, capturing any observations that team members have at that point in time. These field notes can be stored on MediaWiki (as mentioned in Chapter 2, Wikibase is an extension of MediaWiki, which is a simple web-based user interface), on either a central page or one page per person. To ensure these notes are valuable, team members working with the data should record what they worked on, as well as observations, impressions, and anything else noteworthy at the end of each work session. These notes can capture snapshots of the data at a given time, which will come in handy if a team decides to add a data point at a later date, when the original source may no longer be available (e.g., if a website is removed). An advantage of using Wikibase and MediaWiki is that they also record different versions of their pages when changes are made, meaning that content that is deleted, accidentally or otherwise, can easily be retrieved at a later date by accessing a past version of a page. This enables different versions of a codebook, for example, to be easily preserved.

These notes can also be a means for identifying new concepts that need to be added to an ontology. For example, a certain story from recent public events may be mentioned across a series of focus groups. While a single mention may not be deemed interesting or relevant, multiple mentions could warrant an entry into the ontology to capture the existence of that story as it was known at that time. Reading through the field notes can bring this repetition to light. Practically, this means that in addition to the focus group data (including entries for the transcripts and each of the participants), researchers could add entries for the story itself (e.g., ‘Man Bites Dog’), and further flesh this out by adding various news articles that discuss the story from different perspectives. This helps the completeness of the ontology in its goals of mapping the domain as best as is possible and maintaining relevance to the project.

Manually Adding to the Wikibase

If team members are manually inputting data into Wikibase, this will likely be the most time-consuming part of the project. It is also the stage that is most likely to contain errors, due to the repetitive nature of adding entry after entry. Some concept entries may contain a high number of attributes that need to be repeatedly added to each item, and it's easy to accidentally miss one. This can cause complications further into the research process if entries are left out of search results when they are needed.

For this reason, we suggest creating exemplars for each concept type at the outset of the project. These are entries that may or may not relate to real data points, but which contain all the anticipated attributes and relations for that concept entry. Then, when adding entries to the wiki, or updating them if the ontology is revised, the exemplar can be opened in a separate window and quickly consulted after completing each entry to ensure its completeness.

MAINTAINING PROJECT ONTOLOGIES

Revise, Revise, Revise

It is essential to revise an ontology throughout a project to ensure it continually meets the needs of all users and the research goals themselves. This is because, if you have built your ontology yourself ahead of engaging in data collection, it is likely that the ontology was not a complete representation of the domain. (After all, if everything about it were already known, there would not be much of a purpose to the project.) Ontologies usually change from being more specific to being broader, in line with the growth of the dataset. This process should be an iterative one involving all team members who will be involved in the coding and data analysis process, if not the full team. At the outset of a project, scheduling a dedicated ontology review meeting, such as every three or six months, can be a good way to keep the process on track.

Between meetings, those working closely with the data and maintaining field notes should note new concepts that they feel could be added to the ontology, including a justification for this and an example of the concept. One reason to add an item could be that a certain topic is repeatedly mentioned; or conversely, another reason could be that a very unexpected outlier topic is mentioned just once, yet it is highly relevant to the research question. Keep in mind that an ontology is not a codebook, however, and significance does not need to be a determining factor: an item may also need to be added simply if it contributes to completing the picture of a particular domain. A new relation could also be added between existing concept entries if one is discovered. Crucially, all ideas should be brought to the relevant team members for

discussion prior to adding or changing items, to ensure agreement and maintain the rigour of the ontology.

Ontologies as Codebooks

An ontology in the Wikibase can also be used as a codebook, in the sense that the digital platform and structure of an ontology is capable of holding a second level of coding. This can be an alternative to, or in addition to, a final revision of the ontology. For example, once the data collection process is completed, a team can put together a list of themes based on their experiences with the data. These themes can be added to the Wikibase via a new relation (such as 'has theme' or 'is 2LC' for 'second-level coding') that is applied alongside the routine entries about a concept entry's attributes. This approach allows the original understanding of data to be preserved while the new coding is added, allowing for a richer ontology that captures the nature of the concept entries, their relation to each other, and, in the form of themes, the relation of the data to the research questions.

EXAMPLES OF PROJECT ONTOLOGIES

Below are two examples of ontologies to illustrate how they can be built (Box 5.1 and Box 5.2). The first example is of a dataset containing objects, as might be used for archival research, but it also serves to illustrate the basics of an ontology and how it can be used for classification even without data. The second example illustrates human data, such as that which might be gathered in interviews or focus groups for social sciences research methods, and the third example depicts online data, as might be gathered for a computational approach. In Chapter 6, we will see additional examples.

BOX 5.1 ONTOLOGY FOR TANGIBLE OBJECTS

Book

- Instance of: paper object
- Title is: The Tale of Genji
- Author is: Murasaki Shikibu
- Publication year: 1021
- Published in: Japan
- Number of pages: 1,504
- Number of chapters: 54
- Genre is: Monogatari (Japanese epic literature)
- Original language: Japanese
- Number of characters: 400
- Read by: adults

Magazine

- Instance of: paper object
- Title is: The Girl's Own Paper
- Started printing: 1800
- Ended printing: 1956
- Published by: The Religious Tract Society
- Published in: Britain
- Published in: Japan
- Contains: serial fiction
- Contains: poetry
- Contains: educational articles
- Contains: answers to correspondents
- Original language: English
- Read by: teenage girls

BOX 5.2 ONTOLOGY FOR HUMAN DATA

Focus Group

- Title: Teachers' focus group 1
- Instance of: focus group
- Date conducted: 19 January 2024
- Location: Poppy Café
- School: Rose High School
- Recording length: 80 minutes
- Transcript length: 100 pages
- Research questions answered: 1
- Research questions answered: 4
- Participants include: Teacher 1
- Participants include: Teacher 2

- Participants include: Teacher 3
- Participants include: Teacher 4
- Topics discussed: workload
- Topics discussed: curriculum
- Topics discussed: student behaviour
- News stories mentioned: student cheating scandal

Topic of Discussion

- Title: Student behaviour
- Instance of: topic
- Dataset: school research
- Description: related to students respecting other people, including staff and students, as well as the school's rules and expectations of appropriate conduct
- Connected to: student cheating scandal

News Stories Mentioned

- Title: Student cheating scandal news story
- Instance of: news story
- Published in: The Rose Local Times
- Journalist: Amber Doe
- Published on: 2 September 2023
- Has topic: student behaviour
- Description: a news story about students cheating on tests in local exams

6. Designing and running collaborative research projects with Wikibase

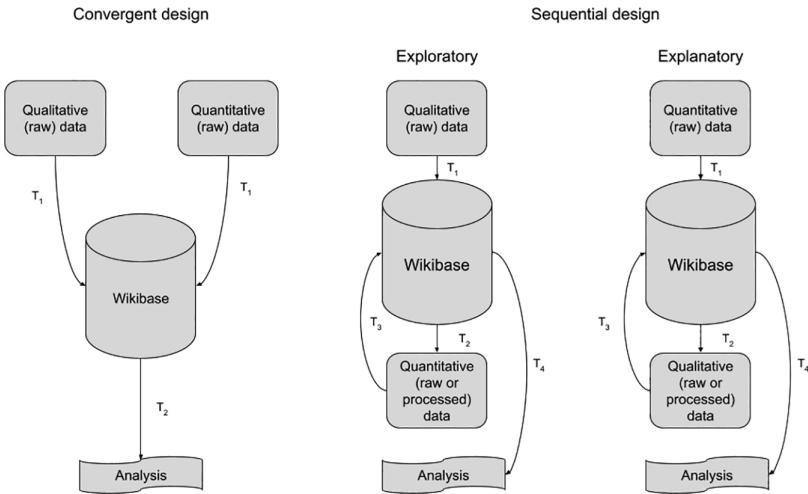
Francesco Bailo, Emily Booth, Jennifer Williams, and Marian-Andrei RizoIU

DATA ENTRY

Data Entry Overview and Sequences

Data entry is the process of adding data to our project's database. The data we enter can be *raw* research data collected from our sources but can also be *processed* data created from the analysis of our database. One of the key advantages of Wikibase is that data can be entered without predefining a comprehensive structure (although a primitive structure usually already exists, defining, for example, the type of data that can be initially created). Unlike traditional database systems that require teams to establish detailed schemas (and ontologies) before any data collection can begin, Wikibase allows researchers to start entering data with only minimal structural requirements. Schemas and ontologies can be defined and redefined over time as the research progresses, allowing the data structure to grow in complexity and enabling teams to develop their understanding of the data organically rather than being constrained by predetermined frameworks (see Figure 6.1).

Within the same project, data entry may involve different data types and streams, requiring separate entry procedures for each. Likely, in a mixed-methods research project different researchers will be responsible for curating the entry of different types and streams. For example, qualitative researchers might curate the entry of the interviews they conducted, while quantitative researchers might curate the entry of large corpora of digital documents. It is therefore important to plan how to split the work based on discipline expertise and knowledge of how to maintain database coherence as it naturally grows. This involve working out the most appropriate entry procedures and progressively refining the data structure and ontology to accommodate different types of data and critically afford their integration.



Note: Arrows indicate distinct data stream and the timing of the sequence.
Source: Adapted from Creswell and Clark (2018).

Figure 6.1 Three types of design for data entry and analysis

When different typologies of data enter a project database, this can happen either *convergently*, with the different streams being entered independently and without affecting each other, but potentially also *sequentially*, with data entered later being ‘informed’ by data entered earlier. Creswell and Clark (2018) identify three broader types of research design involving the collection and analysis of qualitative and quantitative data (see Figure 6.1). *Convergent design* involves collecting and analysing qualitative and quantitative data streams independently, with integration occurring only when comparing and linking the results from both analyses. Sequential design, instead, involves streaming the results from the analysis of one type of data directly into the creation of the second type. If the qualitative analysis precedes the creation of quantitative data, Creswell and Clark talk about ‘*exploratory sequential design*’, whereas if the quantitative analysis precedes the collection of qualitative data, they refer to ‘*explanatory sequential design*’. For example, in an *exploratory sequential design*, qualitative data such as interviews might be entered first and subsequently processed either manually or through some text analysis software (or a combination of both) to retrieve information such as common patterns or themes that can then be analysed quantitatively. This could involve a sequential data entry with a first batch of qualitative data exported programmatically, processed, and then re-entered, also programmatically. We will explore how

data can be accessed (read and written) programmatically later in this chapter. Instead, in an *explanatory* sequential design, quantitative data such as survey results can be programmatically entered into the database, while qualitative data is created by conducting follow-up interviews with a selective sample of survey respondents. In this example, interview notes and transcripts can be entered into the database and then linked to the survey responses of the same participants.

Although these typologies are useful general categories, the sequential design can often be, in practice, significantly more complicated, as it can involve sequences where data is transformed from qualitative to quantitative multiple times through the interventions of people and machines. With the development of research tools and methods involving large language models (LLMs) to support the creation of *quantitative* information from qualitative data (e.g., through sophisticated information retrieval and classification), but also the creation of synthetic *qualitative* data from a mix of qualitative and quantitative sources (see Hayes, 2025), the potential for very complex and diverse sequences and workflows is certainly going to increase. In this sense, the flexibility of Wikibase in accommodating the entry of very diverse streams and sequences makes it a lasting resource, even in the context of rapid methodological developments.

Data Entry in Practice

To provide a practical example of how Wikibase could accommodate the entry of complex data, let's consider the example of a mixed-methods research project involving different types of data (qualitative and quantitative) distributed across different data streams. Some of these streams will then enter the database convergently or independently from one another, while others will do so sequentially. Let's consider research into the political communication of Members of Parliament (MPs) to understand how they construct their messages as they switch between their parliamentary platforms and their social media platforms. The research involves the collection and analysis of multimodal data from different sources, with both qualitative and quantitative components: (1) official transcripts of parliamentary debates, (2) video recordings of the same debates, (3) textual and (4) visual content published by the MPs through their social media accounts with the corresponding metadata, and finally, (5) video recordings of interviews with a selected number of MPs and (6) corresponding transcripts. Notably, although these sources are far from unusual, as all are regularly accessed by researchers, they can still generate complex multimodal data, including a mix of formats such as text, audio, video, and still images, which is either coherently accommodated by the data structure and ontology or necessarily excluded from the research. In addition, the project will produce

intermediary processed data for further analysis resulting from a (7) thematic and (8) sentiment analysis of all the texts and an entity analysis of the texts where (9) people and (10) organisations are tagged in the texts.

Data types

Let's first review the types of data the project will need to work with, while qualifying its structured and unstructured components. We distinguish between raw data, which is directly collected from the field, and processed data, which is instead created through the analysis of raw data.

Raw data

- **Parliamentary data:** The parliamentary data is made of text of the (1) transcripts and the corresponding (2) video recordings, which are both inherently unstructured. Yet, this data, depending on the parliament's official archiving services, can come with a number of structured layers. In this case, we assume that the transcripts and the corresponding video also provide structured information, such as the date of the speech, a unique identification for speakers and their parties, the length of the speech in minutes, and a unique identification linking the speech to a specific parliamentary debate, which is therefore linked to other speeches.
- **Social media data:** Similarly to parliamentary data, social media data also usually come with structured and unstructured layers. In this case, we assume the unstructured data to be the (3) text of the posted message and (4) still images embedded in the posting. The structured layer of the data is instead made of the unique identifier of the posting account, the unique identifiers of all the accounts tagged or mentioned in the posting, the number of reactions and views to the posting, the identifier of other postings that the posting linked to directly (e.g., because it is a reply to another post or because it mentions another post), and finally, the date and time of publication.
- **Interview data:** During the interview, the research team collects unstructured data comprised of the (5) video recording, the (6) transcript of the interview, and some limited structured information: the date of the recording, the name of the interviewee, and the name of the interviewer.

Processed data

After collecting textual data, it is very common for social scientists to produce additional layers of data, either qualitative and semistructured or quantitative and structured, through the systematic analysis of the texts. Results from this systematic analysis are then often used as input data for additional analysis, as in our example. That means that processed data must re-enter the database as coherently as possible with what is already there to facilitate further analysis.

- **Thematic data:** A common way to add structure to raw documents is through thematic coding, where researchers identify and label recurring themes that emerge from the content. There are many ways to identify themes, and a comprehensive review of the different methods is certainly out of the scope of this chapter. For the sake of this example, let's define a thematic analysis that involves manually inspecting the documents while labelling them using a set of (7) predefined themes when human coders deem that this is an appropriate description of the document.
- **Sentiment and entity data:** Sentiment and entity data are instead produced using dedicated software and so without human intervention. Sentiment analysis and named entity recognition (NER) are natural language processing (NLP) tasks afforded by a large (and growing) list of software. A review of how these tasks can be implemented is again out of the scope of this chapter. For the purpose of constructing our example, we will use a lexicon-based approach. A lexicon is a domain-specific dictionary of terms, with a lexicon-based approach consisting of tagging a text with the terms contained in a lexicon. For our sentiment analysis and NER tasks, we will employ three dictionaries: a dictionary of words associated with their degree of positivity or negativity (or neutrality) estimated on a scale from -10 to +10, a dictionary of public figures, and finally, a list of notable organisations. The resulting sentiment data (8) will be a quantity value associated with each document (its overall 'sentiment'), resulting from summing the single value of each term from the lexicon that is matched in the document. Similarly, the resulting entity data will be a label associating each document with people and organisations identified in their texts.

Let's now see how all this data can be entered into Wikibase.

Data entry design

Designing our data entry will require, first, understanding how our data can effectively be represented in Wikibase. Second, and based on this representation, we will design the procedures and sequences to actually enter the data into Wikibase. As we design our Wikibase, we will refer to concepts discussed in Chapters 4 and 5, and we will assume that Wikibase has been installed following the instructions presented in Chapter 3.

For this project, we will define a simple ontology using linked data principles. In Wikibase, there are just two critical *entity* types to understand: *items* (which represent the concepts or things in our domain) and *properties* (which define the relationships between items or between items and their attributes). What information should be mapped onto items and what onto properties is, of course, part of the design of our research project and its procedures. Natural

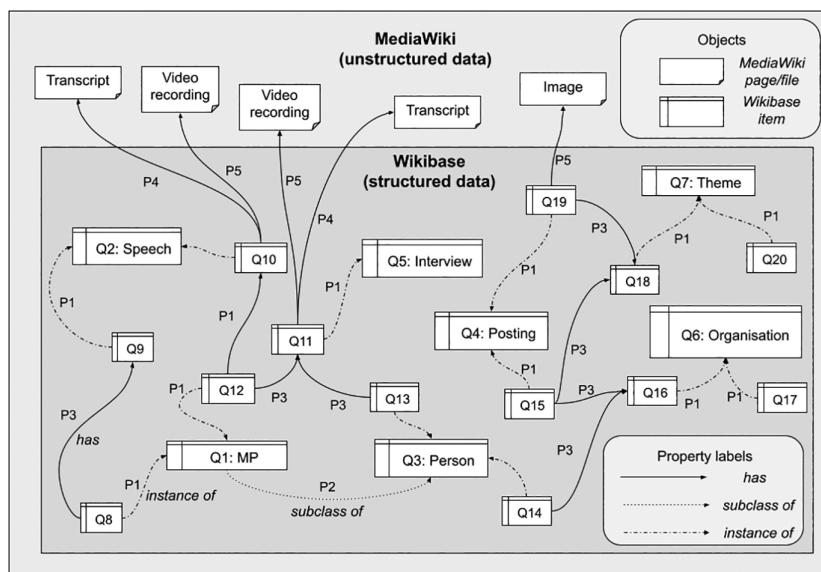
items in our project investigating the political communication of MPs are, clearly, MPs, but also all the documents on which we base our research. The first step is then to create a new item for each MP, which will therefore be assigned a unique identifier, and one new item for each speech and for each social media posting, as we explained in Chapter 4.

But do the *unstructured* textual, video, and visual data pertaining to speeches and posts need to be entered as distinct Wikibase entities (i.e., items)? That is probably not recommended. Indeed, our Wikibase installation is formally an extension of MediaWiki, a powerful collaborative software that offers more appropriate ways to accommodate unstructured data files such as long texts, videos, or images. Once we have an item representing a speech or a social media posting, we can easily link it to a MediaWiki page containing the full text of the transcript, the files of the video recording, or any other file we want to store in our knowledge base and link to our items. This is particularly useful if we want to collaboratively annotate our texts, for example, during the coding process. MediaWiki offers a sophisticated but intuitive interface to work on texts, including the possibility of adding annotations and links to other pages or items within the knowledge base. Would it also be appropriate to create an item for each interview? Yes, each interview of MPs conducted by the research team could be represented as speeches or posts, therefore, by a Wikibase item linked eventually to the transcript of the interview and to its stored video recording.

Items

Figure 6.2 illustrates a simplified ontology for this project that demonstrates the full capabilities of Wikibase and MediaWiki. There are two types of objects in the diagram: *items* defined within Wikibase and *pages* defined within MediaWiki (every time we upload a file to our site, we also create a corresponding page). Items are assigned a unique identifier by Wikibase, while pages (or files) are identified by their unique URL. Item identifiers are persistent—meaning they remain unchanged throughout the project's duration—and in Wikibase they consist of the letter Q followed by an incremental number (e.g., Q1, Q2, Q3).¹ Yet, what can be freely changed (even multiple times) without compromising the database and its schema is the label of an item. In this sense, persistent identifiers define the schema of our knowledge base, whereas labels define its ontology.

¹ If you are wondering why Wikibase uses the letter Q for item identifiers, it is because one of the early developers of Wikidata (and Wikibase) met his future wife, Qamarniso, at the Wikimania 2011 conference, where the project was presented (Vrandečić et al., 2023).



Note: A simple example of a schema and ontology for a knowledge base of a project researching the political communication of Members of Parliament.

Source: Authors.

Figure 6.2 Schema and ontology example

Properties

In addition to the two types of objects, Wikibase items and MediaWiki pages, the diagram in Figure 6.2 also defines the relationships linking our objects. These are the *properties* we define within Wikibase and that we use for characterising our items. As in the case of items, we also need to distinguish the role that properties play in terms of our schema and our ontology. From a technical schema perspective, properties are simply direct links that allow Wikibase's query engine to relate one item to another. Once you create a new property, this will be assigned (as with items) a persistent identifier prefixed by the letter 'P' (e.g., P1, P2, P3). However, in the context of developing your project ontology, property labels represent the meaningful relationships these connections have for researchers—so if the meaning changes, the labels should change accordingly (or new properties should be created). In Figure 6.2, we define six different properties. It is likely that any project will have more than six properties, but these can be considered as the minimum number of properties for a 'working ontology'. First, we define a relationship, 'instance of' (P1) that

can link two items while defining one as belonging to the class defined by the other item. In our examples, we have seven classes. This follows the practice of dividing an ontology into individuals and classes that define, using meaningful labels, the different types of individuals that are to be described by the ontology. Again, the distinction between ‘class’ and ‘individuals’ only makes sense in the context of ontology specification because at the level of Wikibase schema, these are all items. Second, we define a relationship ‘subclass of’ (P2), which helps us construct a hierarchy among classes. In our example, we use it only once, defining the class ‘MP’ as a subclass of ‘Person’.

Third, we define a very general property ‘has’ that simply links individuals of any class to other individuals when we want to link them. From an ontology perspective this could be defined as the single property ‘has’; in the Wikibase schema it is instead necessary to create more than one distinct property. At the level of Wikibase’s data model, which is what we need to refer to when we design our data schema, the properties linking outside the Wikibase domain must necessarily be different from the properties linking within the Wikibase domain. Indeed, in Wikibase what defines a property and how it is used is not only its label but also the property’s *data type*—that is, the objects or values that the property will associate with the item we want to describe. So far, the property ‘has’ (P3) has been used to link Wikibase items to other items. That is, in our schema, that property is defined by ‘Data type = Item’. Yet, Wikibase allows us to specify properties pointing to many other data types in addition to ‘Item’. In our example, we are pointing to two types of resources. As mentioned, our interview transcripts are stored in wiki pages on our site and uniquely identified by the page URL. We can therefore create a property P4 linking an interview item to its transcript and labelling ‘has’ (although you might want to be more descriptive and use instead ‘has transcript’ as the label) while setting ‘Data type = URL’. Then, since we plan to have media files describing our interviews but also our posts, we can create a different property P5 also labelled ‘has’, but this time setting ‘Data type = Media file’.² As before, we might want to be more descriptive, so instead of a generic property ‘has’ linking to a generic media file stored in our knowledge base, we could have different properties for ‘has image’, ‘has video’, or even ‘has video transcript’. How many properties to have is therefore mostly defined at the level of ontology, with only marginal limitation imposed at the level of schema by the Wikibase data model.

But let’s consider how this is used in practice.

² The list of data types allowed by Wikibase is constantly expanding and evolving. We refer the reader to the documentation for more details: <https://www.mediawiki.org/wiki/Wikibase/DataModel#Datatype>.

Describing items through properties

In Figure 6.2, we identify some individuals (that is, items of some class) with a persistent identifier using the same numbering convention used by Wikibase. We note first that all the class items have been assigned Q1–Q7, whereas the individuals belonging to these classes have been assigned Q8–Q20. As mentioned, from a schema perspective, class items and their individuals are exactly the same type of item, and the difference can be appreciated only in the ontology; in fact, we create both types using the same process, through the Wikibase special page ‘Create a new Item’ (`wiki/Special:NewItem`), generally accessible from a link on the left-hand side navigation panel. Consequentially, it makes no difference how identifiers are distributed among class items and their individuals.

In the diagram in Figure 6.2, Q8 is an individual of class (i.e., *instance of*) *MP* that also *has* a link to a second item (Q9), which we identify as an instance of *Speech*. In other words, we have entered the item Q8 by creating a new item in our knowledge base to represent a member of parliament (a label to the item Q8 will possibly identify the name of the MP), and we have also linked it to their speech, Q9.

MP Q8 -> Has Speech -> Q9

Also in the diagram, we note another MP entry, Q12, which is yet to be linked to any speech, but it is linked through a property ‘has’ to item Q11, an individual of class *Interview*. In other words, based on the knowledge base, we can identify interview Q11 as the interview conducted with MP Q12.

MP Q12 -> Has Interview -> Q11

What else do we know about interview Q11? First, we note another incoming connection defined through the property ‘has’ from individual Q13 of class *person*. To keep the ontology behind the diagram simple, we haven’t specified a subclass for this person yet, but we could imagine that eventually we will decide to add an item, ‘interviewer’, and define it as a subclass of Q3, *person*. Second, we note that the item Interview Q11 (similarly to Speech Q2) has two outgoing properties connecting it to two objects in the MediaWiki domain: a transcript and a video recording of the interview. These relationships are described through the properties P4 and P5, as they link respectively to a Wiki page containing the text of the transcript, identified by the page URL, and to a media file stored on the website.

Interview Q11 -> Has P4 -> Transcript URL

Let's now move to describe one of the two postings we have defined in the diagram. For the sake of keeping the diagram simple, we have not related this posting to an MP (but of course this could be achieved using the property 'has' P3). The posting Q19 has two properties: it is associated with the theme Q18 through property P3 and with a media file, an image embedded in the post, through the property P5. We can also imagine a posting to be described by additional properties, not mapped in Figure 6.2, linking it to different data types. For example, since the textual message of a social media post is generally short, we can decide that instead of storing it as an unstructured MediaWiki page, we can add it as an attribute of a new property, 'has message', defined by 'Data type = String'. This would likely also be the approach if we want to describe the date and time of an item; we can define a property with 'Data type = Point in time'. Or if we want to add some quantitative description to an item, such as the number of interactions or the number of reply comments, we can use 'Data type = Quantity'.

But let's consider the item Q18 describing the theme of posts Q19 and Q15 through the property 'has' or the item Q16 describing the mention of an organisation in post Q15. The data entry design we have discussed so far has involved a *convergent* design for the preliminary entry of raw data from our sources into the project's knowledge base. But of course, one of the main advantages of Wikibase is that it supports extremely complex *sequential* design, through which our source data is processed into new data, which can then also be entered in the knowledge base for supporting further analysis. Since the data of the knowledge base is layered both as structured and unstructured information, it affords different kinds of exploration and navigation in support of our analysis. The labelling of postings through the identification of themes or entities, such as those presented in Figure 6.2, can be entered back into the knowledge base. Similarly, social media postings and speeches can be further described through sentiment analysis, using an additional property not illustrated in the diagram, such as 'has sentiment', pointing to a numerical data type. These analytical tasks can be conducted either manually or programmatically by exploring and navigating the data already in the knowledge base.

We will see now how a sequential design for data analysis and data creation might look with Wikibase.

EXPLORING LINKED DATA

Manual Navigation and Discovery

Once the data has been entered into the knowledge base, there are fundamentally two ways to navigate it. A rather traditional approach would be to do a text-based search of the entire knowledge base using the search box at the top of each page on the site. As expected, this feature will return a list of pages and items where the search terms have been found. But this approach will clearly not take advantage of the graphical (that is, linked) nature of the information contained in Wikibase. Since our data has been entered through the definition of items and the relations among these items, it is natural to see this structure as a graph, with items as nodes and relations as the edges. That means that our knowledge base can be navigated from any item, following the incoming or outgoing links connecting that item with the rest of the knowledge graph.

Item-to-item navigation

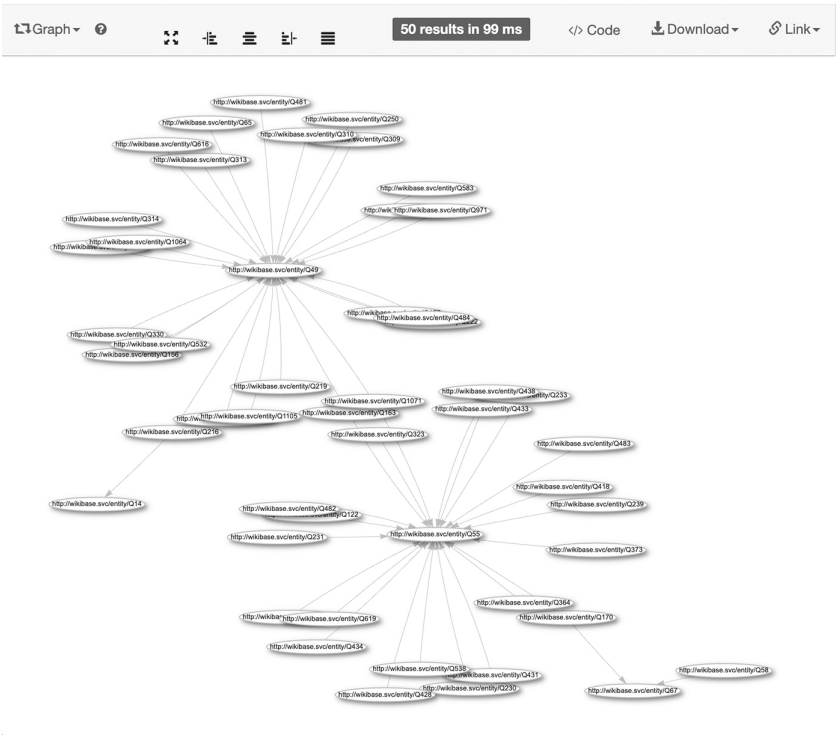
The most natural way to explore linked data in Wikibase is through direct navigation between connected items. Each item page displays all its properties and their associated values or items as clickable links. For example, when viewing an MP item (Q12) from our research project example, researchers can click on any linked speech (e.g., Q10), interview, or social media posting to navigate directly to that item's page. This creates a seamless browsing experience where patterns and relationships emerge organically through exploration. But notably, the links defined in a Wikibase knowledge graph are therefore direct, in our example, from MP Q12 to their Speech Q10 to the corresponding transcript. But what if I want to identify the MP linked the Speech Q10? As introduced in Chapter 4, for links running within Wikibase, thus excluding links pointing to media files or MediaWiki pages, it is possible to reverse-navigate a link by clicking on 'What links here' among the tools listed in the navigation panel on the left-hand side of any item or property page. When we do this from a property page, we reach a list of all items using this property. When we do this from an item page, we list all the items pointing to this item.

The advantage of item-to-item navigation clearly emerges when our knowledge base grows in complexity as we start to add layers of structure. Let's imagine that we have already completed the computer-assisted NER task across our documents, as mentioned above. While browsing an MP's speech about rural development, you click on the item representing an organisation mentioned in the document and discover through 'What links here' that this same entity (i.e., organisation) appears in documents from eight different MPs over three months. Following these connections reveals that MPs mentioning

this organisation also frequently reference related organisations, indicating a cluster of agricultural stakeholders actively engaging with parliamentarians during a period of drought policy discussions. This navigation-driven exploration helps you understand the ecosystem of organisations involved in rural policy debates and identifies which MPs are most connected to agricultural advocacy groups, revealing patterns of sectoral engagement that wouldn't be apparent through simple thematic coding or party-based analysis alone (see Figure 6.3).

Visual relationship mapping

In addition to manually navigating links in both directions, Wikibase also provides a series of tools to visualise the graphical structure of our items and their relationships. Figure 6.3 shows how a visualisation of 50 relationships might



Source: Authors.

Figure 6.3 Connections among items in the Wikibase knowledge graph

look. However, as the knowledge graph grows beyond a few dozen items, full visualisations quickly become an indecipherable spaghetti bowl of connections that obscure rather than illuminate patterns. Therefore, it becomes essential to focus visualisations through precise queries that extract meaningful subsets of the data (something we will explore in the following section on SPARQL querying), allowing researchers to examine specific relationship clusters rather than attempting to display the entire network at once.

Collaborative Annotation and Enrichment

The combination of features offered by MediaWiki and Wikibase offers extremely robust and flexible tools to collaborate on a research project, enabling, for example, sophisticated collaborative annotation that goes beyond simple data entry. Indeed, enabling and facilitating collaborative editing at scale is the foundational goal that has guided the development of this software over the last two decades. Exploring all these features is outside of the scope of this book.³ Yet, it is still useful to briefly describe how researchers might integrate some of these features in their design.

Tagging text and linking structured and unstructured data layers

As mentioned, it is probably good practice to store long, unstructured textual data in a regular MediaWiki page and not as an item structured in Wikibase. Indeed, except for short texts (e.g., social media messages) that can be added as a property of an item in Wikibase, long texts are more easily and comfortably accessed through a traditional web page. Moreover, once the text is in a MediaWiki page, it is possible to collaboratively annotate it in a very intuitive way. Let's consider our example. Our research team has just concluded entering the raw data into the knowledge base (the convergent phase, as discussed above), and now the team of qualitative researchers is tasked with identifying the critical themes emerging from the interviews and from the speeches (which again are stored as regular web pages created using MediaWiki). A way to proceed would be to tag the documents for everybody to see directly from the relevant pages. Each MediaWiki is, of course, fully editable through intuitive user interfaces, which are constantly refined. So, once a researcher wants to add a comment, maybe because they have identified a good candidate for a theme, it is possible to just click on 'Edit' at the top of the page and add a comment

³ For a reader interested in a comprehensive exploration of the features of MediaWiki, we suggest the second edition of *Working with MediaWiki* (Koren & Robbins, 2017), which was updated in 2022 and is fully available online at <https://workingwithmediawiki.com/>, but also the constantly updated documentation at <https://www.mediawiki.org/wiki/MediaWiki>.

where we want it to appear in the text. It is just as important for the researchers to define a consistent way to tag the document so that it is clear during the NLP tasks what part of the document is not to be considered as source text. A very simple way to identify in the document where a new theme seems to emerge is to add an internal link with a double-square bracket. For example, in

Interview with Dr. Sarah Mitchell, Environmental Policy Researcher

Date: 15 March 2024

Interviewer: Research Team Member

...

Interviewer: What about the role of social media in this debate?

Mitchell: It's been fascinating to watch. You see politicians like S. T. and M. R. taking completely different approaches on social platforms. The messaging is very targeted `[[social media targeting (theme)]]` —what they post on professional networks for business audiences is quite different from their posts aimed at community groups.

`'[[social media targeting (theme)]]'` will include an identifiable tag in the text which can then optionally be followed by the creation of a new page with the same name (which will allow the research team, through the feature 'What links here') to map all the documents that point there. A more advance form for tagging a section of a page is through the use of templates. Templates are basically pages with information to be included multiple times in other pages and are highly customisable. They are simply created adding two curly brackets `'{{template-name}}'` to the text of a page.⁴

Tagging a document with links or templates can be a first step in a process that ultimately results, after all the documents have been manually coded, in the formalisation of emerging themes within the knowledge base. For instance, the recurring theme 'social media targeting' would be created as a formal theme item in Wikibase, complete with structured properties and relationships to other concepts. This Wikibase item would also link back to a corresponding MediaWiki page (or template page) containing the full thematic definition, coding guidelines, and examples from the corpus. This bidirectional linking facilitates seamless navigation between the structured data layer (where themes can be queried and analysed quantitatively) and the unstructured data layer (where the rich contextual meaning and coding decisions are preserved), enabling researchers to move fluidly between systematic analysis and interpretive understanding of their data.

⁴ For more details on using MediaWiki templates, see <https://www.mediawiki.org/wiki/Help:Templates>.

The collaborative editing functions of MediaWiki are very advanced and can easily accommodate more sophisticated collaborative, potentially computer-assisted, workflows on the source materials.

Talk pages for research documentation

Another useful feature that the research team could use is the discussion (or talk) page associated with each Wikibase item and, more generally, each page within a MediaWiki site. These pages align with open-science principles by promoting transparency and reproducibility in research workflows. Following the Open Science Framework's emphasis on collaborative research practices and preserved documentation (Foster & Deardorff, 2017), these discussion spaces can be used to asynchronously discuss materials and project development while creating a transparent record of analytical insights that emerge during exploration. Individual researchers can be tagged in discussion pages to facilitate collaboration, echoing OSF's collaborative features that enable controlled access and contributor management. These pages are particularly important for documenting coding decisions and noting data quality issues, supporting the open-science principle of research transparency. For complex mixed-methods projects, these discussion spaces become invaluable repositories of methodological knowledge that inform both ongoing data collection and the eventual publication of findings, creating the kind of preserved, accessible documentation that promotes reproducibility and enables other researchers to understand and build upon your team's work.

Watch lists, templates, and notification systems

Finally, another useful feature for large teams requiring systematic work division is the combination of watch lists and template-based flagging systems to coordinate expertise-specific responsibilities. Researchers can create watch lists to monitor changes to specific items or properties of interest, enabling precise division of labour based on methodological expertise or content domains. This functionality proves particularly useful when different team members have clearly delineated responsibilities—for example, qualitative researchers might watch for updates to interview transcripts and thematic coding, while quantitative team members monitor the addition of sentiment analysis results and statistical annotations of selected items.

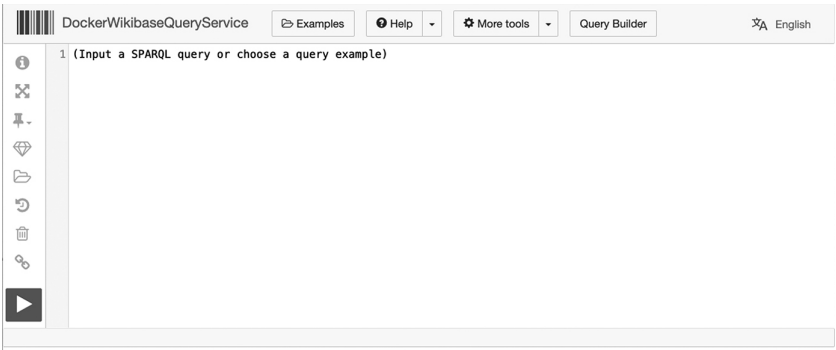
Additionally, teams can implement template-based flagging systems either in regular pages or in discussion pages using custom templates like `'{{needs-coding}}'`, `'{{requires-verification}}'`, or `'{{pending-analysis}}'` to systematically mark content requiring specific types of expertise (remember that each template corresponds to a page that is linked back to where the template is used). These templates can include researcher assignments (e.g., `'{{needs-coding|assigned=researcher2|deadline=2024-07-15}}'`) and automatically populate

tracking categories, creating clear visual indicators of task distribution across the knowledge base. Combined with watch lists configured to monitor these template categories, this approach ensures that work flows efficiently between team members with different specialisations whilst maintaining oversight of project-wide progress and preventing tasks from being overlooked in large, complex collaborative research projects.

QUERYING LINKED DATA WITH SPARQL

Introduction to SPARQL for Research

Item-to-item navigation, crawling the links of our graph, can provide powerful qualitative insights into our data, either structured or unstructured. Yet this approach is clearly limited by our capacity to ingest—at least qualitatively—only a small number of items. If we want instead a comprehensive appreciation of our data based on the very structure that we have defined as we entered the data, we need a different, quantitative approach. SPARQL (SPARQL Protocol and RDF Query Language) is the syntax developed over the last 20 years to access linked-data information and serves as Wikibase’s interface for systematic data retrieval and analysis. In Wikibase, we can use SPARQL through the Wikibase Query Service (also known as WDQS), as seen in Figure 6.4, which, as shown in Chapter 3, should be installed independently from MediaWiki and Wikibase and can then be accessed through a dedicated website (that is, not



Source: Authors.

Figure 6.4 *The Wikibase Query Service interface with the field where to insert the SPARQL query*

outside of the MediaWiki website, although depending on your setting, you can have both sites within the same domain).

Unlike traditional database queries that operate on fixed table structures such as SQL, SPARQL leverages the graph-based nature of linked data to enable flexible, relationship-driven queries that adapt to evolving research questions. In this section we provide a quick, practical introduction to SPARQL, but we refer the reader to other resources for a more comprehensive introduction (an excellent introduction is DuCharme, 2011).

Basic Query Structure

In a SPARQL query users specify the relationships they want to find within the knowledge graph. Every query begins with a ‘SELECT’ clause defining what information to return, followed by a ‘WHERE’ clause containing the graph patterns to match.

For example, to find all the speeches by MPs in our knowledge base we can run this query:

```
PREFIX wd: <http://www.example.something/entity/>
PREFIX wdt: <http://www.example.something/prop/direct/>
SELECT ?mp ?mpLabel ?speech ?speechLabel
WHERE {
  ?mp wdt:P1 wd:Q1 . # MP is instance of MP class
  ?mp wdt:P3 ?speech . # MP has speech
  ?speech wdt:P1 wd:Q2 . # Speech is instance of Speech class
  SERVICE wikibase:label { bd:serviceParam wikibase:language 'en' . }
}
```

This query demonstrates the basic triple pattern (subject-property-object) that forms the foundation of SPARQL. Let’s break down some of the syntax. First, the *variable names* in a query are identified with the ‘?’ prefix. The variable names after the ‘?’ sign do not need to match anything specific in the data and are set in the query and are only to be used within this query (i.e., we can pick whatever name we want). Second, the prefix ‘wdt:’ precedes the property that defines the triple—and in this case the identifier of the *property* must match what is in our knowledge base. Third, similarly, the prefix ‘wd:’ precedes the identifier of one of the *items* contained in our knowledge base.

Before proceeding, let’s examine the prefixes ‘wd’ and ‘wdt’ in more detail (though other prefixes can be defined according to your project’s needs). These prefixes function as shortcuts or abbreviations that replace longer URL strings,

which specify the exact location of properties (P) or items (Q) on the Internet. While the specific letters ‘wd’ and ‘wdt’ are not mandatory, using them provides a practical advantage: when working in the interface shown in Figure 6.4, you can simply hover over these prefixes to see the label of the property or item as defined in your Wikibase instance.⁵

To implement these prefixes effectively, you must specify what they represent within your particular domain context. For instance, if your Wikibase operates under the domain `http://www.example.something`, the prefixes would serve as shortcuts for:

```
wd: http://www.example.something/entity/ (for items)
wdt: http://www.example.something/prop/direct/ (for properties)
```

Naturally, you would replace ‘www.example.something’ with your actual domain name.

Now we can explain the other lines of the query below the ‘PREFIX’ definition (which will not repeat in the other examples). The first line specifies (i.e., ‘SELECT’) what we want to select from the result of the query, indicating the variable names we want to use. Notably, here, the variable names are not defined but referenced; how these variable names will map onto our knowledge base schema is instead defined in the rest of the query. The second and the last line define the boundaries of the proper query with ‘WHERE’ and the opening and closing curly brackets. On the third line, ‘?mp wdt:P1 wd:Q1.’ specifies that the variable ?mp is scoped by the set of items in our knowledge base that are linked to the item Q1 (or the class ‘MP’ in our knowledge base) through the property P1 (or ‘instance of’). The query could stop here and would simply return all the items in the knowledge base that belong to the class ‘MP’. With the other lines, we filter further these results. With the line ‘?mp wdt:P3 ?speech.’, we ask the query to return items of class MP (‘?mp’) that are related through the property P3 (or ‘has’) to *something*, which we assign to the variable name ‘?speech’. Note that we don’t specify here what this *something* should be, and, in fact, if we stop here, this *something* would be interpreted by the query engine as *anything*. But we scope the meaning of ‘?speech’ in the following line: ‘?speech’ must only be something that is related to the item Q2 (the class ‘Speech’) through the property ‘instance of’, as seen, identified with P1. And this concludes the query, which should therefore be intended

⁵ As you might have guessed, ‘wd’ stands for ‘Wikidata Item’ and ‘wdt’ stands for ‘Wikidata Property’ in reference to the project that Wikibase was originally designed to serve.

as a request for returning all the instances of relationships where an item of class ‘MP’ is related to an item of class ‘Speech’ through the property (and only through the property) P3 ‘has’. The final line in the query ‘SERVICE wikibase:label { bd:serviceParam wikibase:language 'en' . }’ automatically retrieves human-readable labels for items, so that can then be ‘SELECT’ed with ‘?mpLabel’ and ‘?speechLabel’.

Integrating SPARQL with Research Workflows

SPARQL queries can be saved, shared among team members as a simple link, and integrated into regular research workflows. For example, results from SPARQL queries can be regularly consulted to understand how many items or how many specific relationships are present in the database. For example, let’s assume that we want to monitor during the coding phase of our project our many social media posts that are associated with each theme we have in the knowledge base. The following query does exactly that:

```
SELECT ?theme ?themeLabel (COUNT(?post) AS ?postCount)
WHERE {
  ?post wdt:P1 wd:Q4 . # Posting is instance of class social media posting
  ?theme wdt:P1 wd:Q7 . # Theme is instance of class theme
  ?post wdt:P2 ?theme . # Post has theme
  SERVICE wikibase:label { bd:serviceParam wikibase:language 'en' . }
}
GROUP BY ?theme ?themeLabel
```

Here we have added some additional SPARQL syntax elements. For the three lines of the ‘WHERE’ statement, simply select all the posts that have any theme. Importantly, this will return a record (i.e., a row) for each relationship post-theme, so the same post associated with more themes will appear over multiple records, and so will the same theme that appears associated with multiple posts. The ‘(COUNT(?post) AS ?postCount)’ as part of the ‘SELECT’ statement simply returns an extra column, named ‘?postCount’ in our result table, with the ‘COUNT’ of the occurrences of the post, while how to ‘GROUP BY’ posts for the count is defined by the last line, which indicates that the count must be the result of grouping posts by ‘?theme’ and ‘themeLabel’, which is the label specified for each theme item.

SPARQL query can also be used to programmatically retrieve data from the knowledge base either to produce tables or visualisations or to be used in some

computer-assisted tasks. We will provide some examples of what these tasks might look like in the next section.

PROGRAMMATIC ACCESS TO DATA

While the manual data entry and SPARQL querying described above provide powerful tools for research, Wikibase also supports programmatic access to its data that can significantly enhance research efficiency and scale. This programmatic approach allows researchers to automate repetitive tasks, integrate with external tools, and handle larger datasets than would be practical through manual methods alone. There are three main ways to access the Wikibase structured data programmatically:

1. By programmatically accessing the query engine (WDQS) for retrieving ('SELECT') and writing ('INSERT') data using SPARQL.
2. By exporting the entire database using a resource description framework (RDF) format to maintain all the triple statements contained in the knowledge base (e.g., item Q1 linked to item Q2 through property P2) and then having it read programmatically by software.
3. And finally, by accessing the Wikibase API, also leveraging the data structure for retrieving or adding information.

Independently from the method of choice, programmatically accessing the data affords a wide and growing range of sophisticated computer-assisted or fully automated analytical tasks. At the time of writing, the most promising opportunities in this field for social science researchers possibly involve network, text, and image analysis. But computer-assisted tasks can also involve error detection in complex schema (e.g., identify misspecification of properties or relationships). A few of these tasks, before moving on to explore an example based on our research project scenario, are outlined in some more detail.

Automation Opportunities

Manual data entry can be tedious and fraught with human error. Thus, the ability to automate some data entry, analysis, and error detection is a significant benefit to using Wikibase. Below are selected illustrative examples of how automation can enhance research workflows.

Automatically assigning properties

Consider the task of categorising hundreds of social media posts by platform. Rather than manually checking each URL, a simple automated script can

examine URL patterns and automatically assign the correct platform property. For example, posts from ‘twitter.com’ or ‘x.com’ get tagged as Twitter, ‘facebook.com’ links become Facebook posts, and ‘linkedin.com’ URLs are classified as LinkedIn content by relating each item to the corresponding platform.

In practice, this can be obtained either by automatically extracting data and metadata of social media posts from the knowledge base (if present) or directly on the source data before entering it into Wikibase for the first time (in fact, for large datasets, manually entering the data is not practical). Social media data and metadata can then be processed with a script to extract platform information before eventually entering the following statement into the knowledge base:

posting Q19 -> has platform -> Twitter Q42

Similarly, parliamentary transcripts often contain standardised metadata that can be automatically extracted—debate dates, speaker identifiers, and session types can be programmatically parsed and entered as properties, eliminating hours of manual data entry whilst reducing transcription errors.

Image classification for visual content analysis

Social media posts frequently contain images that carry meaning that can help us understand ‘human experience and behaviour’ (Chen et al., 2023, 850). It is therefore not surprising that the analysis of these images has been attracting significant scholarly attention. Yet until recently, since the analysis was conducted manually (Chen et al., 2023), the sample of images was necessarily small. But modern image classification tools, and especially generative AI technologies such as LLMs, can now effectively analyse these visual elements and suggest relevant tags and annotations with a higher accuracy than human coders (Törnberg, 2024). An automated system could, for example, send images from MPs’ social media posts to the API of an LLM with a prompt asking the model to identify those containing ‘crowd scenes’ (suggesting public events), ‘formal settings’ (indicating official meetings), or ‘policy documents’ (showing legislative materials). These automated classifications can then be reviewed by researchers and formally added as properties, enabling analysis of how visual messaging strategies vary across different political contexts.

Network analysis

Network analysis represents a powerful research tool that can return important insights not visible through other tools and methods, especially when the dataset is large. An automated workflow could extract all the textual documents

contained in the knowledge graph social media posts from Wikibase, identify the organisations mentioned in their speeches, social media posts, or during interviews using NER processes discussed earlier while then re-entering the information as knowledge base statements linking MPs to organisations through their textual documents. If one MP mentions a regional development authority in a speech, and another MP references the same organisation in a tweet, this creates a potential network connection. Automated scripts can detect these co-mention patterns across thousands of documents, enabling researchers to construct networks where MPs form nodes, and shared organisational mentions create edges between them.

These automatically generated networks can then reveal important structural patterns through community detection algorithms that identify clusters of MPs who consistently reference similar organisations, suggesting coordinated policy positions or shared influence networks. Such analysis might uncover coalition structures that span traditional party boundaries or reveal how certain organisations serve as bridging entities between different political groups. The resulting network structures can illuminate influence pathways, information flows, and hidden coordination patterns that would be impossible to detect through manual analysis alone, transforming the entity recognition data from simple lists into meaningful relationship maps that reveal the underlying structure of political communication networks.

Automated error detection and quality control

Automation excels at identifying inconsistencies and gaps in large datasets. How error detection should be designed is, of course, highly dependent on the type of project and data and must be informed by the experience of researchers with similar situations. For example, a quality control script could regularly scan the knowledge base and flag speeches that lack date properties, social media posts missing sentiment scores, or MP profiles without party affiliations. More sophisticated error detection might identify logical inconsistencies—such as MPs having social media posts dated before their parliamentary service began or sentiment scores that fall outside expected ranges. Additionally, automated systems can detect potential duplicate entries by comparing text similarity across documents or identifying MPs with multiple profiles based on name variations, alerting research teams to data quality issues before they compromise analysis results. Programmatic approaches can handle routine tasks efficiently whilst freeing researchers to focus on higher-level analytical and interpretive work that requires human expertise and judgement.

Retrieval augmented generation for intelligent knowledge synthesis

Retrieval augmented generation (RAG) represents an emerging approach that combines the structured knowledge, such as that stored in a Wikibase

instance, with the natural language capabilities of large language models to enable sophisticated query and summarisation workflows. Rather than requiring researchers to manually construct complex SPARQL queries or navigate through hundreds of individual items, a RAG system can accept natural language research questions and automatically retrieve relevant information from the knowledge graph to generate comprehensive, contextualised responses. For instance, a researcher could ask, ‘What are the main themes discussed by Labour MPs in speeches about climate policy during drought periods?’, and the system would automatically query the Wikibase for relevant speech items, extract connected thematic annotations, identify the appropriate MP party affiliations, and synthesise this structured information into a coherent narrative summary that highlights patterns and connections across the dataset.

This approach proves particularly valuable for large-scale mixed-methods projects where the volume of interconnected data makes manual exploration impractical. An automated RAG workflow could leverage the API to extract relevant subsets of the knowledge graph based on researcher queries, feed this structured context to language models that understand both the semantic relationships and the content of linked MediaWiki pages, and generate research summaries that preserve the nuanced connections between quantitative patterns and qualitative insights. Such systems can identify emerging themes across hundreds of documents, trace how specific policy positions evolve through different communication channels, or generate literature review–style syntheses that draw connections between interview findings, parliamentary speeches, and social media discourse. By maintaining the provenance links back to the original sources within Wikibase, these AI-generated insights remain transparent and verifiable whilst dramatically expanding researchers’ capacity to interrogate complex, multilayered datasets at scale.

Automatic text classification

Text classification has been one of the areas most transformed by recent developments in computational methods and technologies. As vast collections of digitised and born-digital text have become available over the past decade, social science researchers have increasingly incorporated these methods into their research projects. The introduction of the transformer architecture (Vaswani et al., 2017) and its implementation in LLMs has opened up further opportunities for researching large corpora of textual data, enabling researchers to identify patterns and extract insights at scales previously impossible with traditional qualitative methods.

A Case Study: Human-in-the-loop Text Classification

After discussing some opportunities and scenarios, we now briefly summarise how we used Wikibase and human coders to improve the performance of our text classifier for social media data (we will further discuss the design and implementation of this project in Chapter 7). In our social media research project (Kong et al., 2022), we initially used Wikibase to manually code and categorise social media posts about topics like climate change, COVID-19, and vaccination. While developing the project ontology, our team coded as a set of ‘opinions’ specific viewpoints expressed in posts, such as ‘Climate change isn’t real’ or ‘COVID-19 is a scam by elites’. However, manually coding hundreds of posts was extremely time-consuming and couldn’t scale to the millions of posts we had collected in the project.

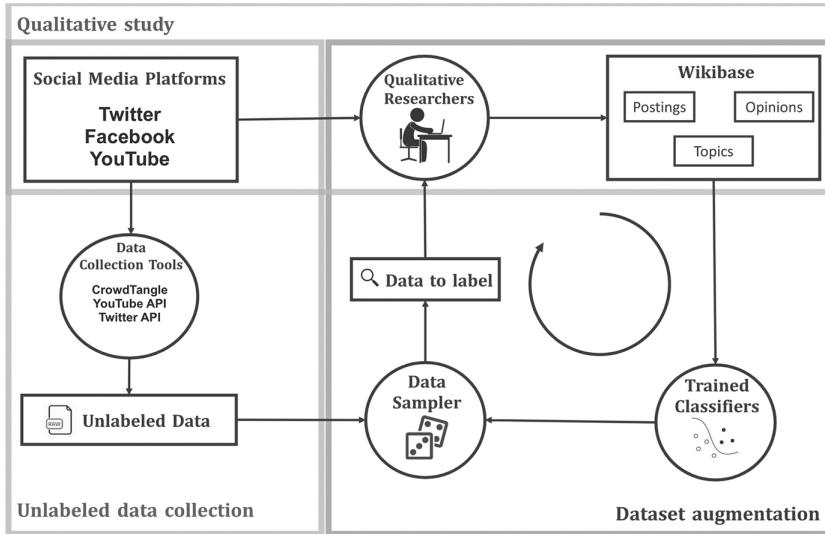
To address this challenge, we developed a human-in-the-loop machine learning approach that used our Wikibase as both the training ground and the destination for automatically classified content. The key insight was that our carefully curated manual annotations in Wikibase could *train* algorithms to recognise similar patterns in much larger datasets.

How the automatic classification worked

Our automated approach involved several interconnected steps (see Figure 6.5) that built upon the structured data in our Wikibase. We began by using SPARQL queries to extract all the manually coded posts and their associated opinion labels from our Wikibase, creating a high-quality training dataset where each social media post was linked to specific opinions our qualitative researchers had identified. This extraction process leveraged the graph-based structure of our knowledge base, ensuring that we captured not just the posts themselves but also their rich contextual relationships to themes, platforms, and temporal markers that our team had carefully established during the manual coding phase.

With this training data in hand, we then developed a text classification algorithm using RoBERTa, a state-of-the-art language model, training it to recognise patterns in the text that corresponded to different opinions. The model learned to identify linguistic features that indicated specific viewpoints, conspiracy theories, or problematic speech patterns, essentially automating the pattern recognition that our qualitative researchers had been performing manually. However, rather than simply training once and deploying the model, we implemented an iterative ‘human-in-the-loop’ approach that maintained the collaborative spirit central to our research design.

This iterative process involved the algorithm classifying new batches of posts, followed by our human experts reviewing a sample of these classifications, correcting errors, and feeding this refined understanding back into the



Source: Kong et al. (2022, 525).

Figure 6.5 The workflow of classifying text involving documents being extracted, classified, and reinserted into Wikibase

Wikibase knowledge base. We continued this cycle through seven iterations until the model's performance stabilised, with each iteration improving both the algorithm's accuracy and our team's understanding of the classification challenges. Once the models were performing consistently well, we applied them to classify over 1.6 million social media posts—a scale that would have been impossible to achieve through manual coding alone whilst maintaining the nuanced understanding that our interdisciplinary team had developed through their collaborative work in Wikibase.

What this achieved

The programmatic access to Wikibase through its API was crucial for this workflow, enabling us to extract training data with complex relationships through SPARQL queries, automatically inject machine-generated classifications back into the same data structure as our manual annotations, track which classifications were human-verified versus machine-generated for transparency, and update classifications in bulk whilst preserving the complete annotation history as we refined our models. Moreover, this process enabled seamless and productive integration of researchers with different skill sets into a unified

workflow. Computer scientists could focus on developing and refining the classification models without needing to engage directly in the nuanced work of opinion mapping, while qualitative researchers and coders could access the automated classification results directly within Wikibase—an environment that had already become familiar through their manual coding work. This division of labour allowed each team member to contribute their expertise most effectively whilst maintaining a shared workspace for collaboration.

7. Collecting, mapping, and modelling social media data with Wikibase

Francesco Bailo, Emily Booth, Marian-Andrei RizoIU, and Jennifer Williams

QUALITATIVE AND QUANTITATIVE SOCIAL MEDIA DATA

Social Media Research as a Natural Venue for Wikibase

Social media research is a good example of a field that can significantly benefit from what is offered by Wikibase. Broadly speaking, social media research is interested in understanding the effects of the widespread adoption of social media technologies. The main reason why social media research projects are particularly well-suited to using Wikibase as their core working platform is the distinctive type of data that is collected and generated. Social media data is inherently multifaceted, combining qualitative content with structured information from content metadata. The scale of data produced by the digital traces left behind by users of social media platforms is immense and unprecedented, both in terms of accessibility—when platforms allow researchers to collect it—and in terms of analytical possibilities. This data enables researchers to focus inward on qualitative content whilst simultaneously scaling outward across quantitative structured data using techniques such as network analysis and natural language processing (NLP). Moreover, social media data is inherently connected in graph form, with posts being linked by users who are themselves linked to other content and communities, creating dense webs of connections that span multilayered geographies.

The possibilities offered by this unprecedented class of data have naturally attracted researchers from multiple disciplines, well beyond the social sciences. According to Web of Science, among the disciplines that have used social media data over the last 10 to 15 years, we find computer science, environmental sciences, medical sciences and epidemiology, geography, business and management, urban studies, ecology, civil engineering and architecture,

economics, linguistics, health policies, ethics, logic and philosophy, mathematics, physics, law, oceanography, zoology, art, and history. Because of this breadth, social media data can play an important role in constructing interdisciplinary mixed-methods research agendas, whilst Wikibase can offer the information and communication infrastructure to facilitate the work of teams operating in this space: from collecting to analysing and publishing the data. Traditionally, researchers' methodological skills—whether qualitative or quantitative—have been the defining characteristic for team selection and project participation. However, Wikibase opens opportunities for both approaches without being constrained by existing skill sets (the only significant technical barrier in the use of MediaWiki and Wikibase is probably its installation). In fact, it creates a platform where qualitative and quantitative researchers can meet, collaborate, and leverage each other's expertise within a unified analytical framework.

What Kind of Data Is Social Media Data?

Social media data encompasses multiple interconnected layers of information that emerge from user interactions on digital platforms. At its most basic level, social media data includes textual content—such as posts and comments—that represents users' thoughts, opinions, and communications. This textual data often contains rich qualitative information about attitudes, beliefs, and experiences, but also structured elements like hashtags, mentions, and URLs that can be analysed quantitatively.

Beyond text, social media data incorporates multimedia content, including images, videos, and audio files. Each piece of multimedia carries both explicit content (what is shown or said) and implicit metadata (technical specifications, creation time stamps, geolocation data). This creates multiple analytical entry points: researchers might analyse visual content qualitatively for symbolic meaning whilst simultaneously extracting quantitative data about posting patterns or engagement metrics.

Interaction data forms another crucial layer, capturing how users engage with content through likes, shares, comments, reactions, and follower relationships. These interactions create network structures that can be mapped and analysed through social network analysis and graph analysis, revealing patterns of information flow, influence, and community formation. Crucially, interaction data also preserves *contextual information*—the surrounding conversations, thread structures, reply chains, and cross-platform references that give meaning to individual interactions. For example, a 'like' might represent agreement, sarcasm, or social obligation, depending on the contextual framework within which it occurs.

Critically, social media data is incredibly reactive to events as it is captured and recorded by always-on sensors. Unlike traditional survey data or archival materials that are mostly static pictures, even if very detailed, social media data exhibits rapid, dramatic shifts in volume, content, and sentiment in response to breaking news, political developments, natural disasters, or viral phenomena. This reactivity creates additional analytical dimensions: researchers must account for baseline patterns versus event-driven spikes, understand how different events trigger different types of responses, and capture the temporal dynamics of how conversations evolve from initial reactions through sustained discussion to eventual decline. The reactive nature means that social media data can serve as a real-time barometer of public response but also challenges researchers to develop methodologies that can distinguish signal from noise (including inauthentic content produced by bots or large-scale coordinated behaviour) and contextualise momentary reactions within longer-term patterns.

WHY INTERDISCIPLINARY RESEARCH IS HARD

Definition of Interdisciplinary Research

Interdisciplinarity is defined as ‘any study or group of studies undertaken by scholars from two or more distinct scientific disciplines’ that is ‘based upon a conceptual model that links or integrates theoretical frameworks from those disciplines, using study design and methodology that is not limited to any one field, and requires the use of perspectives and skills of the involved disciplines in all phases from study design through data collection, data analysis, specifying conclusions and preparing manuscripts and other reports of work completed’ (Aboelela et al., 2007, 340–41). In other words, interdisciplinarity requires academics to leave the comfort zone defined by their disciplinary scholarly traditions to meet others in a space that belongs to no single discipline—a new intellectual territory constructed by integrating values and approaches from different perspectives.

Why Pursue Interdisciplinary Research?

If interdisciplinary research is so challenging, why do academics embark on it in the first place? We suggest there are two main reasons that might require or justify an interdisciplinary approach. These two reasons are not mutually exclusive; in fact, they often coexist. First, researchers might realise that a specific research problem of interest escapes the boundary developed by a single discipline. For example, let’s consider the example of digital misinformation during a health crisis such as the 2020 COVID-19 pandemic. There are,

of course, many perspectives from which to tackle this particular research problem (we can define it somewhat more precisely with ‘Why does misinformation emerge and flourish across social media platforms during a health crisis?’). If we are political scientists, we immediately see the political implications of fringe actors trying to score political points while discrediting ruling institutions and, more generally, elites. If we are public health scholars, we instead see the implications of people engaging in potentially dangerous behaviours by refusing vaccines or social distancing norms. If we are sociologists, we see the structure of social networks that encourage these behaviours. Finally, if we are computer scientists, we see the arrangement of computer networks and algorithms that select and diffuse content through platforms that contain potentially harmful information. While each discipline can independently provide interesting and relevant perspectives on the research problem, no single discipline can aspire to answer it comprehensively. The problem is simply too complex and operates across multiple dimensions. An interdisciplinary approach therefore encourages different disciplines to contribute toward developing an integrated understanding of the problem and offers the most comprehensive answer possible.

A second reason for pursuing interdisciplinary research is that the research problem is rooted in complex, multimodal, and multidimensional data. Comprehensively analysing this data might therefore require the involvement of different analytical capabilities (e.g., because the data combine qualitative dimensions with quantitative dimensions and operate at a large scale) but also the use of different interpretative perspectives. Again, social media data exemplifies this kind of dynamic. Social media data is produced at an unprecedented scale while still offering many qualitative dimensions, such as rich textual descriptions and images. Moreover, social media data requires interpretation within a context shaped by the affordances that emerge from the interaction between technological and human dimensions. The study of these social media affordances is necessarily an interdisciplinary endeavour.

In short, interdisciplinary research, while challenging, allows us to come closer to capturing the real-world complexity of social phenomena, their causes, and impacts.

The Main Challenges of Interdisciplinary Research

After having to justify the need for an interdisciplinary approach, let’s discuss, more specifically, the challenges that an interdisciplinary team might expect to face.

First, different disciplines might produce (or possibly be the product of) very different *worldviews*. Creswell and Creswell write that ‘individuals develop worldviews based on their discipline orientations and research communities,

advisers and mentors, past research experiences, and cultural experiences. Individual researchers' beliefs based on these factors will often lead to embracing a strong qualitative, quantitative, or mixed methods approach in their research' (2022). Some of these worldviews are hard, if not impossible, to reconcile. For example, a digital ethnographer might contend that people's subjective meanings resulting from experiencing social media platforms cannot be compressed into aggregated data from the digital traces left behind by hundreds, thousands, or often millions of users. The same digital ethnographer will not dismiss the relevance of computational methods to create *some* knowledge, but that knowledge will be different from the knowledge that only emerges from the analysis and understanding of subjective (and necessarily individual) meanings and experiences. We can refer to this challenge as the epistemological challenge.¹

Closely linked to the epistemological challenge is the methodological challenge. Methodology is the rationale for choosing among the methods that are available to the researcher and that are compatible with the kind of data we want to research. In this sense, our methodology derives from our epistemology and allows us to link it in practice to our data to produce new knowledge. That means that our methodology is necessarily constrained by our epistemology (and worldview) as much as by the methods and data that are available to us.

The epistemological and methodological challenges are arguably the most difficult to address because they have been hardwired into respective disciplinary traditions over decades. We do not argue that Wikibase can magically resolve these tensions; however, as we will demonstrate, it can help contain these differences whilst creating a shared space for producing knowledge through interdisciplinary research. This is achievable because Wikibase effectively addresses two additional challenges that, whilst not primary, remain significant barriers to collaboration.

Technical barriers and communication challenges are often the first challenges encountered by researchers involved in an interdisciplinary project. Technical barriers emerge from a lack of experience with handling some types

¹ It should be noted that it is sometimes recommended in debates about research design and a researcher's worldview to explicitly address and differentiate between *ontological* ('What's out there to know about?') as well as *epistemological* ('What can we (hope to) know about it?') questions (Hay, 2002, 64). This is because it might still be possible to have the same epistemological position without sharing the same ontological position. Yet, we think that the most practical challenge for the purposes of designing a project's knowledge base when it comes to differences in terms of worldview is indeed the epistemological challenge, since it most directly affects the data creation process.

of research data or from the technical tools that are required to deal with such data. For example, a computer scientist might have no experience in dealing with qualitative data such as field notes produced through a digital ethnography and wonder how these notes—which can be effectively stored as MediaWiki pages—enter the research workflow and how they are analysed. Conversely, a qualitative researcher might wonder how to technically deal with thousands or (sometimes) millions of documents. Storing, accessing, and analysing these documents clearly requires technical (i.e., computational) skills that are outside of the experience of a qualitative researcher.

Communication challenges are instead more subtle and can emerge during the entire life cycle. They can involve a misunderstanding of expectations about what the project should deliver in terms of insights and knowledge but can also involve a terminological mismatch around words that can have very specific meanings in different disciplines. For example, the act of ‘coding’, and consequentially the act of creating a codebook, clearly has different meanings in the context of the social sciences and computer sciences. And so, the word ‘significant’ might have a different connotation when it is mentioned in the context of a statistical analysis or in the context of a qualitative analysis. But let’s also consider the process of data preparation or data ‘cleaning’, for example, in the context of preparing texts for further analysis. Data preparation might involve screening data with the purpose of excluding some of it because it is irrelevant (e.g., stripping HTML tags from a webpage) or transforming data to make it compatible with the project methodology and methods (e.g., translating social media posts across languages or averaging responses of a survey into an index). But disciplinary standards and protocols might differ significantly, and data preparation can have very different meanings across disciplines, even when using very similar tools.

Let’s consider the task of analysing a corpus of texts with NLP in the context of computational, quantitative text analysis. Different disciplines may have fundamentally different approaches to preparing and ‘cleaning’ text before applying the same text modelling algorithms (such as latent Dirichlet allocation). In essence, disciplines vary in their answers to crucial questions: which parts of the text contain the richest and most informative content for analysis, and which parts should be excluded to improve analytical accuracy? Consider the treatment of stop words—common words like ‘the’, ‘and’, or ‘but’ that appear frequently across texts. One of the most influential textbooks on natural language processing notes that ‘sometimes, some extremely common words that would appear to be of little value in helping select documents matching a user need are excluded from the vocabulary entirely. These words are called stop words’ (Manning et al., 2008, 25). Removing stop words has long been considered standard practice in NLP (see also Bird et al., 2009, 60–61). However, a linguist employing stylometry to understand an author’s distinctive style and

tone might consider stop words essential to their analysis. For stylometric analysis, these seemingly mundane words often carry the most diagnostic information about authorial identity and writing patterns. Where computer scientists see noise to be filtered out, linguists see crucial stylistic fingerprints that must be preserved. Such an example effectively demonstrates how interdisciplinary collaborations face significant challenges when clear communication about integrating different disciplinary approaches is lacking throughout the project life cycle—not merely during goal-setting phases. Without such integration, each discipline will assume its own practices represent the standard to follow.

To this end, interdisciplinary projects require continuous dialogue about protocols, terminology, and methodologies to ensure clear progress and effective collaboration. While ideally these discussions would occur upfront, methodological differences often emerge only when they become directly relevant to the work at hand. Maintaining an accessible record of decisions and project-specific definitions as they develop helps preserve consistency and prevents confusion later. While convergence on shared understanding is important in any project, it becomes especially critical in interdisciplinary work, where team members bring distinct approaches and vocabularies that must be reconciled to address the unique collaborative challenges these projects present.

DESIGNING PROJECT ONTOLOGIES IN PRACTICE WITH WIKIBASE

This section draws directly from our experiences with two multiyear interdisciplinary research projects. The first project was funded by Facebook, now Meta, to research online problematic speech across different platforms. The second project investigated Internet-mediated influence operations in the Pacific and was funded by the Defence Innovation Network. Both projects involved social scientists with different methodological expertise and computer scientists. Both projects demonstrate how Wikibase can be leveraged to address in practice the challenges of researching social media data as well as the challenges of interdisciplinary research identified earlier in this chapter.

We discuss how Wikibase afforded the development of the two projects' ontologies, which played a critical role in giving coherence to the interdisciplinary approach while defining the structure of the data. As discussed in Chapter 5, defining an ontology is critical because it provides the blueprint for progressively adding structure to unstructured and semistructured data and, thus, provides an interface for coherently integrating the different disciplines into a single working space. A well-defined, comprehensive project ontology and the collaborative work required for its compilation represent the most effective way to address both the epistemological and communication challenges outlined above. Wikibase facilitates this collaborative work and enables

clear communication of its results, as the project ontology remains accessible even as it evolves and becomes intuitively navigable by the entire research team. Below, we explore two examples of ontology development using two of our own real-world interdisciplinary projects.

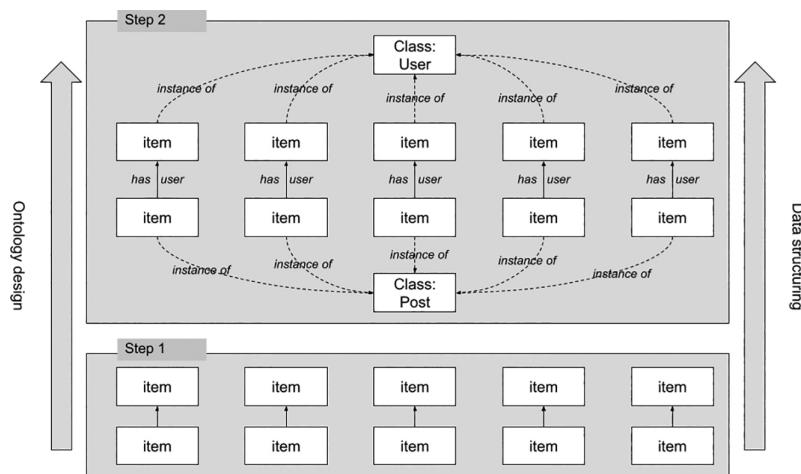
Fundamental Ontological Structures for Social Media Data

Both projects began with a preliminary mapping exercise through qualitative observation of the social media space. Only a very preliminary schema and ontology must be defined for these observations to become part of the knowledge base. This preliminary structure could easily replicate the inherent structure of social media content and data (e.g., with individuals identified as ‘social media posts’ and ‘users’ and very simple relationships such as ‘has’). This will allow the team to quickly move into the data collection phase before a more comprehensive discussion about the design of the project ontology.

Here, we briefly return to the difference between schema and ontology (as discussed more extensively in Chapter 5). Data schemas define the structural relationships of data for efficient storage and retrieval, whilst ontologies encode both meaning and relationships in ways that enable a shared understanding between humans and computational systems. In this sense, entering structured information into Wikibase requires only a basic schema—one that can be remarkably simple. At its most fundamental level, we need only establish that item A relates to item B. This initial schema doesn’t require meaningful definitions of what items A or B represent, nor does it need to specify the nature of their relationship. The ontological layer can be developed subsequently, as the research team establishes how the relationships can be best represented to extract meaning that is most valuable for meeting the aims of the project.

For example, as illustrated in Figure 7.1, after entering numerous items into our knowledge base and establishing their connections, we can retrospectively define the property ‘instance of’ to specify the relationship between an item and its broader category. This property provides meaningful classification by identifying what type of entity the item represents. Through this process, we simultaneously enhance the complexity of our schema whilst creating an ontological framework that imbues our data structure with human-comprehensible meaning.

In our two projects, the two-step process described in Figure 7.1 happens multiple times through an *iterative* process. As mentioned, as a very preliminary step, in each project our researchers were tasked to collect observations of social media channels identified as potentially relevant. This first preliminary mapping exercise was conducted with the goal of qualifying the social media space of interest by collecting exemplars of social media posts and interactions. The first exemplars were then entered into Wikibase without a detailed



Source: Authors.

Figure 7.1 The ontology emerges from a simple data schema

ontology. In fact, at this stage, during the first project, the ontological layer did not exist yet, and our data was only ordered by a very basic schema, with items (later to be identified as instances of social media posts) being simply defined by a 'label' (e.g., 'Facebook post') and with the textual content of the post and possibly its URL stored in the description field of the item.²

As the knowledge base increased in size with the progressive collection and addition of new social media posts, we decided to add a layer of structure by capturing what we determined to be the most important metadata of social media posts: the user who posted it, the date of publication, and the URL linking to the post on the Internet. This required defining two new classes for the individuals of our ontology, *social media posts* and *users*, as well as new relationships such as 'instance of' to link an item to another item specifying its class, 'has date' to link a post to a datetime value, 'has author' to link a post to an item of class user, and 'has text' to link a post to a text value. These relationships (or properties in Wikibase) allowed our data to start reflecting an *ontology*, as more meaningful statements linking our data could be finally formulated. In our second project, this preliminary step was skipped as we imported some of the definitions of the ontology developed early in the first

² In Wikibase, an item label and description are part of its fingerprint and do not involve a statement linking the item to some other entity through a property.

project. That is, in our second project, even before we started collecting data from social media, we set up a working ontology for data collection so that our researchers could enter social media data and its corresponding metadata in a single step, instead of returning to the same items later with additional information.

This difference across our two projects in terms of how the data collection was implemented in the very preliminary phase of each project illustrates the flexibility of Wikibase to adapt in the face of different levels of existing knowledge about the project's data and its ontology. Wikibase enables researchers to start a project without predetermined assumptions about data structure or the knowledge base. The data schema and ontology emerge naturally as the project develops. How this will happen depends entirely on the perspective, expertise, and goals of the research team. In an extreme version of this scenario, an ontology might not be required and therefore will never emerge, with the team only relying on a data schema to structure its data. Under a different scenario, an ontology (or a schema) might instead be already available, either because the team has experience with that type of data and project or because the project is going to use an established ontology and therefore align its data to some existing standards (which could eventually facilitate linking the project data to other existing data repositories). We believe that the most common scenario sits somewhere in the middle: it is likely that some reference ontology might already exist (and facilitate, at least early on, the data collection), but it is also likely that the team will find it necessary to adapt it as more data and more knowledge about the data (and project) become available. This midway scenario is also what we experienced in both of our projects. If the ontology providing some preliminary structure to social media data quickly emerged from social media metadata, a tailor-made ontology was instead developed to capture and structure the content of what was discussed in the collected posts.

Developing an Ontology About Textual Data Content

After a first round of data collection and after reaching a sufficient degree of diversity in what was captured by the data, both of our projects faced the challenge of developing an ontology to map, first, the content presented in the documents (i.e., social media posts), but also a method to relate these posts meaningfully and coherently to their context.

Qualitative and quantitative topic analysis

In our first project we investigated the intersection of misinformation and hate speech within Australian social media discourse (Bailo et al., 2024; Johns et al., 2024; Kong et al., 2022). In the first phase, we developed a preliminary thematic ontology mapping the content of the posts against only four wide topic

areas, while also ensuring that all collected data connected to at least one of the following thematic areas: climate change, LGBTQ+ issues, immigration, and vaccines. In parallel to collecting and entering the data in Wikibase, our researchers also compiled in MediaWiki a set of detailed field notes. These field notes captured details such as the pages visited and posts reviewed, as well as impressions of how the content fit into the overarching goals of the page or group. For example, some politically organised groups' reasons for protesting COVID-19 vaccines shifted during the two-year duration of the project, and the field notes were able to capture these changes while adding context from timely news events. Recurring opinions and rhetorical patterns documented in project field notes were systematically added as properties, functioning as first-level codes that could capture specific forms of climate change denial, anti-immigrant rhetoric, anti-vaccination stances, and anti-LGBTQ+ discourse.

As social media observation progressed, it became evident that our preliminary ontology required expansion to accommodate the nuanced ways that hateful and misinformative content manifested within these discussions. Upon reaching data saturation at the end of the first year of the project, the team developed higher-level thematic categories that distinguished between different forms of problematic speech—misinformation, hate speech, and propaganda—whilst capturing the targeting strategies and narrative structures characteristics of each. We also added a layer of coding that captured the types of hate speech that were deployed, such as racism, antisemitism, and sexism, which allowed us to map how different forms of discrimination overlapped and when they were likely to co-occur. This iterative process of ontological refinement enabled us to examine not only what constituted hate speech within each topic area, but also how misinformative and hateful content reinforced each other across different domains of Australian online discourse.

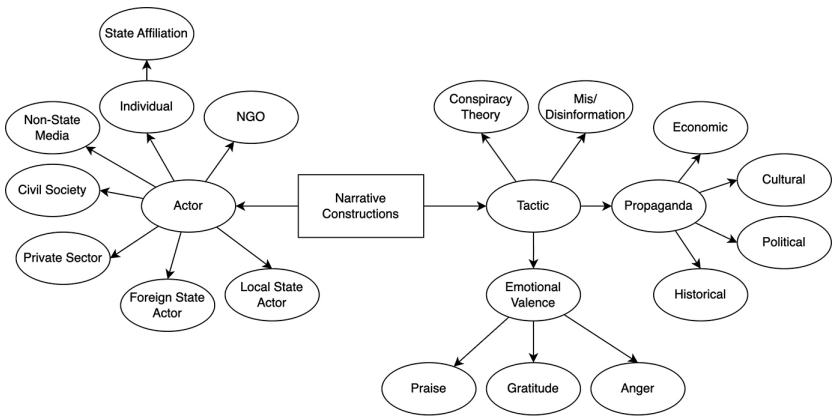
Multi-axial iterative codebook development

In our second project, we analysed online political discussions across three Melanesian nations: Vanuatu, Solomon Islands, and Papua New Guinea. Following the methodology from our first project, we developed an initial ontology to capture essential details of posts and their associated comments, then iteratively expanded it to better reflect the thematic landscape we observed. We adopted this exploratory, grounded theory approach because this region consists of small social media environments under unique sociocultural and political circumstances. The region generally remains understudied, and applying theoretical frameworks developed for WEIRD (Western, educated, industrialised, rich, and democratic) nations or large-scale social media platforms risked distorting both our data and our understanding of the broader environment. Wikibase proved ideal for this flexible, iterative methodology. Our coding process was developed in two phases.

Initially, in the first phase, codes were developed during the data collection process to build a codebook based on the broad thematic content of the posts and comments. However, during this process, we recognised that understanding what users were saying provided only partial insight into these social media interactions. Thus, our research interests had expanded to also understanding the *who* and *how* of these political discussions that take place.

Thus, in phase two, to address these new research questions, we introduced a second coding axis focused on the characteristics of social media content—what we termed ‘narrative constructions’ (see Figure 7.2 for an exemplar narrative construction codebook). This multi-axis approach allowed us to examine both thematic content and communicative characteristics simultaneously. Not only could we categorically aggregate what was being said, but also by which actors, how they employed certain tactics of persuasion, and the veracity of the claims made.

Thematic codes and narrative constructions were initially captured by our Wikibase ontology under the same ‘has theme’ property. However, we later coded narrative constructions under an independent ‘has narrative construction’ property to enable more practical dimensions of analysis without requiring manual post-processing to separate themes and narratives. This change in coding strategy exemplifies the iterative ontological process by which we structure the dataset based on our research goals and knowledge of the data.



Note: This codebook has been simplified for this exemplar.
Source: Authors.

Figure 7.2 Exemplar of narrative constructions codebook

For an even more descriptive range, narrative constructions could be captured under several individual properties. For example, using the codebook in Figure 7.2, data could be represented in the ontology with a single ‘has narrative construction’ property or with a combination of ‘has tactic’, ‘has emotional valence’, and ‘has actor’. The level of descriptive range you employ will depend on the nature of the project and the specificity of the research question. The more properties specified in the ontology, the more time and effort are required to enter and code data. Thus, a cost-benefit analysis and the timeline for the project should be considered when deciding on the number of thematic and narrative properties to include. The flexibility provided by Wikibase makes perpetual ontological expansion easy and tempting, but possibly counterproductive for short deadlines and intuitive data coding and analysis.

After recoding, we then extracted this dataset (containing just under 700 individual items and associated metadata) into a single CSV file with columns pertaining to the item’s name, its geography, and associated themes and narrative constructions for further analysis and visualisation in R. Using this table format, we grouped items by their geography, themes, and narratives to examine (1) which themes occurred most often, (2) how these themes were generally communicated and by who, and (3) whether these characteristics differed between geographies. We note that this analysis can be done natively in Wikibase using SPARQL, but we chose to do so using the `ggplot2` and `ggpubr` packages in R for the purposes of generating specific publication-quality aggregated visualisations.

Wikibase’s flexibility allowed us to work with multiple, complex axes of thematic coding to answer several research questions in a large dataset in a way that many traditional qualitative tools could not facilitate within the same time frame. Typically, qualitative analysis can be difficult to visualise given the lack of data structure and the traditional, manual method of analysis. Translating this form of data into a machine-readable format in Wikibase allowed us to visualise our qualitative data in new ways to suit our research.

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